StandardEyes — Statistical Reference and Walkthrough

Comparative outcome analysis for refractive surgery · version 1.1.0

Companion document to the main manuscript. Part A states what the software does statistically for every option you can choose, including how its behavior changes with more than two groups. Part B walks through a complete analysis of the example dataset, screen by screen and sheet by sheet.

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Every screenshot and figure in Part B is an unmodified capture of the software, or of its output, running on the example dataset distributed with the app (Example Data\Example_Refractive_Dataset.xlsx). The dataset is simulated: it contains no patient information, its generating model and true effects are written in its own README sheet, and its numbers illustrate the software's behavior rather than any clinical result. Every number quoted from an example comes from the release run of StandardEyes 1.1.0 on the seven example settings files.

PART A · STATISTICAL REFERENCE

A.1 The data model

The software expects **one spreadsheet row per eye, per surgical procedure**. Everything else follows from that convention.

Role you assign	What it is used for
Patient / cluster ID	Links the two eyes of one patient. It is the clustering unit of every clustered test (tables, GEE, regression, figures, astigmatism) and of the patient bootstrap, and the denominator of patient-level statistics.
Eye / case ID	Row identity; used for traceability (the *Astigmatism vectors* sheet names each eye by it).
Group source column	The column whose values define the groups being compared.
Reference group	The comparator. Every effect estimate, matching pairing and adjusted model is expressed *relative to this group*.
Patient-level parameters	Variables that describe the patient, not the eye (age, sex, smoking, surgeon, a per-patient treatment setting).

Patient-level parameters. If a patient contributes two eyes, their age appears twice. Summarizing it per eye inflates the sample and understates the standard error. A variable flagged as patient-level is collapsed to one row per patient (per patient and group) before it is described or tested: the first recorded value is kept, patients whose eyes disagree are named once in the run log, and the row is tagged **[per patient]**. As soon as any variable is patient-level, the column headers print both denominators, e.g. `Reference (n = 226 eyes; 136 patients)`. Such a variable is also never the outcome of an eye-level GEE (A.9).

Missing patient identifiers. A row whose patient ID is blank is given its own token, `(no ID) row <n>`, so that eye counts as its own patient everywhere: tables, GEE, figure *P* values and the bootstrap. The number of such rows is written to the run log and the Manifest sheet. The tokens are given after the inclusion rules have run (n is the row's position in the original sheet), so a rule that excludes rows without a patient ID still works. When no patient-ID column is mapped at all, the summary tables and the figures use the classical tests, and GEE and the adjusted models treat every row as its own cluster and say so in the sheet note, rather than refusing to run.

Reading the spreadsheet. The sheet is read by R's `readxl` package, and the interface types the columns it shows in the same way (a column is numeric, text, date or logical according to the cells it holds), so the live counts and previews see the same values as the analysis. A cell formatted as a date inside a numeric column is a data-entry error whose serial number would otherwise pass numeric filters and distort means: it is set to missing, and the run log names the column, the number of such cells and an example.

Column names in the output. Internally each column is given a plain ASCII key, which the model formulas need, but everything you read uses the header **as typed in your file**, including non-Latin scripts, accents, spaces and symbols: the *Parameter* labels of every table, the GEE and regression sheets, the within-group sections, the PSM covariate lists, the balance-plot labels, figure text that names a column, the run-log lines, and the headers of the filtered CSV and of the *Matched_Data* sheet. A header that occurs twice is kept apart as the data preview shows it (Age, `Age.1`).

Retreatments. A second procedure on the same eye is simply another row. A treatment-number column can therefore be used as a filter (`== 1`) to restrict the analysis to primary procedures, which is the usual choice.

A.2 Refraction standardization (plus → minus cylinder)

Exports frequently mix plus- and minus-cylinder notation. When enabled, each nominated sphere/cylinder/axis triple is transposed to minus-cylinder form by the standard rule:

new sphere = sphere + cylinder · new cylinder = −cylinder · new axis = axis ± 90°

1. It is applied to **every triple you nominate** (preoperative refraction, treatment parameters, postoperative refraction), never guessed from column names. Once the preoperative and postoperative sphere, cylinder and axis roles are mapped (*Column roles*, Special outputs tab), those two sets are filled in; any other triple you add yourself.
2. Only rows recorded in plus notation are altered; rows already in minus form are left alone, including their axis. A transposed row's new axis is written in the clinical range 1–180. An axis outside 0–180 in an untouched row is reported in the run log and left as recorded.
3. The summary tables, filters, models and the mEYEstro / AstigMATIC tabs then see one consistent notation. In the filtered CSV and *Matched_Data* the transposed columns carry the suffix `_minus_cyl`, because their values are no longer the recorded ones.

The astigmatism vectors (A.14) do **not** depend on this switch: they convert every row by the sign of its own cylinder, so a refraction typed as −2.00 × 180 or as +2.00 × 90 gives the same vector with the switch on or off.

A.3 Visual acuity: notation, logMAR analysis and Snellen display

Declare the notation. One setting, *VA notation in your file* (Variables & Types tab), states how every acuity column is recorded: the four VA roles (preoperative and postoperative UDVA and CDVA) and every column listed for logMAR analysis.

Notation	Example	How it is read
Decimal (default)	1.0, 0.5, 0,8	as written; a comma decimal is accepted; a value ≤ 0 is impossible
logMAR	0.00, 0.30, -0.10	decimal = 10^(−logMAR); values outside −0.5 to 3 are impossible
Snellen fraction	20/40, 6/12, 20/25-2, 6/9+1	20/x and 6/x alike; logMAR = log10(b/a) + 0.02 per letter missed ("−k"), − 0.02 per extra letter ("+k")
ETDRS letter score	85	logMAR = 1.7 − 0.02 × letters (0–100)

- **Off-chart entries** (CF, HM, LP, NLP and their spelled-out forms) become missing and are counted in the run log. They are never given an invented acuity; the mEYEstro tab shows the recorded label in its Snellen column for those eyes.
- For every acuity column the run log states how many values were read, how many could not be parsed (with examples) and how many were off-chart. **The run stops with a clear message** when the declared notation is contradicted: more than half of the non-blank cells unreadable, any decimal value ≤ 0, or a logMAR value outside its range.
- **Decimal acuity declared as ETDRS letters** stops the run when more than half of the column's analyzed numeric cells are not whole numbers or its largest value is below 5. The interface also warns, under *VA notation in your file*, when the mapped columns look like another notation.
- In Excel a typed Snellen fraction such as 6/12 becomes a date; keep Snellen columns formatted as text.

Filters see your own units. Exclusions and filter rules are applied to the values *as recorded in the file*, before any conversion.

logMAR is the default. *Analyze visual acuity in logMAR* is on by default. A settings file without a column list converts the four mapped VA role columns; a list that is present (even an empty one) is used as given. Each listed column is analyzed as logMAR = −log10(decimal acuity), so lower values mean better vision. A file already in logMAR is analyzed as recorded and never converted twice. With the switch off, the column is analyzed as decimal acuity.

How an acuity cell reads. A logMAR acuity row of a summary table reads

`20/20 (0.00 ± 0.05 logMAR); range 20/16 to 20/32`

- the first value is the **Snellen geometric mean**, 20/(20 × 10^(mean logMAR)); then the logMAR mean ± SD; then the range from the best to the worst eye, in Snellen;
- **the Snellen rule:** a logMAR value within 0.005 of a 0.1-logMAR chart line prints that line's standard label (20/10, 20/12.5, 20/16, 20/20, 20/25, 20/32, 20/40, 20/50, 20/63, 20/80, 20/100 … 20/1000); any other value prints 20/ followed by 20 × 10^v rounded to a whole number (one decimal below 10). So a mean of −0.04 logMAR reads 20/18, and no mean or range ever carries a chart-letter suffix such as "20/20+". The lettered notation appears only where a single recorded acuity is echoed back (the mEYEstro tab);
- "logMAR" is written with a lower-case l everywhere.

The figure headers use the same form without the range. **The figures and the export tabs always work from decimal acuity** derived from the declared notation, whatever the logMAR setting: the acuity panels bin decimal acuity at the standard Snellen cut-offs, so the logMAR setting does not change them. A decimal-to-Snellen table is embedded, so no external workbook is needed.

The Overview sheet names the columns analyzed in logMAR, the Manifest records the *VA notation* and the *logMAR conversion*, and the filtered CSV and *Matched_Data* mark converted columns with a `_logMAR` (or `_decimal`, for acuity text read as decimal) suffix.

A.4 Variable types and the tests of the summary tables

Every variable you place in Baseline, Preop / Intraop or Outcome is given one of four types. The type sets both the presentation and the test.

Type	Reported as	Patient-clustered test (default)	Classical test, two groups	Classical test, more than two
Continuous	mean ± SD	Gaussian GEE of the value on the group	Welch *t*-test	Welch ANOVA
Ordinal	median [IQR]	Gaussian GEE on the ranks	Wilcoxon rank-sum	Kruskal–Wallis
Dichotomous	n/N (%)	binomial GEE of the event	χ^2, or Fisher exact	χ^2, or Fisher exact
Categorical	n/N (%) per level	two levels: binomial GEE; more: joint Wald test on stacked level indicators	χ^2, or Fisher exact	χ^2, or Fisher exact

Means, SDs and medians are printed with two decimals and percentages with one.

When the tables use the patient-clustered tests

With the default *Summary-table tests* = **auto** (GEE tab), the *P* value of a per-eye row of the summary tables and of the Publication Table comes from a **patient-clustered GEE** when both of these hold:

- a patient-ID column is mapped and *Cluster p-values by patient* (Special outputs tab) is on;
- at least one patient contributes **two of that row's analyzed eyes** (the check is made per row, on the eyes with a value).

The models are the ones behind the clustered *P* values of the figures (one implementation), so a table row and the figure of the same outcome give the same *P* value. Every group term is tested jointly, so with more than two groups the *P* column holds one omnibus test. The test uses the small-sample correction of A.9 (a *t* reference for one comparison, *F* for a joint test) and an exchangeable working correlation, refitted with an independence working correlation where the exchangeable fit is not accepted (A.9); the stacked level indicators of a categorical variable always use an independence working correlation. The Test column names all of it, e.g. *GEE, exchangeable working correlation, clustered by patient (robust SE (Mancl-DeRouen), t reference)*.

The **classical tests** in the table above run instead:

- when no patient-ID column is mapped, or patient clustering is switched off;
- when every patient has a single analyzed eye in that row;
- for a **[per patient]** row of an unmatched cohort (it already has one row per patient);
- for every unweighted row when *Summary-table tests* = **classical**. The figures then stay clustered; the Publication Table, the figure legend files and the Manifest say that a table and its figure can then give different *P* values.

Details of the classical tests:

- Welch forms are used, not Student's *t*-test or Fisher's ANOVA, because refractive groups usually differ in size and spread. Welch ANOVA is undefined when a group has zero variance; three or more groups are then compared by Kruskal–Wallis, and the Test column says so.
- Ordinal variables use R's `wilcox.test` (exact distribution for small samples without ties, otherwise a continuity-corrected normal approximation) or Kruskal–Wallis.
- Categorical tests switch automatically. Empty rows and columns of the contingency table are dropped first. If every expected count is at least 5, χ^2 is applied without continuity correction; otherwise Fisher's exact test. A table too large for the exact algorithm gets a Monte-Carlo Fisher test (20,000 replicates, fixed seed), labeled so.

When a clustered model cannot be fitted

A clustered GEE that was fitted but failed (a group with no events or only events, which has no finite estimate; no convergence; a numerically singular covariance) is replaced by a **permutation test over patients**, never by an eye-level test. Group labels are permuted across patients with each patient's eyes kept together; the eyes of a patient who has eyes in two groups (a contralateral design) exchange their labels within the patient, so that pairing is kept. The statistic is the between-group sum of squares of the eye means (numbers, ranks or 0/1 outcomes) or the Pearson χ^2 statistic of the level-by-group table (categorical variables); B = 9999 permutations with a fixed seed, *P* = (count of permuted statistics ≥ observed + 1) / (B + 1). The Test column reads *Permutation test over patients, 9999 permutations (GEE not fitted)*, the Interpretation column names the cause, and the run log counts such rows.

No test is run, and no *P* value printed, when the data cannot support one: a single group with data, a constant variable, a group with fewer than two usable values ("Cohort B has n = 1"), or, for a clustered test, a group whose eyes all belong to one patient, data below the minimum of every clustered model, or too few

patients for the *t* reference (A.9). The Test column then reads *Not tested* (or *Not applicable (...)*), and the Interpretation column says *Not estimable* with the reason. In the Publication Table such a cell reads **NE**, explained in a footnote. A test that could not be performed is never reported as "no difference".

Interpretation column. It reads "Statistically significant (between groups)." or "Not statistically significant (between groups)." ("among groups" with more than two). A clustered or weighted GEE *P* value from a group of 2 to 9 patients adds the caution "Caution: Cohort B has 4 patients - robust SE can be too small.", repeated in a Publication Table footnote.

The test actually used is printed beside every *P* value, in the Test column of the summary sheets and, for the Publication Table, in its footnote "Tests behind the P values", which lists each test with the variables it was run on.

Matched cohorts

A matched cohort is described and tested as follows (matching: A.8):

- every model of a matched cohort uses an **independence** working correlation with robust standard errors clustered by patient; matched sets are not used as clusters. This is stated in every matched sheet;
- with **unit matching weights**, per-eye rows get the clustered tests above, and a [per patient] row gets the independence GEE on one row per patient, the same model as its GEE column, so the row prints one *P* value;
- with **unequal matching weights** (any weight different from 1), every cell is weighted (weighted mean ± SD, weighted counts and proportions, weighted median [IQR]) and every test is a **weighted Wald test**: a GEE of the variable on the group, weighted by the matching weights, independence working correlation, robust SE clustered by patient (on the ranks for an ordinal variable; on the stacked level indicators for a categorical variable with more than two levels). Its estimate is exactly the difference of the weighted means (for a proportion, the odds ratio of the weighted proportions) printed in the same row. A [per patient] row gives each patient the mean weight of all its eyes in the cohort. A group with no events (or only events) has no weighted test, and the cell says which group; there is deliberately no unweighted fallback. The group headers add the effective sample size ESS = $(\Sigma w)^2 / \Sigma w^2$, e.g. `Reference (n = 137 eyes; 94 patients, ESS = 122)`, and a note row above each matched table says that it is weighted.

A.5 Dichotomous "positive" level

For every dichotomous variable you choose which level is the event (Sex = F, Complication = 1, and so on). It sets the reported n/N (%) and the direction of the odds ratio in every model. Defaults are guessed from the data, and a guess that is only alphabetical is flagged for you to check. The chosen level is matched regardless of case, surrounding spaces and the usual spellings of a logical value (TRUE / True / 1 / 1.0). A level that cannot be found is reported in the run log, and one documented default is then used in the tables and the models alike.

A.6 Filters — inclusion rules

Rules are combined with AND. Sixteen operators are available:

Operator	Meaning	Typical use
in, notin	value is / is not in a chosen set	treatment type
==, !=	equal / not equal	treatment number = 1

Operator	Meaning	Typical use
<, <=, >, >=	numeric comparison	age ≥ 18
between	inclusive numeric range	follow-up 150–210 days
abs_le, abs_ge	absolute value ≤ / ≥	cylinder magnitude ≤ 5.00 D regardless of sign
abs_between	absolute value in range	cylinder 1.00–3.00 D either sign
contains, not_contains	text match	free-text surgery description
isna, notna	missing / not missing	require a recorded postoperative refraction

A row with a missing value is kept by every rule except *is missing* / *is not missing*. Rules act on the values as recorded in the file (after the plus → minus transposition of A.2, before any logMAR conversion). As each rule is added, the number of rows remaining is shown **per group**, so the effect of every criterion on every arm is visible before anything is run. The Manifest lists the rules in words.

A.7 Subsets

Nominating a subset column (typically the procedure) makes the software run a **complete, independent analysis for each subset**, a full PRK analysis and a full LASIK analysis from one configuration. Three workbook modes are offered: one workbook per subset (default); one per subset plus one of all enabled subsets pooled; or the pooled workbook only. Each run has its own workbook and figure folder, named after the subset (the pooled run is named `Pooled`, a run without subsets `All`). Subsets are not a grouping variable: they partition the data, and the group comparison is repeated within each partition. Rows outside every enabled subset are excluded.

A.8 Matching

Matching is performed by `MatchIt` against the reference group. Each non-reference group is matched to the reference **separately**, giving one pairing per group.

Parameter	Options	What it does
Enable matching	on / off	If off, the matched branch of the output is skipped.
Matching covariates	any baseline variables	The variables matched on (with a propensity-score distance, the variables of the propensity model). A row must have **all** of them to be eligible.
Method	nearest (default), exact, cem, subclass	Nearest = greedy nearest-neighbor matching on the distance. Exact = keeps the eyes whose covariate values exactly equal those of an eye in the other group (needs discrete covariates). CEM = coarsened exact matching. Subclass = strata of the propensity score, eyes weighted within each stratum. These are the four methods the bundled R library runs.
Distance	logit of the propensity score (default), propensity score on the probability scale, Mahalanobis, robust Mahalanobis	For nearest and subclass only. The default is the propensity score from a logistic model, matched on its **logit** (MatchIt distance = "glm", link = "linear.logit"). The probability scale is the same model on the probability scale. Mahalanobis distances ignore the propensity score; subclassification cannot use them.
Ratio	integer (default 1)	nearest only. Controls matched per treated eye.
Replacement	on / off (default off)	nearest only. Whether a control eye may serve more than once.
Caliper	numeric (default 0.2; blank = none)	nearest with a propensity-score distance only. Maximum acceptable distance.
Standardized caliper	on / off (default on)	The caliper in standard deviations of the score **on the scale of the chosen distance**: by default 0.2 SD of the logit of the propensity score (Austin 2011).
Match order	largest, smallest, random, data, closest, farthest	nearest only. largest and smallest need a propensity score.

Parameter	Options	What it does
Random seed	integer (default 20250601)	Fixes random match order and ties, so identical settings give identical matched cohorts in the run and in the live preview.

The interface grays out every setting the chosen method does not use. The PSM sheet, the Manifest and the balance plot name the propensity-score scale, and the PSM sheet gives the caliper both in SD and on the score's own scale, e.g. "0.2 SD of the logit of the propensity score (= 0.321 on that scale)". A matching covariate that is not found in the file is listed as "not found in the file (ignored)", with a WARNING line in the run log; when none is found, matching is skipped and the Manifest says so.

Balance. Standardized mean differences (SMD, via `cobalt`) are reported before and after matching for every covariate, binary ones included, against the conventional 0.1 threshold: the difference in means (in proportions) divided by the standard deviation of the treated (non-reference) group, cobalt's convention for the average treatment effect in the treated. The *Propensity score (logit)* row gives the balance of the score itself. The balance plot (love plot) shows the same numbers: no title in the image, gray circles before and black triangles after matching, the covariates by their headers as typed in the file; it is written in the chosen figure formats with a legend file beside it.

Matching weights. MatchIt gives every matched row a weight. The weights are all 1 only for 1:1 nearest-neighbor matching without replacement; they differ from 1 with replacement, with k:1 matching (k > 1) when some treated eyes find fewer than k controls inside the caliper, and generally with exact, coarsened exact and subclassification matching. Weights that are not all 1 are carried into the matched tables and figure *P* values (A.4, A.13) and, when *Use matching weights in matched GEE* is on, into the matched GEE and regression. The Manifest records "Matching weights: non-unit (min–max)". The within-group sections and the mEYEstro / AstigMATIC tabs list the matched rows unweighted. The matched analysis estimates the average treatment effect in the non-reference (treated) group.

Live matched-count preview (nearest-neighbor only). The interface runs the same MatchIt call as the analysis on a slim copy of the data and reports, per pairing: group rows → rows with complete covariates → rows matched, the reference pool, the controls used and the patient IDs of unmatched eyes. The three usual causes of lost eyes are a sparse covariate, too tight a caliper, or too high a ratio.

> **Caution.** Matching addresses *measured* confounding only. It cannot repair unmeasured confounding, and it never converts an observational comparison into a randomized one.

A.9 GEE — every parameter

Generalized estimating equations (Liang and Zeger 1986) account for the correlation between the two eyes of one patient. Ordinary tests treat eyes as independent, which understates standard errors and inflates the type I error rate whenever fellow eyes are present.

Parameter	Options	What it does
Run GEE on	none, unmatched, matched (default), both	Which cohort(s) the models are fitted on.
Main GEE	on / off	The unadjusted clustered comparison: outcome ~ group.
Adjusted GEE	on / off	outcome ~ group + covariates.
Adjustment covariates	any variables	The covariates of the adjusted model; a variable is never adjusted for itself.
Use matching weights in matched GEE	on / off (default on)	Carries unequal matching weights into the matched models, so they describe the same design as the weighted matched tables.

Parameter	Options	What it does
Add GEE columns to tables	on / off	Copies the group effect (β or OR with 95% CI) and its *P* value into the summary tables of every cohort GEE is run on.
GEE source shown in summary tables	main, adjusted	Which model supplies those columns.
GEE small-sample inference	md (default), none	The small-sample correction below.
Summary-table tests	auto (default), classical	A.4.
P value style	ama (default), classic	A.12.

Estimation. Models are fitted with `geepack::geeglm`, the patient as the cluster. Continuous outcomes use the Gaussian family and give a mean difference β; dichotomous outcomes use the binomial family and give an odds ratio. Ordinal and categorical outcomes are not modeled on the GEE sheets (no effect estimate); their *P* values in the summary tables come from the tests of A.4, and the sheet says so.

Small-sample correction (Mancl and DeRouen 2001). Every GEE-based test and interval (the table tests, the GEE and regression sheets, the weighted tests of the matched tables, the clustered *P* values of the figures, and the astigmatism tests that use a GEE) uses the Mancl-DeRouen bias-corrected sandwich covariance, $B^{-1}[\Sigma_i D_i' V_i^{-1} (I - H_i)^{-1} r_i r_i' (I - H_i)^{-T} V_i^{-1} D_i] B^{-1}$, with $B = \Sigma_i D_i' V_i^{-1} D_i$, r_i the residuals and $H_i = D_i B^{-1} D_i' V_i^{-1}$ the leverage of patient i, and a **t reference with df = patients − model coefficients** for a single comparison and its 95% CI (estimate ± t(0.975, df) × SE, exponentiated for an odds ratio), or an **F reference** (F = W / q on q and patients − coefficients df) for a joint test of q coefficients. The labels say "robust SE (Mancl-DeRouen), t reference" or "..., F reference", and the GEE sheets give the df of every term. A model with no more patients than coefficients has no reference distribution and is reported as "Not estimable - too few patients for this model (4 patients, 4 coefficients)". A covariate level held by a single patient has a degenerate variance; its SE, *P* and CI are not reported. With none, the uncorrected sandwich covariance and a normal (χ^2) reference are used.

Working correlation, and the independence refit rule. Unmatched models use an **exchangeable** working correlation, but an exchangeable fit is kept only if it converged, its estimated working correlation is **below 0.95**, and **one further Fisher-scoring step moves no coefficient by more than 0.01 of its standard error**. An outcome that is almost constant within patients makes the exchangeable estimating equations ill-conditioned; such a fit is therefore refitted with an **independence** working correlation (the robust standard errors stay clustered by patient) and labeled "independence working correlation (the outcome is almost constant within patients)", with the reason in the Model column (e.g. "estimated working correlation 0.984 (0.95 or more)"). If the refit also fails, the model is not estimable, with the reason. Every matched-cohort model uses the independence working correlation, weighted or not, and so do the stacked level indicators of a categorical table test. Every fit, whatever its working correlation, is accepted only if it converged. Sheets, tables, figures, the run log and the Manifest name the working correlation that was used.

Per-patient models. A variable flagged as patient-level is never the outcome of an eye-level GEE. On an unmatched cohort it is modeled on one row per patient (per patient and group: the outcome of the patient's first analyzed eye, numeric covariates averaged over the patient's eyes) by an ordinary linear regression (model-based SE, *t* reference on the residual df) or logistic regression (model-based SE, Wald *z* reference), labeled "per patient (one row per patient)". On a matched cohort the same rows are fitted by an independence GEE with the patient-clustered robust SE (weighted by each patient's mean matching weight when the weights are used), the model of the matched table's test of that row.

Minimum data — one rule for every clustered model. The GEE sheets, the adjusted regression, the clustered and weighted table tests and the clustered or weighted figure tests all apply the same rule: a model is fitted only when it has at least **6 usable eyes**, at least **3 patients** overall, and more usable eyes than

its coefficients + 2. On top of that, every compared group needs at least two usable eyes and two patients. Below these limits nothing is fitted and every comparison of that outcome says why, e.g. `Not estimable - too few eyes for this model (n = 5)`, `Not estimable - Cohort B has 1 patient`.

A group with no data for an outcome. Each outcome is fitted on its complete cases; a group left without a usable row is dropped from that outcome's model and gets its own row, `Not estimable - Cohort B has no data for this outcome`. The reference group is never replaced silently: without reference data, every comparison of that outcome reads `Skipped: not estimable - Reference has no data for this outcome`.

Few patients. Robust standard errors can be too small when a group has few patients. A comparison in which a group has a single patient is not estimated; one in which a group has 2 to 9 patients is reported with a caution ("Caution: Cohort B has 4 patients - robust standard errors can be too small with few patients") in the Note column of the GEE and regression sheets, in the Interpretation column of the summary tables, under the figure's *P* values, and once per run as a WARNING line in the run log, which the interface lists under "Check before using the results". Without a patient ID the clusters are eyes, and these texts count eyes.

Why a table and a figure can show different *P* values. With the defaults (patient ID mapped, clustering on, *Summary-table tests* = auto), a per-eye summary-table row and the figure of the same outcome print the same clustered *P* value. They differ when the figure tests another quantity (the SEQ accuracy panel tests postoperative SEQ minus the target when a target is mapped), when the tables use the classical tests, or on a matched cohort with unequal weights when *Use matching weights in matched GEE* is off (the figures always use such weights). With more than two groups, the figure's headline *P* is the omnibus GEE test, and its "vs reference" lines come from the same model as the GEE columns, Holm-adjusted as there.

When you do not need it. If the dataset contains one eye per patient, the clustered model adds nothing; the tables then run the classical tests automatically.

A.10 Regression (covariate-adjusted models)

Parameter	Options	What it does
Enable regression	on / off	Produces the adjusted-effect columns and the *Regression (adjusted)* sheet.
Run on	unmatched (default), matched, both	Which cohort the adjusted models are fitted on.
Covariates	any baseline variables	Entered beside the group: outcome ~ group + covariates.
Suggested covariates	automatic	Ranks baseline variables by their largest standardized mean difference against the reference group and suggests those at **0.10** or above, plus the mapped age and sex variables. This quick ranking uses the pooled SD of the two groups on the filtered, unmatched cohort, so its values can differ slightly from the balance table of A.8.

Estimation is by GEE, with the small-sample correction, working-correlation rule, per-patient models, minimum-data, no-data and few-patient rules of A.9.

What the sheet reports.

- The adjusted effect (β or OR with its 95% CI) and *P* value, merged into the outcome table as *Adjusted effect (95% CI)* and *P (adjusted model)*, followed by **Adjusted model n (eyes)**, the complete-case eyes of the adjusted model (with the number of events for a dichotomous outcome).

- **The crude estimate on the same eyes**: the model with the group alone, refitted on the adjusted model's complete-case eyes (*Crude effect, same eyes (95% CI)*, *Crude P, same eyes*), so the effect of adjustment is never confounded with a change of sample.
- With more than two groups, *P (Holm)* and *Crude P (Holm), same eyes* (A.12).
- The full models below, every term with its estimate, SE, statistic, exact *P*, df, effect and CI. In a model of the group alone the intercept reads "Mean = 181.63 (95% CI, ...)" or "Odds = 0.14 (95% CI, ...)" for the reference group; in a model with covariates it is named "Intercept (Reference, all covariates at 0)" without an effect. Group comparisons print two decimals; a covariate's per-unit effect keeps two significant digits (up to four decimals), so a small but real effect never prints as 0.00.

One effect style is used in every table: `β = -0.21 (95% CI, -0.32 to -0.11)` and `OR = 3.23 (95% CI, 1.48 to 7.03)`.

Choose covariates by imbalance, not by their effect on the result. Adding a covariate because it is still imbalanced is correct; adding it because it changes your *P* value is fishing. This is why the suggestion ranks by standardized mean difference.

Safeguards.

- **Constant covariates are dropped**, with a note: a covariate that takes a single value in the filtered cohort (a frequent consequence of a rule such as "primary treatments only") would otherwise make the model fail.
- **Small-sample and zero-event situations are flagged.** With fewer than about ten events (or eyes) per parameter the Note column carries a caution (e.g. "only 28 eyes in the smaller outcome class for 4 parameters"). A dichotomous outcome with zero events (or only events) in one group gives complete separation; that group's comparison is reported as not estimable, with the reason, and left out of that outcome's model, so the other comparisons are still estimated. An odds ratio for the group, or for a categorical covariate, outside 1/10,000 to 10,000 is not reported either. Report the counts and an exact test for such rows.

A.11 The recommended strategy: match, then adjust

Because both are available, **regression adjustment after matching** (described as doubly robust by Ho et al. 2007) can be run in one pass: match on the measured confounders, then adjust in the outcome model for whatever remains imbalanced. The adjustment removes residual imbalance that the matching left behind, and the matching makes the result less dependent on the form of the outcome model. It is not a doubly robust estimator in the AIPW / TMLE sense.

The workflow: run matching → read the post-matching standardized mean differences → put any covariate still above 0.1 into the regression covariates → report the adjusted estimate on the matched cohort. Its standard errors are clustered by patient (fellow eyes), not by matched set.

A.12 *P* values and multiplicity

P value style. One formatter prints every *P* value of the tables, sheets, figures and run log. The default style (*ama*) follows the journal's rule: two decimals; three when *P* < .01 or when two decimals would turn a value below .05 into ".05"; no leading zero; "<.001" and ">.99", never 0 or 1; an exact half is rounded up (.065 prints .07). Examples: .27, .03, .008, <.001. The *classic* style prints three decimals in the tables and four on the figures. The GEE and regression sheets keep the unrounded value in a *P (exact)* column.

Multiplicity: one policy.

- **Summary tables:** the GEE columns and the adjusted columns compare each non-reference group with the reference. With two or more estimable comparisons in a variable, their *P* values are **Holm-adjusted** across those comparisons and tagged **[Holm]** (Holm 1979). With two groups there is a single comparison and no tag.
- **Regression sheet:** *P (Holm)* and *Crude P (Holm), same eyes* beside the nominal *P*, with more than two groups.
- **Figures:** each "vs reference" line is Holm-adjusted across the panel's comparisons and tagged [Holm] when there are two or more.
- **Astigmatism sheet:** the Holm families of A.14.
- **Not adjusted:** the omnibus *P* values, single comparisons, and the per-term *P* values of the GEE sheets and full models. The **95% confidence intervals are never adjusted**, so a CI can exclude the null while the Holm-adjusted *P* is .05 or more; the Publication Table footnote and the Manifest say so.

With several groups and several outcomes the number of tests grows quickly. Pre-specifying a primary outcome is the cleanest solution.

A.13 Figures

The standard graphs

The **StandardEyes figures** are drawn by the R engine from your data. Every cohort appears on one chart in its own color, however many cohorts there are. The metrics and bins are those of the published standard graphs for refractive surgery (Waring et al. 2011; Dupps et al. 2011; Reinstein et al. 2017 for the intraocular-lens set). With all panels of the Full set and a stability panel, the composite holds eleven panels:

Panel	Content	Bins and labels
A	Cumulative uncorrected distance visual acuity (postoperative UDVA beside the reference CDVA)	20/12.5, 20/16, 20/20, 20/25, 20/32, 20/40 … "20/x or better"; header: Snellen geometric mean (logMAR mean ± SD) of each series
B	Uncorrected vs corrected distance visual acuity (efficacy)	3 or more worse, 2 worse, 1 worse, same, 1 or more better (Snellen lines); efficacy index
C	Change in corrected distance visual acuity (safety)	seven categories: Loss 3 or more, Loss 2, Loss 1, No change, Gain 1, Gain 2, Gain 3 or more; axis "Change in Snellen lines of CDVA"; safety index
D	Spherical equivalent refraction accuracy	< -2.00, -2.00 to -1.51, -1.50 to -1.01, -1.00 to -0.51, -0.50 to -0.14, -0.13 to +0.13, +0.14 to +0.50, +0.51 to +1.00, +1.01 to +1.50, +1.51 to +2.00, > +2.00 D
E	Spherical equivalent attempted vs achieved	scatter with y = x, ±0.50 D and ±1.00 D lines and each group's regression line
F	Defocus equivalent accuracy (cumulative)	≤ 0.25, ≤ 0.50, ≤ 0.75, ≤ 1.00, ≤ 2.00 D
G	Spherical equivalent stability (when follow-up visits are mapped)	mean ± SD SEQ per visit; axis "Mean ± SD SEQ (D)"
H	Refractive astigmatism, postoperative beside preoperative	≤ 0.25, 0.26 to 0.50, 0.51 to 0.75, 0.76 to 1.00, 1.01 to 1.25, 1.26 to 1.50, 1.51 to 2.00, 2.01 to 3.00, > 3.00 D
I	Target induced vs surgically induced astigmatism (eyes with TIA ≥ 0.25 D)	scatter as panel E
J	Correction index (eyes with TIA ≥ 0.25 D)	≤ 0.15, 0.16 to 0.30, 0.31 to 0.60, 0.61 to 0.90, 0.91 to 1.10, 1.11 to 1.60, 1.61 to 2.10, 2.11 to 2.50, > 2.50 (bins closed on the right)
K	Angle of error (eyes with TIA ≥ 0.25 D)	< -45, -45 to -35 … -5 to 5 … 35 to 45, > 45° (bins include the edge nearer zero); signed AoE and absolute AoE in the header

Bin labels carry two decimals and the ≤ / > signs; a value on a bin edge is snapped before binning so it falls in the bin its label names. Percentage axes end at 100. Value labels are sized to the panel and left out, never overlapped, where they would collide (the legend file then says so); 0% labels are not printed. The number of eyes per cohort is printed in a box beside each panel. The angle of error is undefined (missing) for an eye with no SIA.

Panel sets. The *Panel set* selector follows the published set sizes:

Set	What it contains
Full	every panel
Refractive	drops DEQ, stability and the vector panels (I, J, K)
Cataract	also drops safety and attempted-vs-achieved

Include astigmatism panels drops the refractive-astigmatism and vector panels from any set; the double-angle and semicircle plots have their own switches. Before you run, the interface states which panels the configuration will produce.

Reference acuity of the efficacy panels. Panel A's second series, the efficacy bars and the efficacy index use preoperative CDVA for the Full and Refractive sets and postoperative CDVA for the Cataract set, and the legend, axis title and header name it.

Intended target. An optional *Target SEQ* role holds the intended postoperative spherical equivalent of each eye (monovision, a myopic cataract target). When it is mapped, SEQ accuracy is postoperative SEQ minus target, attempted is preoperative SEQ minus target, and the defocus equivalent is |postoperative SEQ − target| + |postoperative cylinder| / 2; an eye with a blank target cell is taken as aimed at plano, and the run log counts such eyes.

Stability over time. Each follow-up visit is mapped by its columns (a ready SEQ column, or sphere and cylinder from which SEQ = sphere + cylinder/2 is computed); nothing is guessed from column names. Each visit is plotted at the median day observed; nothing is carried forward; points with fewer than three eyes are dropped; the number of eyes at each time point is printed beneath the axis. The "% changed by more than x D" statistic is paired: only eyes measured at both nominated visits contribute. Stability is an extra panel of the Full set.

Statistics on the panels. With two cohorts the header prints the pairwise test and names it, e.g. `GEE (clustered): P = .008 (d = 0.32)`, where d is Cohen's d on the pooled SD, computed per eye and unweighted, whatever test gave the *P* value. With more cohorts it prints an omnibus test plus each cohort against the reference, Holm-adjusted and tagged [Holm]. The tests follow A.4 and A.9: patient-clustered GEE wherever a patient has two analyzed eyes (with the small-sample correction and the working-correlation rule), the classical test with one eye per patient or without a patient ID, a permutation test over patients (labeled "GEE not fitted") where a clustered GEE was fitted but failed, and no *P* value, with the reason ("P not computed: Cohort B has 1 patient"), where the data cannot support one. On a matched cohort whose weights are not all 1, the *P* values are weighted GEE Wald tests (independence working correlation) and the composite says "Matched cohort: P values use the matching weights; bars and means show the matched eyes unweighted." A cohort of 2 to 9 patients carries a caution under the *P* values.

Correlated eyes

Two eyes of one patient are not independent. An eye-level test that ignores this rejects a true null hypothesis far more often than its nominal level when fellow eyes are present; the clustered test keeps the level. The script `simulate_fellow_eyes.R` in the `validation` folder of the source code simulates two

groups with both eyes of every patient and no true difference (30, 60 or 120 patients per group; correlation between fellow eyes 0.5 or 0.8; 5,000 replicates per setting; fixed seed). Run with version 1.1.0, the eye-level Welch test rejected at 10.5% to 11.1% against a nominal 5% when the correlation was 0.5, and at 13.7% to 14.9% when it was 0.8; the app's clustered test (GEE with the Mancl-DeRouen correction and a *t* reference) rejected at 4.5% to 5.0% throughout.

Journal figure style

Figure style (Special outputs tab) is **journal** by default, following the journal's figure rules: white background everywhere, no frame around a figure or a section, no section title bands or captions, no title or method paragraph inside the double-angle and semicircle images, no shaded wedges. Kept inside the image: the panel letters, axis titles, the group key, the boxed number of eyes and the short statistics lines above each panel. Text is never smaller than 6 pt at the width the figure is written at. *Annotated* keeps titles, bands, frames and method text inside the images and writes no legend files.

Legend files. In the journal style every figure, composite and balance plot gets `<figure>_legend.txt` beside it: the title (for a composite, the panel letters with their titles), the method lines (the test behind each *P*, the inference of the GEE *P* values, the bin rules, the outcome each header *P* compares, statistics moved out of a panel), then every abbreviation and symbol the figure actually shows, defined, in alphabetical order and separated by semicolons, as the journal asks.

Width, pages and resolution.

- *Figure width (cm)*: blank = each figure's own size. A width lays every figure out **at** that width, so its text keeps its point size; 8.76, 12.69 and 17.77 cm are the journal's one-, 1.5- and two-column widths.
- At a width, a combined figure (the standard graphs, the double-angle and the semicircle sets) has **two panels per row** (one at 8.76 cm), rows about 7.6 cm high at 17.77 cm (the same proportion at the other widths), and all text at least 6 pt. A panel whose statistics lines do not fit its row at that size keeps the first line per cohort and its test line, and the remaining lines go to the legend file.
- A combined figure taller than a **23-cm page** is split into pages `_p1`, `_p2` …, each a complete figure with its own legend file, the panel letters continuing (A–F, G–K). At 17.77 cm the six-panel set is one page with up to four cohorts and two with five or more; the ten- and eleven-panel sets are two pages.
- Formats: TIFF (always included by the interface), PNG, JPEG, PDF and **EPS**, at 50 to 1200 dpi (default 300; the journal asks 300 dpi for photographs and 1200 for line art). PDF and EPS are vector and embed their fonts. PostScript has no transparency, so the semi-transparent points and regions inside an EPS are embedded as images at 1200 dpi or the chosen resolution, whichever is higher.
- Colors are RGB: R cannot write CMYK files, so a journal's CMYK conversion is done outside the app.
- Group colors and display labels are configurable. A low-resolution PNG copy of every composite is written to a `_preview` subfolder and shown in the gallery of the Run & Results tab. The PDF device of the bundled R build cannot shape right-to-left text; when PDF is chosen and a label needs it, the run log recommends PNG or TIFF.

A.14 Astigmatism (JCRS 2025 standard)

The astigmatism analysis follows the standard of the JCRS astigmatism working group (Kohnen et al. 2025, Part I; Koch et al. 2025, Part II).

Vectors in plus-cylinder format

All vector analysis works in **plus-cylinder format**: magnitude M ≥ 0 at the plus axis θ (the steep meridian), doubled-angle components x = M cos 2θ, y = M sin 2θ. Each refraction is converted by the sign of its own cylinder, so the result is the same for a row typed in minus or plus form, with the minus-cylinder switch on or off, and for a target typed in either form. On every double-angle plot **with-the-rule astigmatism (plus axis 90) lies on the left** and against-the-rule (plus axis 180) on the right; every printed "@ θ" is a plus axis, written in the 1–180° notation.

Vector	Definition
PRE	preoperative refractive astigmatism
POST	postoperative refractive astigmatism
SIA	surgically induced astigmatism = POST − PRE
PE	prediction error of refractive astigmatism = POST − target (equal to POST when the target is plano)

The Alpins quantities of the standard graphs keep their definitions and are computed from the same plus-format vectors: TIA (target − PRE), DV (= −PE), correction index (SIA / TIA), angle of error, magnitude of error and index of success (Alpins 1993). Pre-calculated vectors may be mapped instead (magnitude and plus axis for TIA, SIA, DV or PE); each vector is resolved on its own, and when only one of PE and DV is supplied the other is derived. An optional target refraction (cylinder and axis) makes TIA the intended change and PE = POST − target.

The double-angle plots

The double-angle set shows **PRE, SIA and PE**, and **POST** as a fourth plot only when a target is mapped and at least one target is not plano (otherwise POST = PE, and the PE plot is titled "Prediction error (= postoperative refractive astigmatism; target plano)"). Each is written as its own file and in a composite. For each cohort the plot shows the eyes, the centroid (a diamond), the 95% region of the eyes (solid) and the 95% confidence region of the centroid (dashed), and its header gives the centroid (M @ θ and x, y), the total SD, the magnitude mean ± SD, median, range and RMS, and the numbers of eyes and patients, followed by one short line per test, e.g.

`Centroid: P = .82 · Centroid vs 0: Reference P = .07, Cohort A P = .68 · Mean magnitude: P = .08`

with "vs Reference ... [Holm]" lines when there are more than two cohorts. The single-angle semicircle plots of TIA, SIA and DV (Alpins) are kept for familiarity; they are not part of the standard. There are no correction-index polar plots (a ratio has no meridian); the correction-index histogram of the standard graphs stays.

Summary per cohort

For every vector and cohort: eyes and patients; the **centroid** (mean of x and y, as M @ θ and x, y); the **total SD** = $\sqrt{(\sigma x^2 + \sigma y^2)}$ with the population (N) denominator, as Part II defines it (invariant to rotation of the frame); the magnitude mean ± SD, median, minimum, maximum and root mean square; and the **cumulative percentages** of eyes with magnitude ≤ 0.25, 0.50 … 2.00 D (steps of 0.25 D; a magnitude on an interval edge counts as within) and > 1.00 D.

The 95% region of the eyes: depth-based convex polygon

The default region is the **depth-based convex polygon, built as in the WHWK functions** (Wilcox-Holladay-Wang-Koch R functions, the reference tool recommended by Part II; Holladay et al. 2022), because Part II notes that astigmatism data are usually not Gaussian. For one cohort with centroid c:

1. for every eye i away from c, all eyes are projected onto the line through c and eye i;
2. on each line, an eye's one-dimensional depth is the smaller of the shares of eyes at or beyond its position on either side (ties count on both sides);
3. an eye's depth is its smallest depth over all lines (every depth is 0 when an eye lies exactly at the centroid, as in the reference tool);
4. every vertex of the convex hull of all eyes takes the smallest depth;
5. depths are rounded to multiples of 1/n;
6. the threshold is the depth of rank floor(0.05 n + 0.5) in increasing order;
7. the region is the convex hull of the eyes whose depth reaches the threshold.

Because depths tie, the region can hold more than 95% of the eyes; the sheet and the legend file give the count (e.g. "197 of 204 eyes"). Because every hull vertex takes the smallest depth, the rule removes no eye whenever the hull has at least floor(0.05 n + 0.5) vertices, which is common up to about 180 eyes; the polygon is then the convex hull of all eyes, and the sheet and legend file say "the depth rule removed no eye". Below 10 eyes the rule is undefined and the region is the hull of all eyes. When all eyes lie on one meridian (or are identical) the region would be a line: it is not drawn, and the output says "95% region not drawn: all eyes lie on one meridian". A bivariate-normal ellipse is available as *95% region of the eyes* = normal (or both).

The 95% confidence region of the centroid: bootstrap convex polygon

The default centroid region is the **bootstrap convex polygon of the centroid, built as in the WHWK functions**:

1. B = 999 bootstrap resamples (seed 20250602) give 999 bootstrap centroids; **patients** are resampled (all eyes of a drawn patient enter together) when a patient ID is mapped and clustering is on, eyes otherwise;
2. the projection distance of each bootstrap centroid from the cohort's centroid is computed within the cloud of bootstrap centroids: for every line through the centroid and a cloud point, the absolute projection lengths are divided by their interquartile range (the "ideal fourths"), and a point's distance is its largest scaled length over all lines;
3. the round((1 − a) × B) centroids with the smallest distances are kept, with a = 0.05 for more than 120 resampling units and 0.045, 0.04, 0.035, 0.03, 0.025, 0.02 for at most 120, 80, 60, 40, 30, 20 units; the region is their convex hull.

The sheet gives the count (e.g. "949 of 999 bootstrap centroids (bootstrap over patients, 136 patients)"). The Hotelling T^2 ellipse is available as *Centroid 95% confidence region* = Hotelling (with patient clustering, its covariance is the clustered sandwich estimate).

Tests

Astigmatism tests = **robust** (default):

- **Each centroid against zero:** the bootstrap projection-distance test of the WHWK functions. The 999 bootstrap centroids of the confidence region and the null point (0, 0) form one cloud; *P* is the share of bootstrap centroids whose projection distance from the centroid exceeds that of the null point. The smallest *P* is 1/B: with B = 999, *P* = .001 is the bootstrap floor, and the sheet says so. The test needs at least 4 resampling units.
- **Between cohorts, for each vector:**
- *centroid:* a Hotelling-type statistic, $\Sigma_i\, n_i\, d_i'\, S^{-1}\, d_i$, where d_i is a cohort's 20%-trimmed mean vector minus the unweighted mean of the cohorts' trimmed mean vectors and S the pooled 20%-winsorized covariance;
- *precision* (each eye's distance to its cohort centroid) and *mean magnitude* (for PE, the standard's main outcome): 20%-trimmed means compared by Yuen's statistic (Yuen 1974; its Welch-type form for more than two cohorts);
- each calibrated by B = 999 bootstrap resamples of the data recentered on each cohort's trimmed mean, **patients resampled within each cohort** (eyes without patient clustering), seed

20250601. These tests need at least 12 eyes.

- **Trimming and winsorizing** follow the order-statistic definitions of the robust-statistics literature (Wilcox 2017): g = floor(0.2 n) values removed at each end (trimmed mean) or replaced by the (g + 1)-th and (n − g)-th order statistics (winsorizing, component by component).
- **The trimmed / untrimmed rule.** Refractive astigmatism measured in 0.25-D steps often puts most eyes on one value, where the winsorized variance is zero and the trimmed statistic does not exist. A trimmed test is used only when its statistic exists in the data and in at least 95% of the resamples; otherwise the same bootstrap test runs **untrimmed** (ordinary means and covariance) and is labeled "untrimmed bootstrap test (the trimmed statistic is not defined for these data)". Resampling continues until exactly 999 usable resamples (at most 2997 drawn); short of that the test is not computed, with the reason. The sheet, the cohort values shown ("20%-trimmed mean:" or "mean:"), the legend files and the Manifest name the variant that ran.
- **Cumulative intervals between cohorts:** a binomial GEE clustered by patient (exchangeable working correlation with the refit rule; small-sample correction); Fisher's exact test with one eye per patient or without clustering. Where the GEE cannot be fitted, typically a cohort at 0% or 100%, a **permutation test over patients** runs instead (999 permutations, seed 20250601, the eyes of a patient kept together), labeled "(GEE not fitted: a cohort at 0% or 100%)". An interval that every eye (or no eye) meets is not tested.
- **PRE against POST within a cohort (McNemar):** at each interval, the eyes within the interval before and after surgery. Below 25 discordant eyes, the exact binomial test; otherwise the McNemar χ^2 without continuity correction. When a patient contributes two discordant eyes and the tests are clustered, the cluster-adjusted form is used: below 25 discordant eyes the exact sign test on the patients' net changes, otherwise the statistic of Durkalski et al. 2003, $(\Sigma_k\,(b_k - c_k))^2 \,/\, \Sigma_k\,(b_k - c_k)^2$ over patients k (b_k improved and c_k worsened eyes of patient k), on 1 df. Not estimable when the discordant eyes come from one patient.
- **Holm:** within each vector and each comparison (the omnibus, or one cohort against the reference) across the family {centroid, precision, mean magnitude, ≤ 0.25 … ≤ 2.00 D}; each measure across the cohorts compared with the reference (the figures' "vs" lines); the centroid-against-zero *P* across the

cohorts of a vector; McNemar across the intervals of a cohort. The sheet prints the nominal and the adjusted *P* values.

- With more than two cohorts: an omnibus test plus each cohort against the reference.

Astigmatism tests = **gee**: clustered GEE Wald tests (centroids by the joint test of both stacked components, precision and magnitude by a Gaussian GEE, centroid against zero by the joint test of the two intercepts), with the small-sample correction. *Astigmatism tests* = **classic**: one-sample Hotelling T^2, Hotelling T^2 / Pillai MANOVA and Welch tests, eyes treated as independent.

The robust between-cohort tests are robust and cluster-aware, but they are not the exact trimmed-Hotelling variant of Part II (they use a between-group quadratic form weighted by the group sizes); the interface and the legend files say so. *Follow the JCRS 2025 astigmatism standard* (on by default) can be switched off, in which case the figures use two separate univariate tests on x and y without adjustment; the astigmatism sheet keeps the standard's tests.

Matched cohorts. The astigmatism statistics of a matched pairing are computed on the matched rows, unweighted, with an independence working correlation where a GEE is used; the sheet heading says that the matching weights are not applied, so the weighted matched tables can differ. With the gee or classic method, the centroid comparison of a matched cohort with unequal weights is a weighted GEE.

Right and left eyes. When an eye column is mapped, the sheet lists the right-eye and left-eye centroids apart. Left eyes are not mirrored.

The astigmatism sheets, the CSV and the WHWK input files

- **Astigmatism (JCRS 2025)** (and *Astig_M_<pairing>* for each matched pairing): the summary per vector and cohort, the cumulative percentages, the centroid-against-zero tests, the comparisons between cohorts, the McNemar table, the right and left centroids, the vertices of every region, and the method notes, all at full precision.
- **Astigmatism vectors:** one row per eye and vector (PE, SIA, PRE, POST and the Alpins TIA and DV) with group, patient ID, row label, magnitude, plus axis and x, y, ready for external tools.
- **CSV copy:** `<prefix>_<run>_Astigmatism_<time stamp>.csv` beside the workbook holds every table of the astigmatism sheets (main and matched), with columns naming the cohort set and the table.
- **WHWK input files:** with the double-angle plots and *Input files for the WHWK functions* on (the default), the sub-folder `WHWK input` of the run's figure folder holds, for each plotted vector, `<vector>_vector.txt` (two columns per cohort, `<Cohort>.x` and `<Cohort>.y`, plus-cylinder double-angle form) and `<vector>_magnitude.txt` (one column per cohort), tab-delimited, one row per eye, shorter cohorts padded with empty cells, cohort names reduced to letters and digits; a matched pairing adds its name to the file names. A README states what the files are, that the WHWK functions treat eyes as independent (so with both eyes of patients their *P* values ignore the pairing), that their wrap-up functions refuse values beyond 6 D, and that the functions are obtained from https://osf.io/xhe8u/.

A.15 Optional sheets and export files

Option	What it produces
Publication-style table	One manuscript-ready sheet: Baseline / Preop / Outcome stacked under section headers, group columns, *P* values and the GEE and adjusted columns, without the Test and Interpretation columns. Its footnotes: *NE = not estimable* with the reason per variable; *Values:* what each kind of cell holds; *Tests behind the P values*; the few-patient cautions.

Option	What it produces
mEYEstro tabs	One sheet per group laid out for the mEYEstro template: columns E–S (preoperative refraction, vertex distance 12 mm, preoperative CDVA; intended refraction; postoperative refraction, vertex, CDVA and UDVA) are copied into the template; the gray columns A–D (group, decimal acuities) and the blue column T (postoperative SEQ) are for reference. Refraction is in minus-cylinder form, acuity in Snellen (an off-chart eye shows its label); the intended columns come from the mapped targets.
Alpins / AstigMATIC tabs	One sheet per group whose columns A–E (eye as OD/OS, preoperative cylinder and axis, postoperative cylinder and axis, minus-cylinder form) load into AstigMATIC. Eye codes R / RE / Right / 1 and L / LE / Left / 2 become OD / OS; an unrecognized code is left blank and counted in the run log.
Astigmatism (JCRS 2025) sheet	A.14.

Filtered CSV (*Export filtered CSV*, on by default): `Filtered_Dataset_<time stamp>.csv` holds the rows that are **analyzed**, i.e. in a group and, when subsets are defined, in a subset, with a *GroupLabel* column (and *Subset*) leading. It holds the values as analyzed: a header carries a suffix where a column no longer holds what was recorded (`_logMAR`, `_decimal`, `_minus_cyl`); a missing value is an empty cell. *Export filtered rows (CSV)* on the Data & Groups tab writes the same rows and headers without running the analysis, and the sidebar's "in analysis" count is the same number.

Manifest sheet. Every workbook opens on a *Contents* sheet (one row per sheet with a link, its full name and a description) and a **Manifest** sheet that records the conditions of the run: StandardEyes version and build (also for a command-line run) and the run duration; R and package versions; the input file, sheet and checksum; the rows analyzed and group sizes; the filter rules in words; the variables by section and type, the preferred levels and the per-patient variables; the VA notation and logMAR conversion; the matching method actually run, its settings, covariates and whether the weights were unit or non-unit; the GEE small-sample inference, the working-correlation rule, the table tests, the multiplicity policy and the matched-analysis note; the GEE and regression settings and covariates; the astigmatism vectors, test and regions; the random seeds; the *P* value style; the figure settings; how the figure *P* values were computed; and every setting of the run as one JSON text. The workbooks name "StandardEyes" as their author.

A.16 Run options and reproducibility

- **Preview only**: results are held in the app for inspection and nothing is written to the output folder until you choose to save. A temporary copy is kept in `webapp\.runs` inside the application folder and deleted when the next run starts, when the app closes, or, after a crash, at the next start.
- **Output folder / file prefix**: where results are saved and how they are named.
- **Before you run**: the Run & Results tab lists, in words, the settings that decide how the numbers are computed and printed (matching scale and caliper, GEE inference, table tests, *P* style, visual acuity, figures, astigmatism methods).
- **Study presets**: every setting of a study is saved under one name inside the app; **Download config JSON** writes the same settings to one portable file, and **Load config JSON** restores them. A settings file or preset saved by an earlier version still loads; a setting it does not contain takes the current default, and the Manifest records the settings used.
- **Fixed seeds**: matching 20250601; Monte-Carlo Fisher, the permutation tests and the robust astigmatism bootstrap 20250601; the centroid regions and the centroid-against-zero tests

20250602. Each is set locally, so identical settings give identical results.

The settings file drives the statistical engine with or without the interface, from the application folder:

```
Rscript --vanilla engine\run_pipeline.R "Example Data\Settings_02_Two_cohorts_full.json"
```

The example settings files use paths relative to the application folder, so they keep working wherever the folder is copied. Every number is produced by the R engine; that the same settings file drives the interface and the command line is the practical guarantee that the interface does none of the arithmetic.

A.17 Two groups versus more than two

Everything scales to any number of groups, but several behaviors change.

Aspect	Exactly two groups	Three or more groups
Table *P* value (clustered)	GEE test of the group term (*t* reference)	joint GEE test of all group terms (*F* reference) — omnibus
Table *P* value (classical)	Welch *t*, Wilcoxon, χ^2 / Fisher	Welch ANOVA, Kruskal–Wallis, χ^2 / Fisher on the full table — omnibus
What the *P* value means	the two groups differ	*at least one* group differs — it does not say which
Matching	one pairing: group vs reference	each non-reference group matched to the reference separately
GEE / regression	one effect estimate	one row per group versus the reference, each with its own effect, CI and *P*
GEE and adjusted columns in tables	one line, no Holm	one line per comparison inside the cell, Holm-adjusted, tagged [Holm]
Regression sheet	*P*	*P* and *P (Holm)*, crude *P* and *Crude P (Holm), same eyes*
Figures	pairwise statistic, test named	omnibus statistic plus each group against the reference, [Holm]; every group on the same chart
Astigmatism	one comparison per measure	omnibus plus each cohort against the reference, Holm across cohorts

With more than two groups the omnibus *P* value says only that the groups are not all alike; which groups differ comes from the comparisons against the reference. Choose the reference group deliberately (usually the control or the standard technique), because every estimate is expressed relative to it.

PART B · WALKTHROUGH WITH THE EXAMPLE DATASET

B.1 The example dataset

`Example Data\Example_Refractive_Dataset.xlsx` is **simulated** from a documented model with a fixed seed (20261004), by the generator `make_example_data.py` shipped in the `validation` folder. Its *Surgeries* sheet has 585 rows and 57 columns: 320 patients, 550 eyes, 550 primary surgeries and 35 retreatment rows. Patients and eyes carry codes only (PatientID, EyeID).

- **Cohorts:** Reference, Cohort A and Cohort B (a property of the patient); **procedures:** PRK, LASIK and KLEx.
- **Fellow eyes** are correlated through patient effects: about 0.85 between the eyes for preoperative spherical equivalent and 0.5 to 0.75 for the postoperative outcomes; a patient has both eyes treated with probability 0.68.
- **Refraction** on the 0.25-D grid, SEQ = sphere + cylinder/2 exactly, 25% of rows typed in plus-cylinder form (column *Cylinder_Notation*); acuity on whole 0.1-logMAR chart lines, in decimal notation; monovision targets in one eye of some patients aged 45 or more.
- **Measured confounding:** age, preoperative SEQ and pachymetry drive both cohort membership and the outcomes.
- Five follow-up visits with drop-out, retreatment rows, missing values and all four variable types.

The **README sheet** documents the generating model with every parameter, the seed, the SHA-256 of the data, a column dictionary with units, and the **true cohort effects**:

Outcome	Cohort A vs Reference	Cohort B vs Reference
Postoperative SEQ (mean difference given the confounders)	−0.20 D	−0.10 D
Dry-eye symptoms (population-averaged odds ratio)	2.0	1.0 (no effect)
Postoperative cylinder (magnitude, vector)	no effect	no effect
Postoperative CDVA, lines gained or lost	no effect (confounded null)	no effect (confounded null)

It also gives the crude, adjusted and true values in a population of 200,000 simulated patients (crude SEQ difference of Cohort A −0.33 D against a true −0.20 D; crude dry-eye odds ratio 2.56 against 2.0), and the file's own crude and adjusted estimates, with a note on sampling variation: in this file the adjusted dry-eye odds ratio of Cohort B is statistically significant although its true value is 1.0, the kind of chance finding that about 1 in 20 contrasts without a true effect give.

B.2 The seven examples

Seven worked examples are supplied as settings files beside the dataset and as the *Example - …* presets inside the app. All use the same inclusion rules (age ≥ 18, follow-up 150–210 days or not recorded, first surgery, |preoperative cylinder| ≤ 5 D), which leave 531 rows: Reference 226 eyes (136 patients), Cohort A 163 eyes (97 patients) and Cohort B 142 eyes (84 patients). Age, sex, smoking, optical zone and surgeon are flagged as patient-level.

Settings file	Study	Rows analyzed	Workbooks	TIFF figures
Settings_01_Two_cohorts_simple.json	Reference vs Cohort A: the summary tables (patient-clustered *P* values) and the figures; no matching, GEE sheets or regression	389	1	19
Settings_02_Two_cohorts_full.json	Reference vs Cohort A with every step: matching (2:1, caliper 0.2 SD of the logit), GEE on both cohorts, adjusted regression, all figures with the stability panel, within-group tables per patient or per eye	389	2	41
Settings_03_Three_cohorts.json	the three cohorts: omnibus tests, each cohort against the reference Holm-adjusted, all cohorts on every figure	531	2	62
Settings_04_Subsets_PRK_LASIK.json	the analysis of Settings 02 run separately for PRK and for LASIK	341	4	82
Settings_05_Stability_focus.json	the stability panel: five visits, SEQ from a column for visits 1 and 3 and from sphere and cylinder for the others	389	1	20
Settings_06_Figures_only_no_stats.json	figure switches: astigmatism panels and vector plots off, stability on: a seven-panel composite lettered A to G	389	1	8
Settings_07_Full_figure_suite.json	the analysis of Settings 03, presented as the full figure suite with double-angle plots	531	2	62

The counts are those of the release run (journal style, TIFF; the TIFF count includes the balance plots). The walkthrough below follows **Example 02**; steps 10 and 11 and parts of B.4 and B.5 use Example 07.

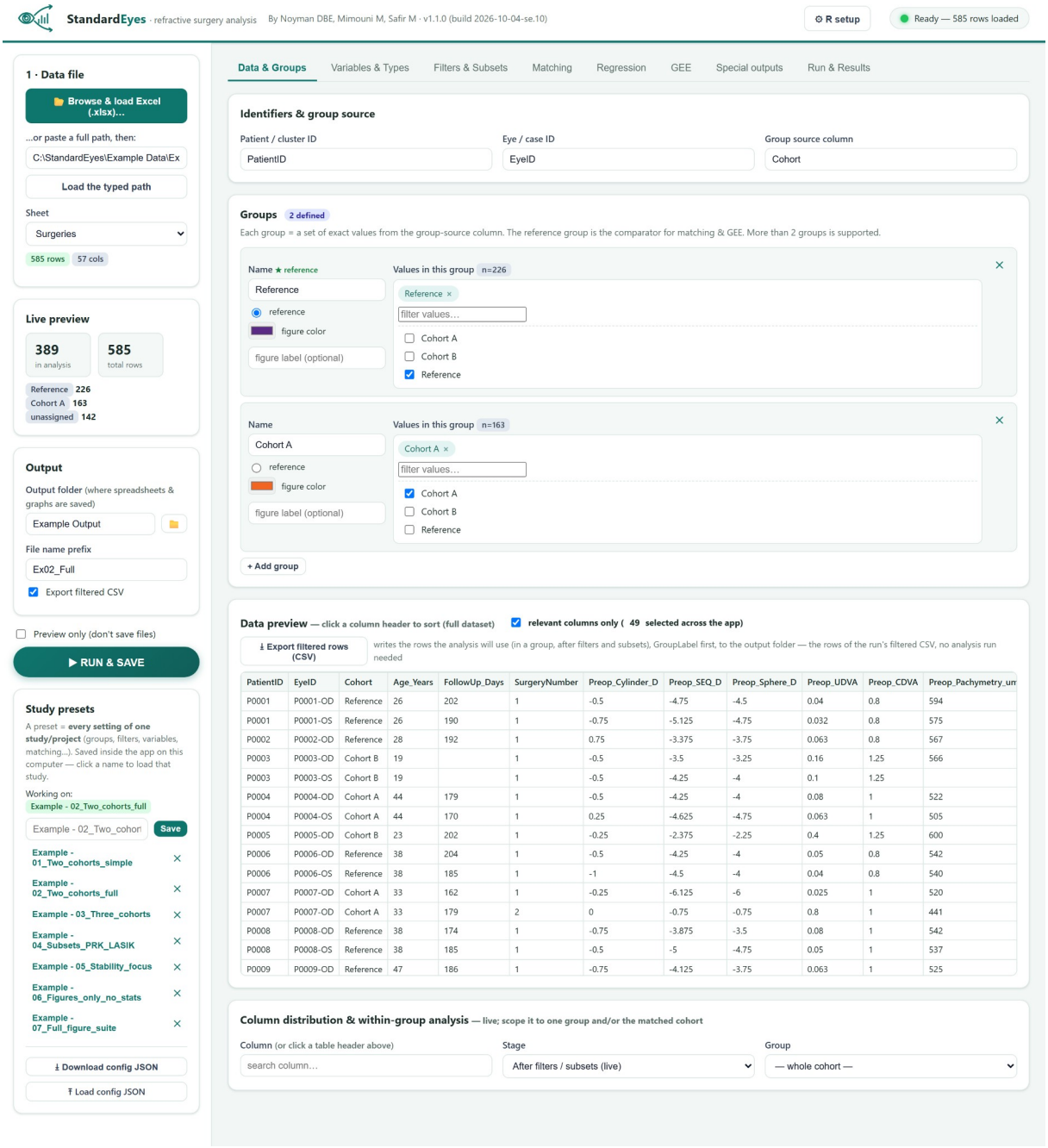
B.3 The interface, step by step

The session loads the example dataset (585 rows, 57 columns) and the *Example - 02_Two_cohorts_full* preset. Steps 1–9 configure and run the two-cohort comparison of Reference and Cohort A; steps 10–11 show what changes with three cohorts, and step 12 shows the R setup screen.

Step 1 · Data & Groups

Load the spreadsheet (browse for it, or paste a full path). The identifiers are then nominated: **PatientID** links the two eyes of one patient and becomes the clustering unit, **EyeID** identifies the row, and **Cohort** is the column whose values define the groups. Each group is built from exact values of that column and given a color; **Reference** is marked as the reference group. Values left unassigned (here Cohort B) are not analyzed. The sidebar shows the rows "in analysis": 389 of 585, the rows that fall in a group and pass the filters.

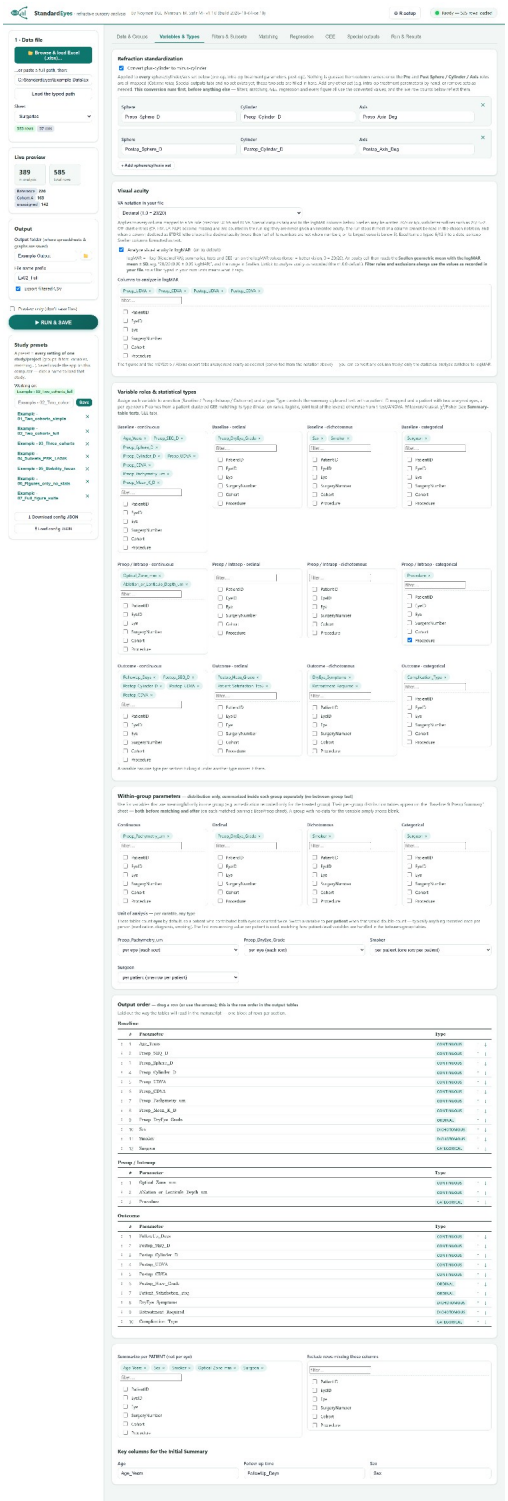
The data preview shows the file as loaded; any column header can be clicked to inspect its distribution, scoped to one group and to the filtered or matched cohort.



Step 2 · Variables & Types

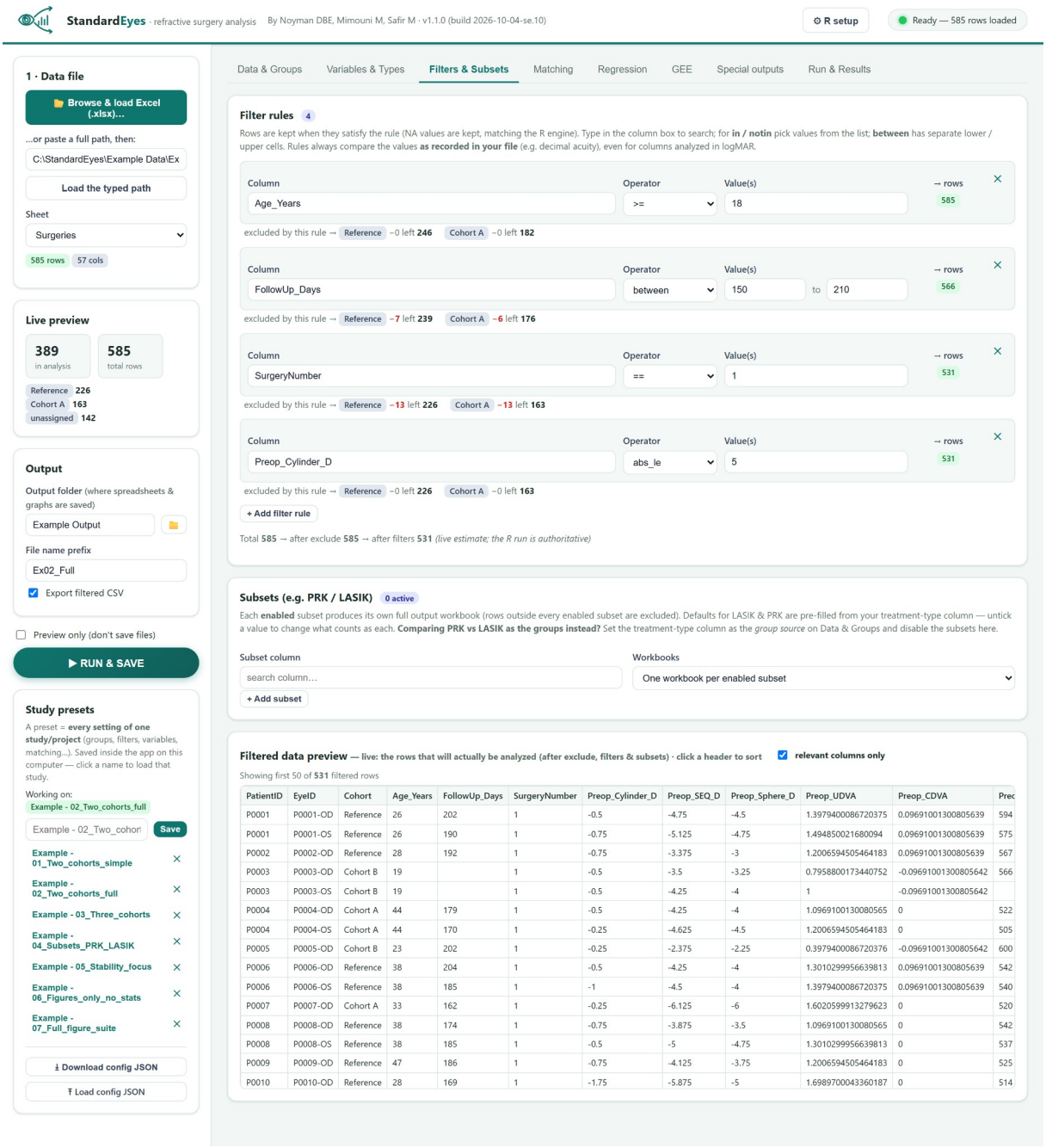
Every variable is placed in **Baseline**, **Preop / Intraop** or **Outcome** and given one of four types. The example uses all four: continuous (age, refraction, acuities, pachymetry, keratometry, optical zone, ablation depth, follow-up), ordinal (dry-eye grade, haze grade, satisfaction 1–5), dichotomous (sex, smoking, dry-eye symptoms, retreatment) and categorical (surgeon, procedure, complication type). Age, sex, smoking, optical zone and surgeon are flagged as **patient-level**.

The same tab holds the two data-preparation steps: refraction standardization (plus → minus, on the preoperative and postoperative triples, A.2) and the *Visual acuity* card, with the notation of the file (decimal) and *Analyze visual acuity in logMAR*, on by default (A.3). The within-group section lists preoperative pachymetry, dry-eye grade, smoking and surgeon, the last two counted per patient.



Step 3 · Filters & Subsets

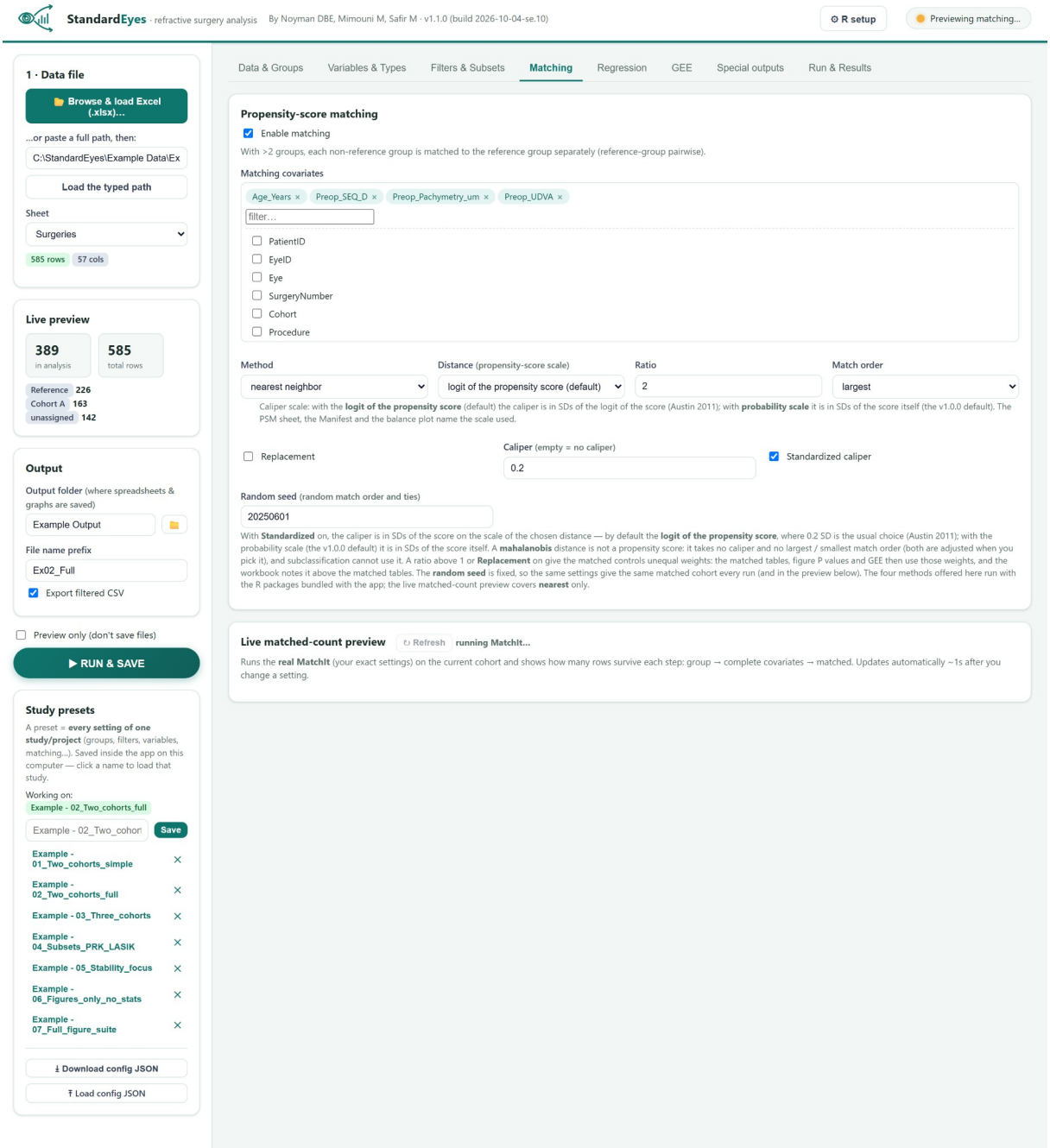
Four inclusion rules: `Age_Years >= 18`, `FollowUp_Days between 150 and 210`, `SurgeryNumber == 1` (which removes the retreatment rows) and `Preop_Cylinder_D abs_le 5`. The count remaining is shown per cohort after each rule; a row with a missing follow-up is kept by the `between` rule. No subset column is set, so one analysis covers every procedure; *Example - 04_Subsets_PRK_LASIK* sets **Procedure** as the subset column and runs a complete analysis for PRK and for LASIK.



Step 4 · Matching

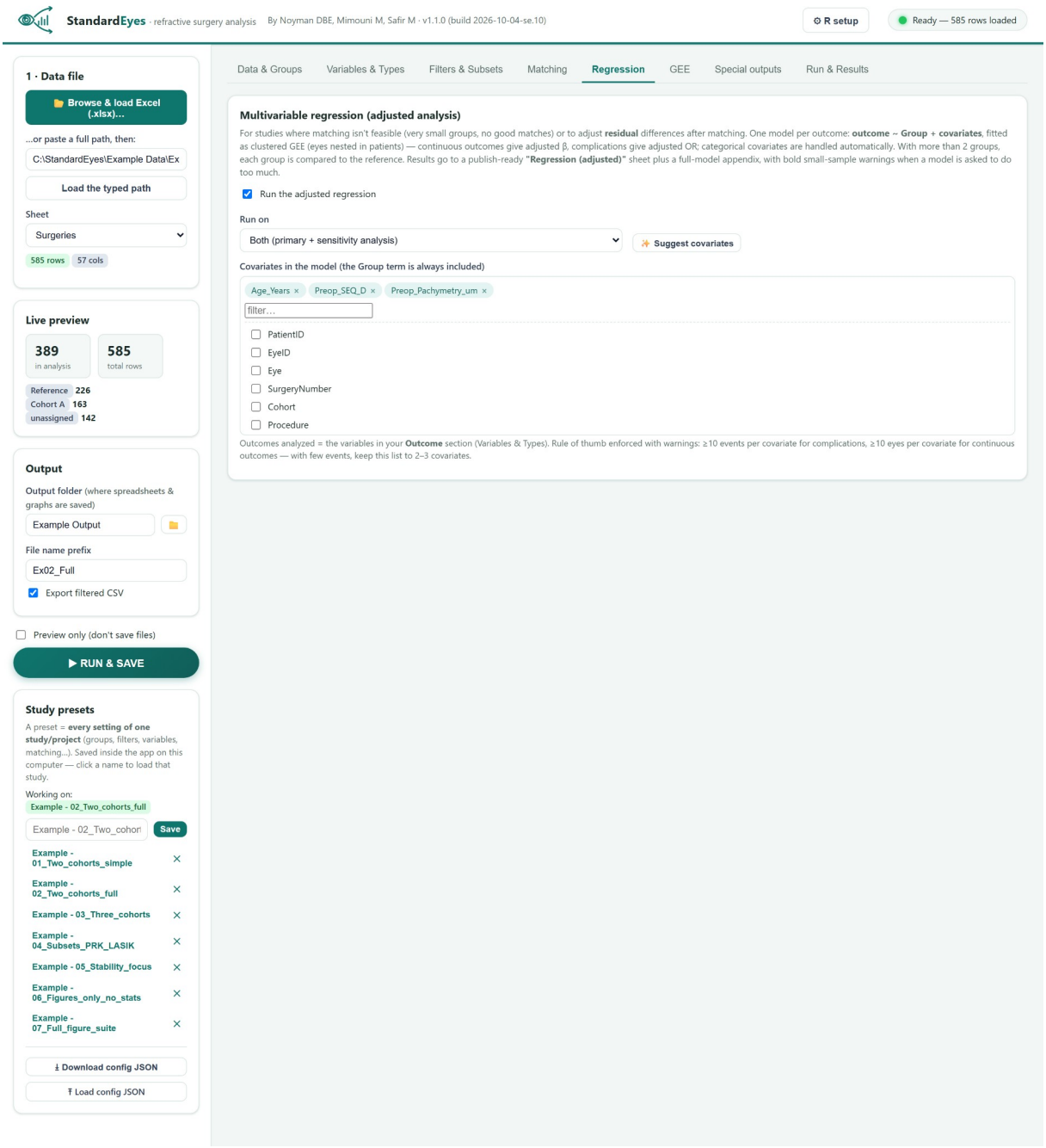
The matching covariates are age, preoperative SEQ, pachymetry and UDVA, with nearest-neighbor 2:1 matching on the **logit of the propensity score**, a caliper of 0.2 SD of that logit (0.321 on the logit scale), no replacement, largest-first order and seed 20250601.

The live matched-count preview runs the same MatchIt call as the analysis. Of the 163 Cohort A eyes, 4 lack a pachymetry value, so 159 have complete covariates; 97 of them find controls inside the caliper and are matched to 137 Reference eyes, and the patient IDs of the unmatched eyes are listed. Because 2:1 matching under a caliper leaves some treated eyes with fewer than two controls, the matching weights are not all 1 (0.706 to 1.41), so the matched tables, figure *P* values and models are weighted (A.4, A.8). The matched dataset (234 rows) with its weights and propensity scores is shown underneath.



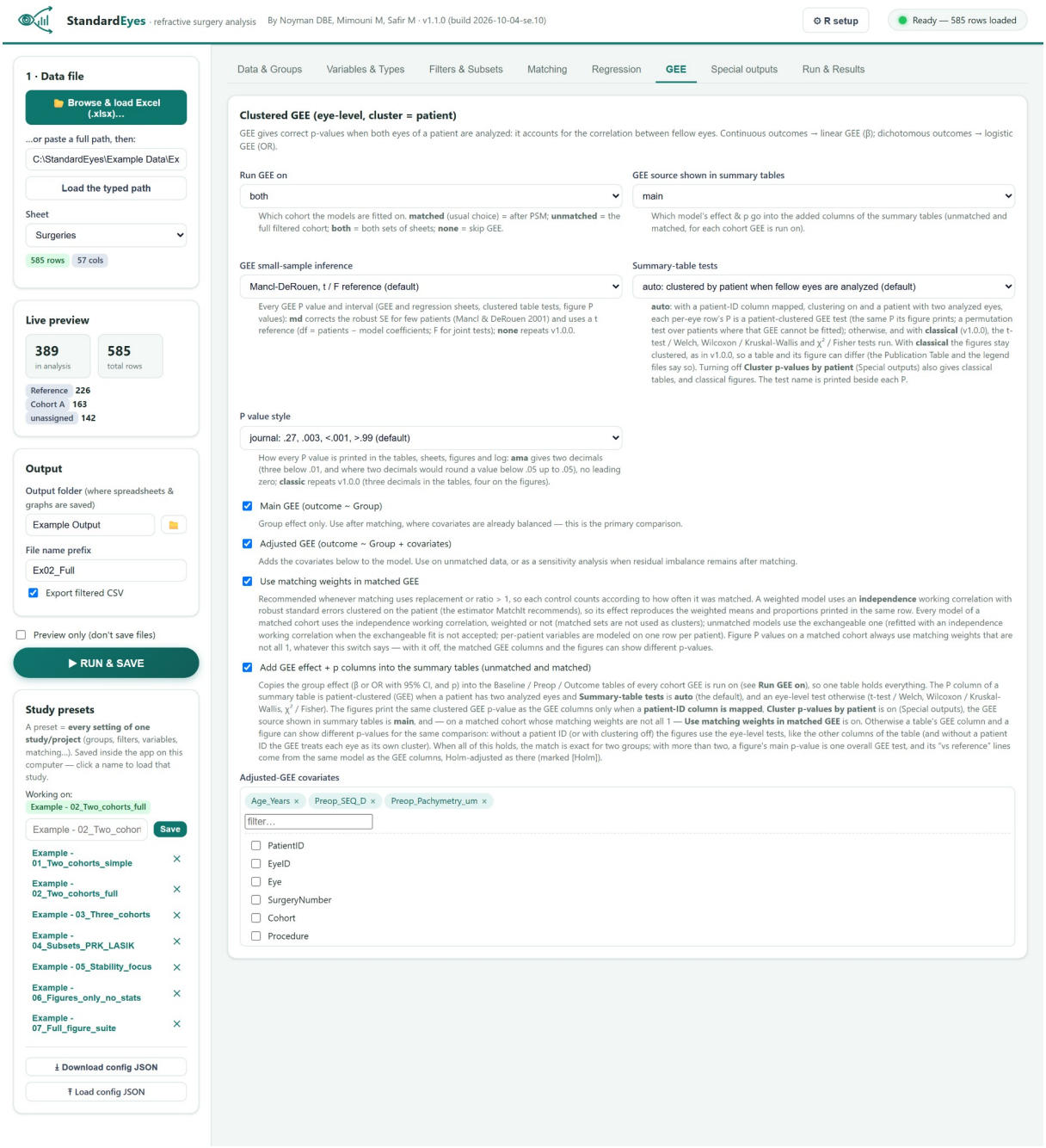
Step 5 · Regression

The adjusted models use age, preoperative SEQ and pachymetry, on both the unmatched and the matched cohort. **Use suggested** ranks the baseline variables by their largest standardized mean difference against the reference group and suggests those at 0.10 or above, so the choice is driven by imbalance rather than by its effect on the result.



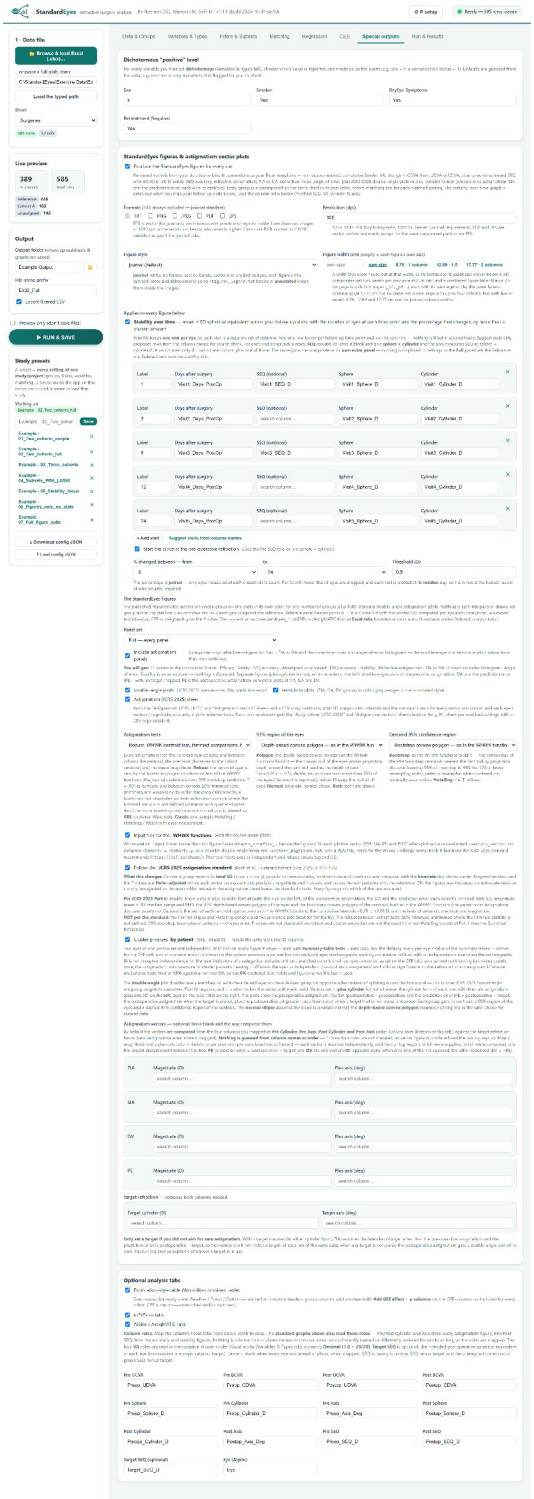
Step 6 · GEE

GEE is run on both cohorts, with the main and the adjusted model, matching weights carried into the matched models, and the main model supplying the GEE columns of the summary tables. The same tab holds the three settings that decide how every *P* value is computed and printed: *GEE small-sample inference* (md), *Summary-table tests* (auto) and *P value style* (ama), all at their defaults.

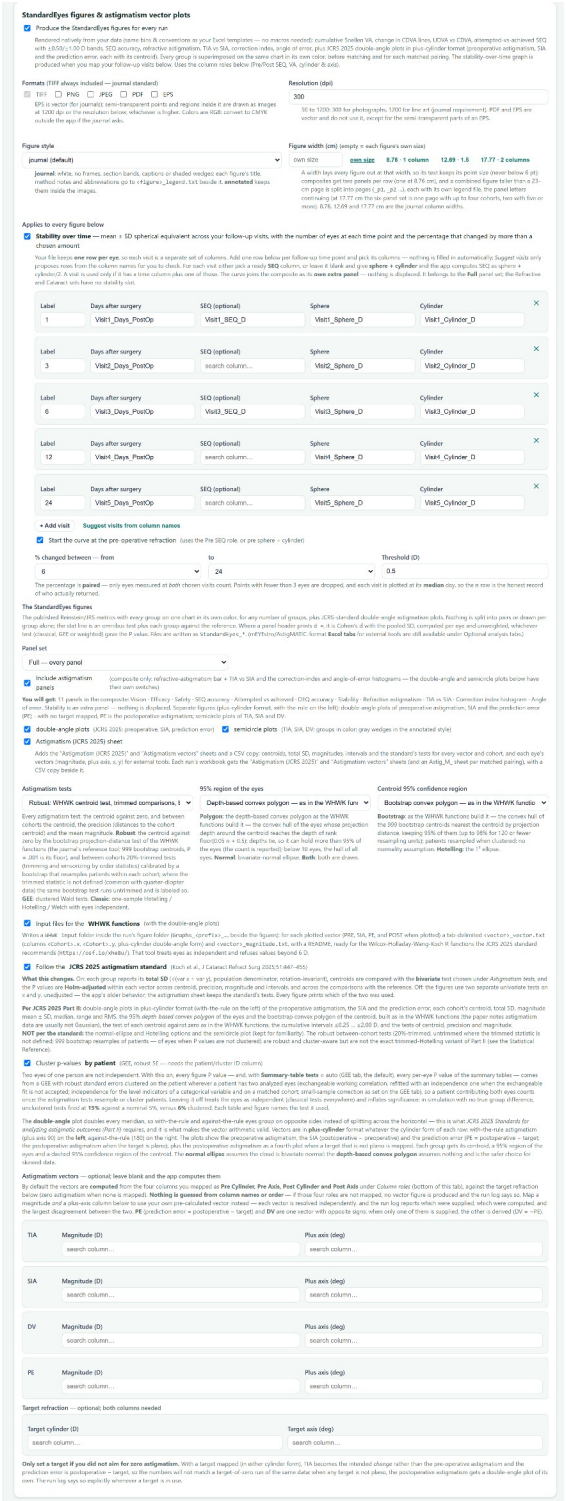


Step 7 · Special outputs

The dichotomous "positive" levels (Sex = F, Smoker = Yes, DryEye_Symptoms = Yes, Retreatment_Required = Yes), the figure options and the optional sheets. The column roles the figures and export tabs need are mapped at the bottom: the four acuity roles (read in the notation declared on the Variables & Types tab), the preoperative and postoperative sphere, cylinder, axis and SEQ, the target SEQ (Target_SEQ_D, which carries the monovision targets) and the eye column. Five follow-up visits are mapped for the stability panel, with the paired change between 6 and 24 months.



The figure and astigmatism block of the same tab: *Figure style* (journal), *Figure width* (blank, each figure's own size), formats and resolution; the double-angle and semicircle switches; the *Astigmatism (JCRS 2025) sheet*; *Astigmatism tests* (robust), *95% region of the eyes* (depth-based convex polygon), *Centroid 95% confidence region* (bootstrap convex polygon); *Input files for the WHWK functions*; *Follow the JCRS 2025 astigmatism standard*; and *Cluster p-values by patient*.



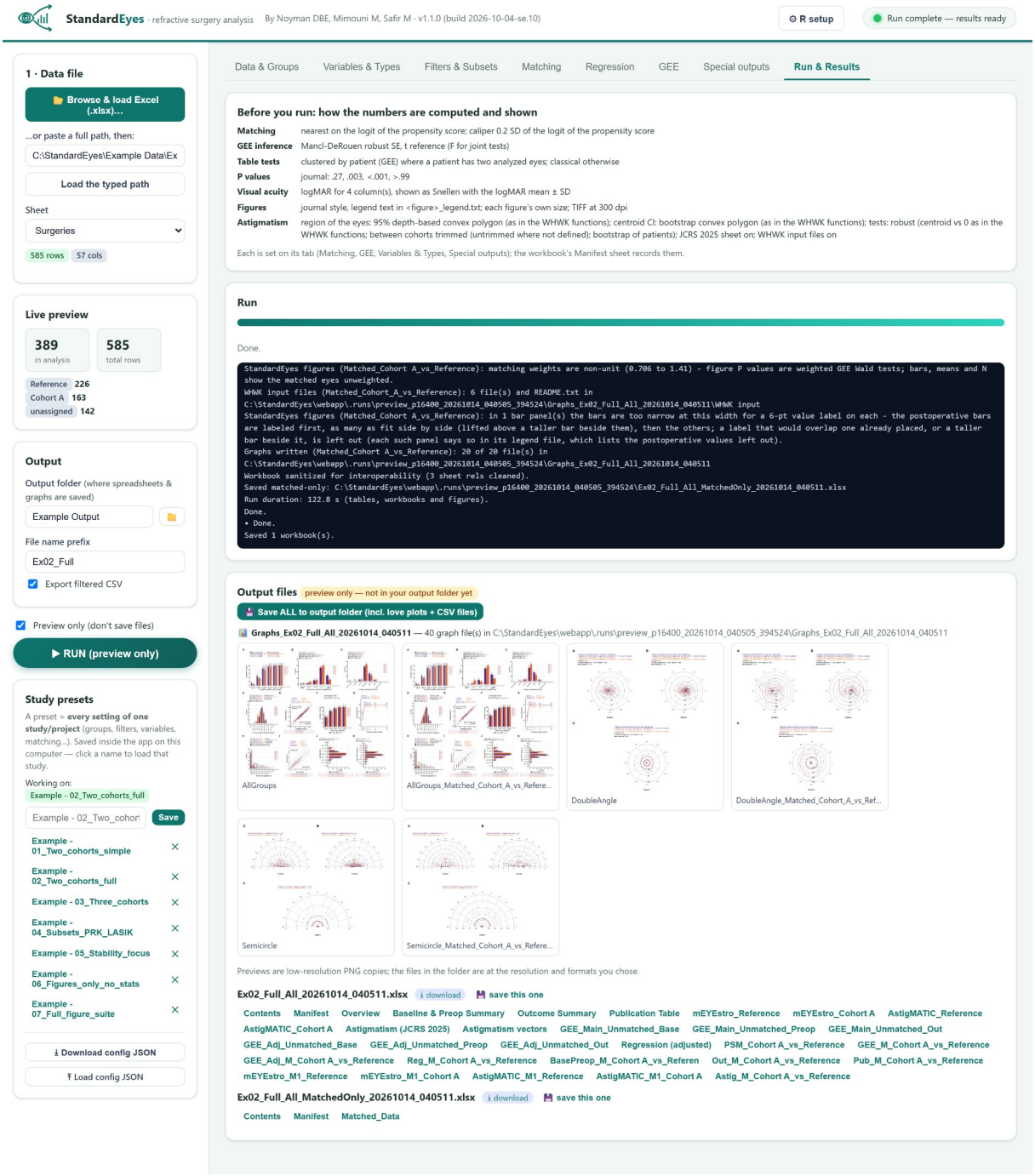
Step 8 · Run & Results

Before you run lists, in words, how this configuration computes and prints its numbers: matching on the logit of the propensity score with a caliper of 0.2 SD, Mancl-DeRouen robust SE with a *t* reference, tables clustered by patient, journal-style *P* values, acuity in logMAR shown as Snellen, the figure style and formats, and the astigmatism methods.

Run with **Preview only** to inspect everything first, or run and save directly. Progress and the R log are shown while the engine works: how many rows each filter removed, how each acuity column was read, every file written, and the WARNING lines, which are repeated under "Check before using the results". A running

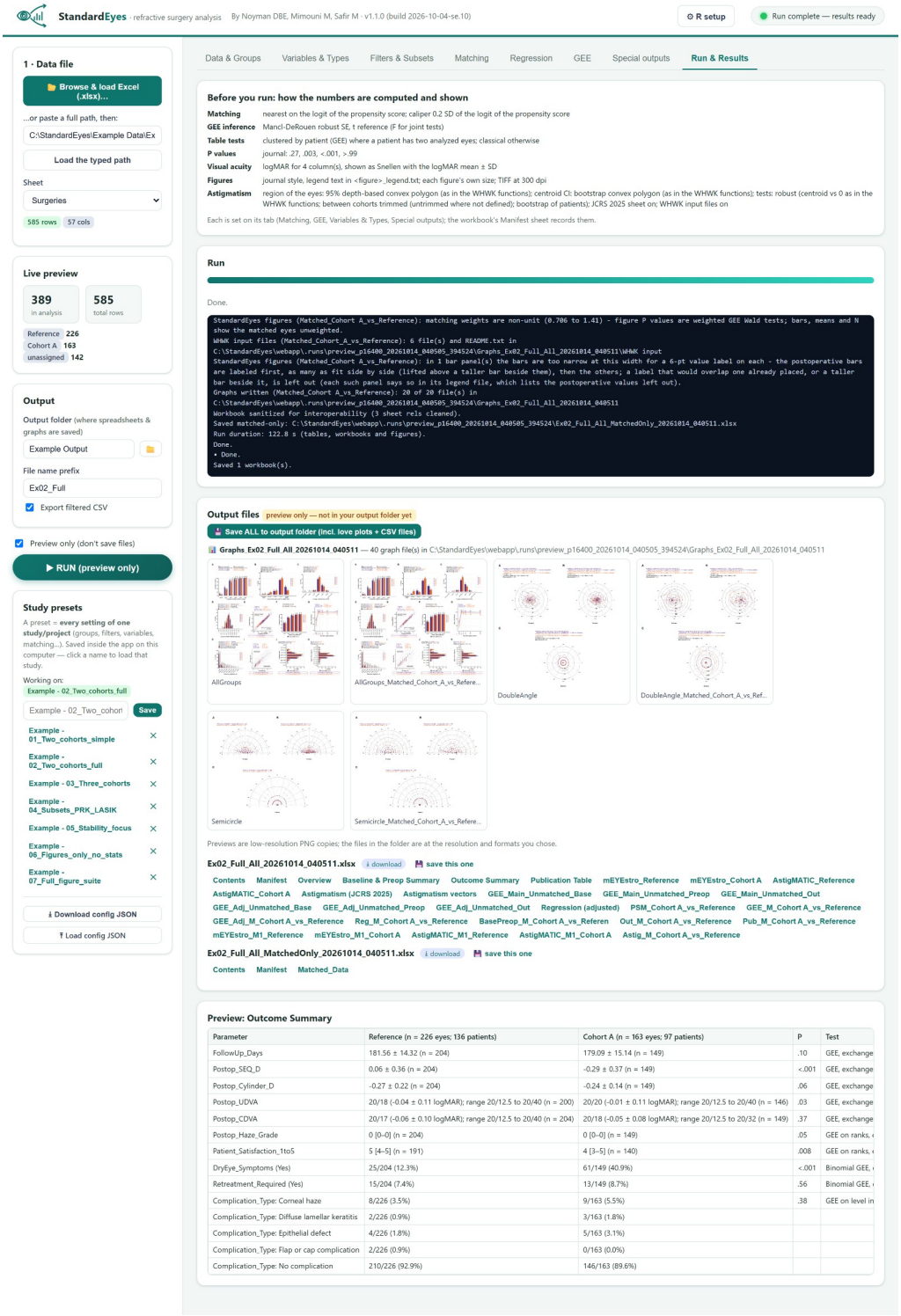
analysis can be stopped with **Cancel**. The figure folder is listed with a gallery of preview images of every composite, followed by the workbook with all its sheets and the matched-only workbook. **Save ALL to output folder** then writes the workbooks, balance plot, figures, legend files, WHWK input files and CSV files in one action.

For Example 02 the figure folder holds 20 figures per cohort (eleven panels, the composite, three double-angle plots and their composite, three semicircle plots and their composite), once for the whole cohort and once for the matched pairing, each with its legend file, plus the `WHWK input` and `_preview` sub-folders; the balance plot is written beside the workbooks.



Step 9 · Browsing the results inside the app

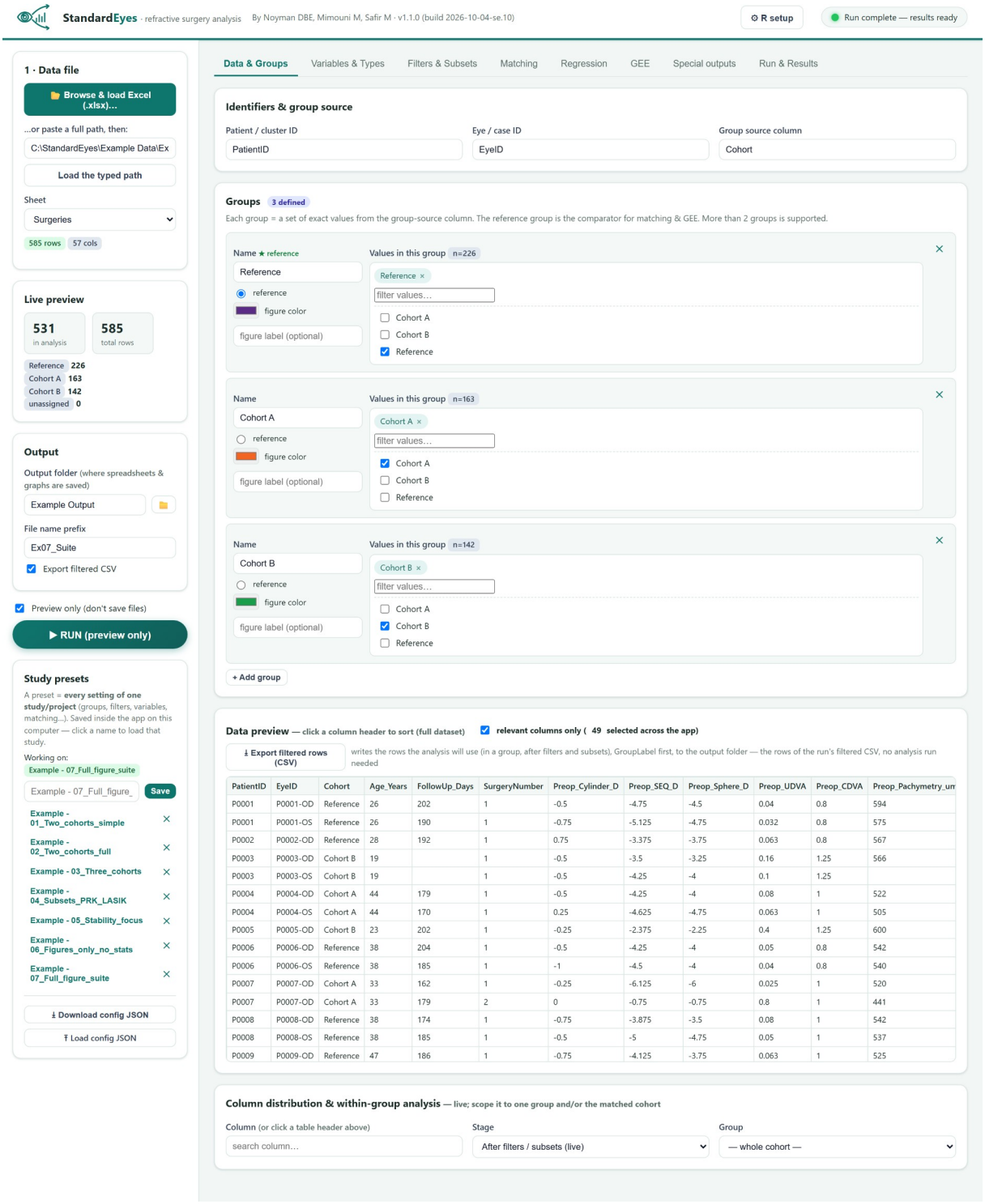
Clicking any sheet name under a workbook opens that sheet in a *Preview* table beneath the list, before anything is saved, so the analysis can be checked without leaving the tool or opening Excel.



Step 10 · Three cohorts

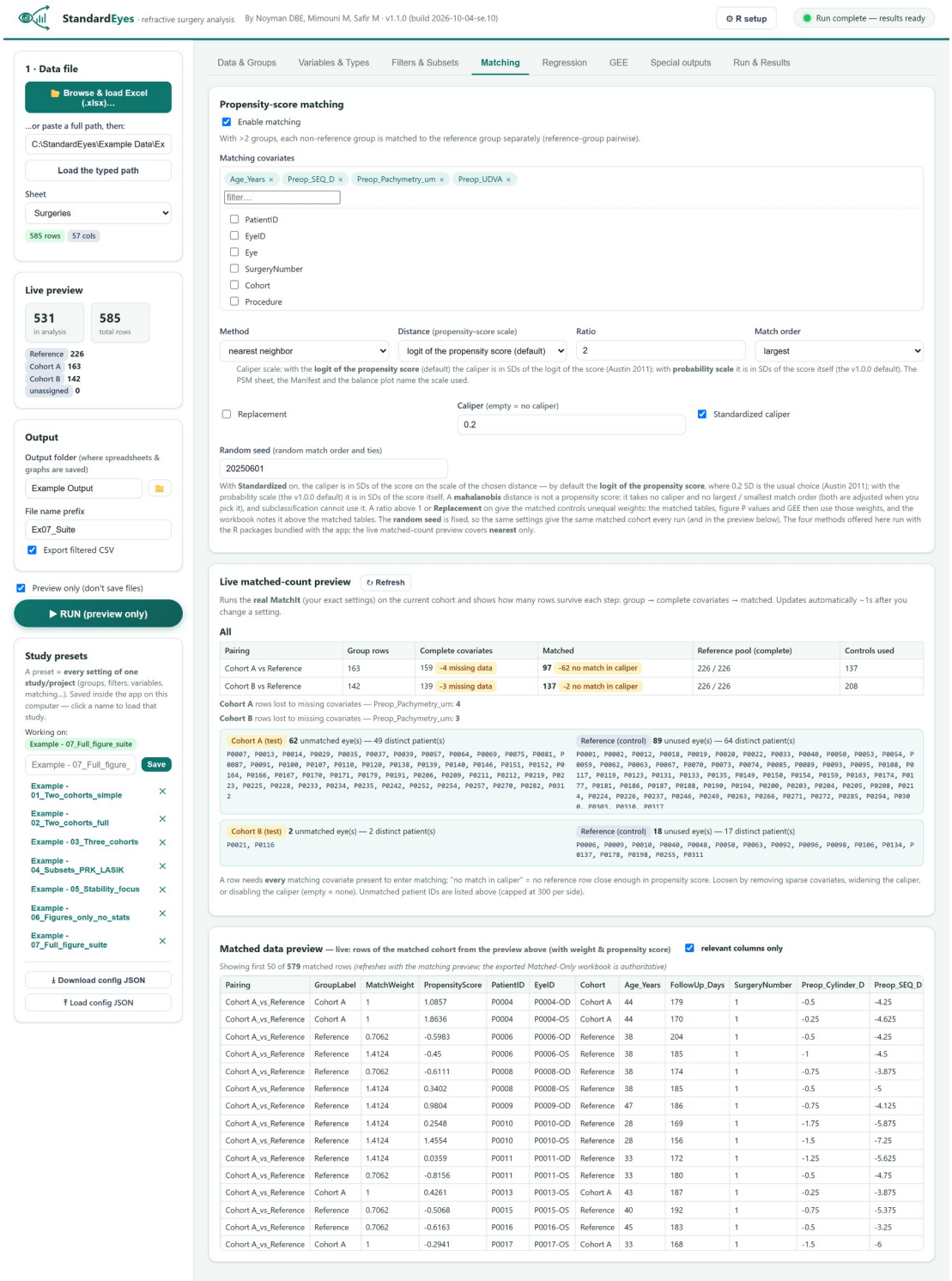
Loading *Example - 07_Full_figure_suite* adds Cohort B: Reference, Cohort A and Cohort B, with Reference still the reference. The sidebar now counts all 531 analyzed rows. Apart from the third group, the file prefix and

the within-group tables (none in this preset), the settings are those of steps 2–7; *Example - 03_Three_cohorts* runs the same analysis.



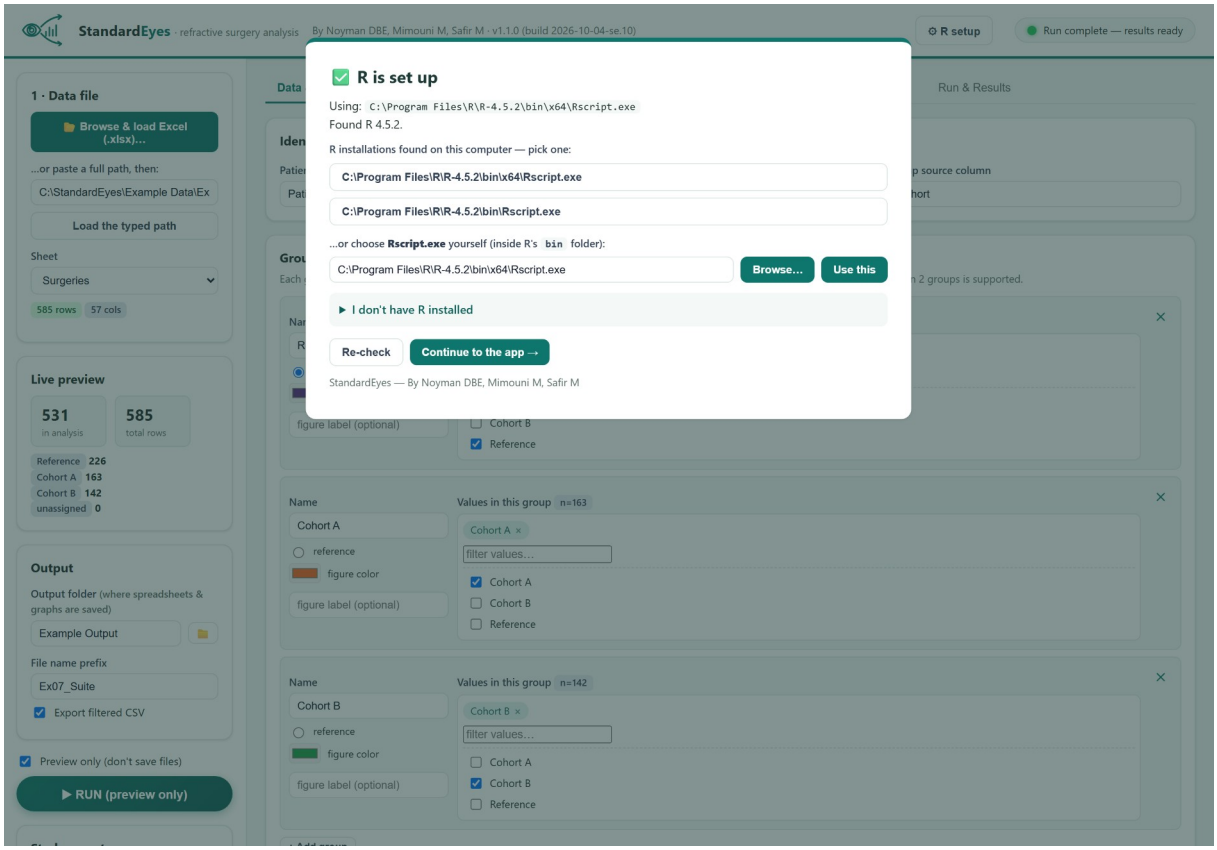
Step 11 · Matching with three cohorts

Each non-reference cohort is matched to the reference **separately**. The preview reports both pairings: Cohort A vs Reference (97 of 163 eyes matched, 137 Reference eyes used) and Cohort B vs Reference (137 of 142 matched, 208 Reference eyes used). The run produces a complete set of matched outputs for each pairing.



Step 12 · The R setup screen

The only external requirement is R 4.5. If it is not found automatically, this screen opens; it can also be reopened from the header. It lists the R installations found, accepts a path chosen with a file picker or typed in, and validates the choice by reporting the R version.

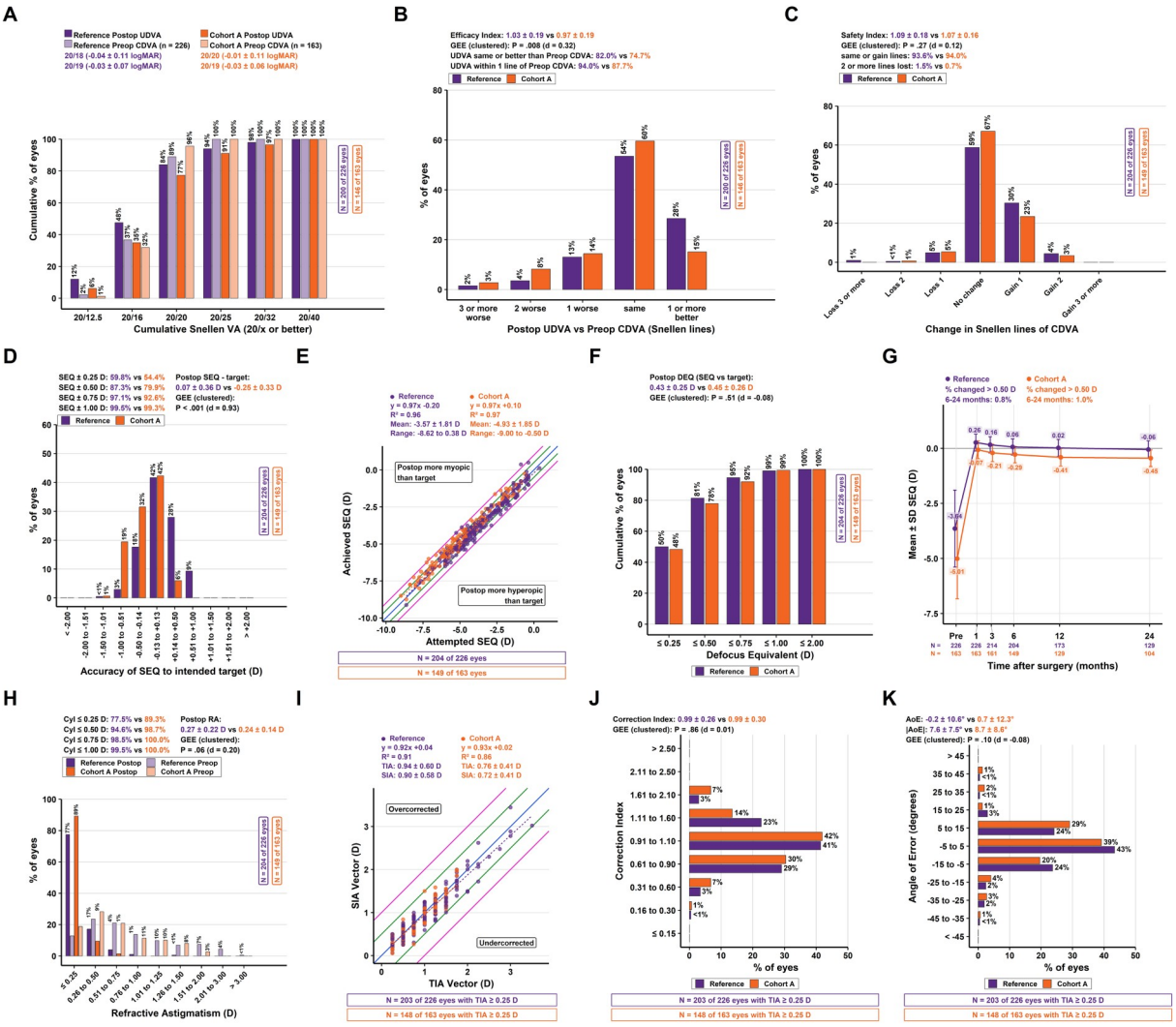


B.4 The output figures

The figures below come from the release run of Example 02 (two cohorts), Example 05 (stability) and Example 07 (three cohorts), journal style, each figure at its own size. Every panel is written both as its own file and inside the composite, each with its legend file.

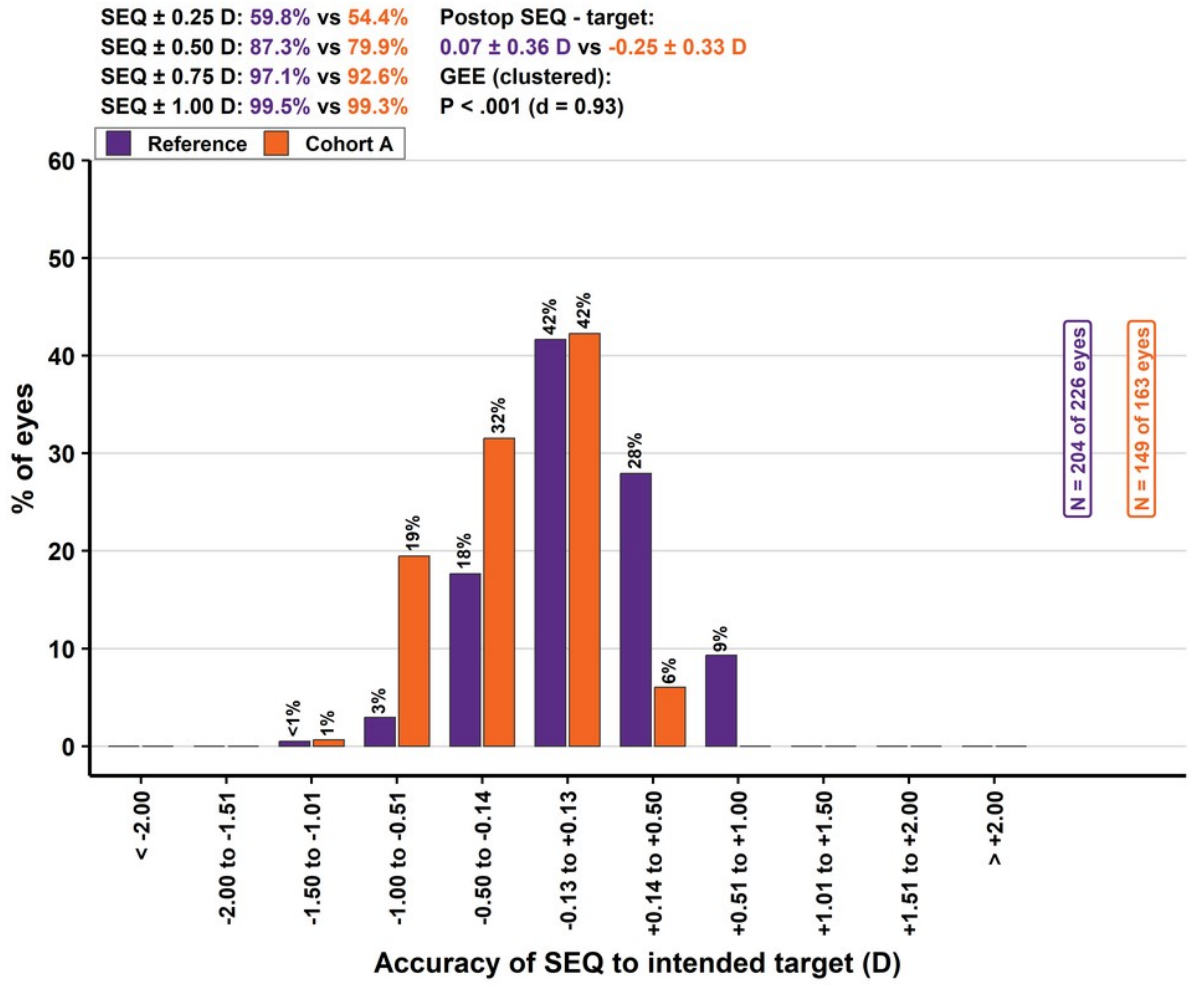
The StandardEyes composite

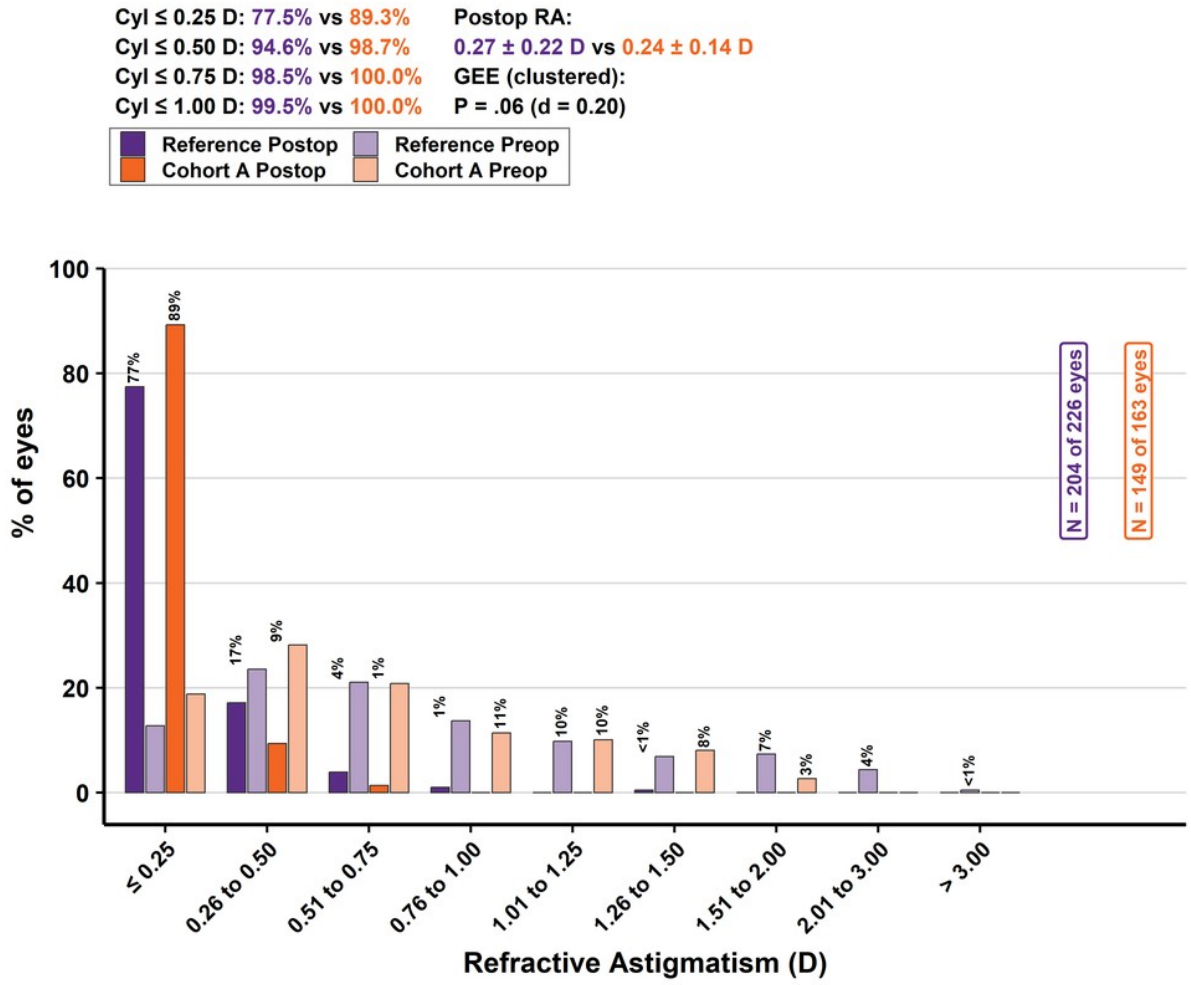
Both cohorts share the axes of every panel, eleven panels lettered A to K (A.13), with the number of eyes per cohort in a box. The headers print the pairwise test. In Example 02 the efficacy index is 1.03 ± 0.19 in the Reference cohort and 0.97 ± 0.19 in Cohort A (GEE (clustered): *P* = .008, d = 0.32), and the paired percentage of eyes whose SEQ changed by more than 0.50 D between 6 and 24 months is 0.8% and 1.0%. Panel titles, the outcome behind each *P* value and every abbreviation are in `StandardEyes_AllGroups_legend.txt`.

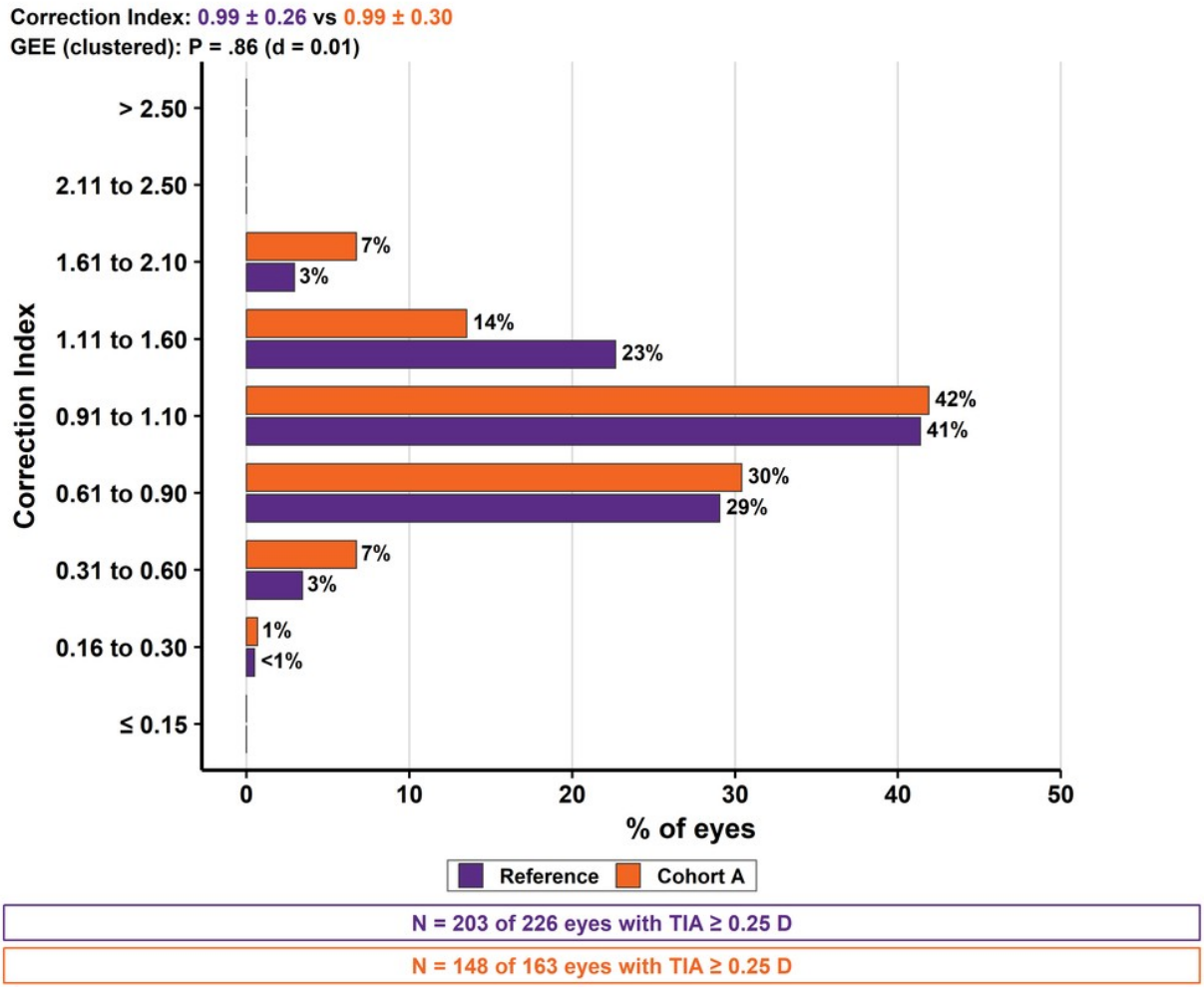


Individual panels

Each panel is also written on its own, at full size. The SEQ accuracy panel bins postoperative SEQ minus the target: 87.3% of Reference eyes and 79.9% of Cohort A eyes are within ±0.50 D, and the mean error is 0.07 ± 0.36 D against −0.25 ± 0.33 D (GEE (clustered): *P* < .001, d = 0.93). The refractive-astigmatism panel compares the postoperative cylinder magnitude (77.5% and 89.3% of eyes ≤ 0.25 D; *P* = .06), and the correction-index panel the eyes with TIA ≥ 0.25 D (0.99 ± 0.26 and 0.99 ± 0.30; *P* = .86).

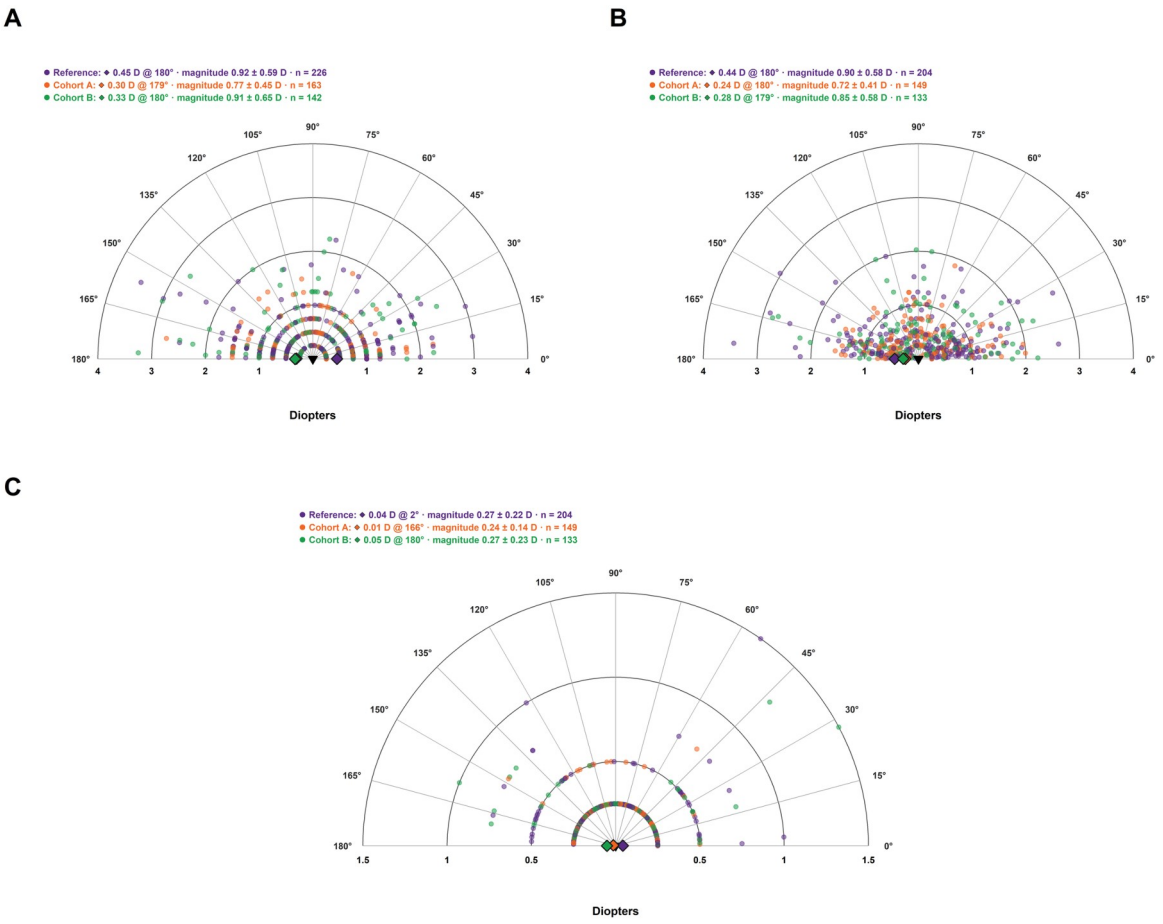

SEQ ± 0.25 D: 59.8% vs 54.4%
SEQ ± 0.50 D: 87.3% vs 79.9%
SEQ ± 0.75 D: 97.1% vs 92.6%
SEQ ± 1.00 D: 99.5% vs 99.3%
Postop SEQ - target:
0.07 ± 0.36 D vs -0.25 ± 0.33 D
GEE (clustered):
P < .001 (d = 0.93)
Reference
Cohort A
% of eyes
60
50
40
30
20
10
0
<1%
1%
3%
19%
18%
32%
42%
42%
28%
6%
9%
N = 204 of 226 eyes
N = 149 of 163 eyes
< -2.00
-2.00 to -1.51
-1.50 to -1.01
-1.00 to -0.51
-0.50 to -0.14
-0.13 to +0.13
+0.14 to +0.50
+0.51 to +1.00
+1.01 to +1.50
+1.51 to +2.00
> +2.00
Accuracy of SEQ to intended target (D)


Cyl ≤ 0.25 D: 77.5% vs 89.3%
Cyl ≤ 0.50 D: 94.6% vs 98.7%
Cyl ≤ 0.75 D: 98.5% vs 100.0%
Cyl ≤ 1.00 D: 99.5% vs 100.0%
Postop RA:
0.27 ± 0.22 D vs 0.24 ± 0.14 D
GEE (clustered):
P = .06 (d = 0.20)
Reference Postop
Cohort A Postop
Reference Preop
Cohort A Preop
% of eyes
100
80
60
40
20
0
77%
89%
17%
9%
4%
1%
1%
11%
10%
10%
<1%
8%
7%
3%
4%
<1%
N = 204 of 226 eyes
N = 149 of 163 eyes
≤ 0.25
0.26 to 0.50
0.51 to 0.75
0.76 to 1.00
1.01 to 1.25
1.26 to 1.50
1.51 to 2.00
2.01 to 3.00
> 3.00
Refractive Astigmatism (D)



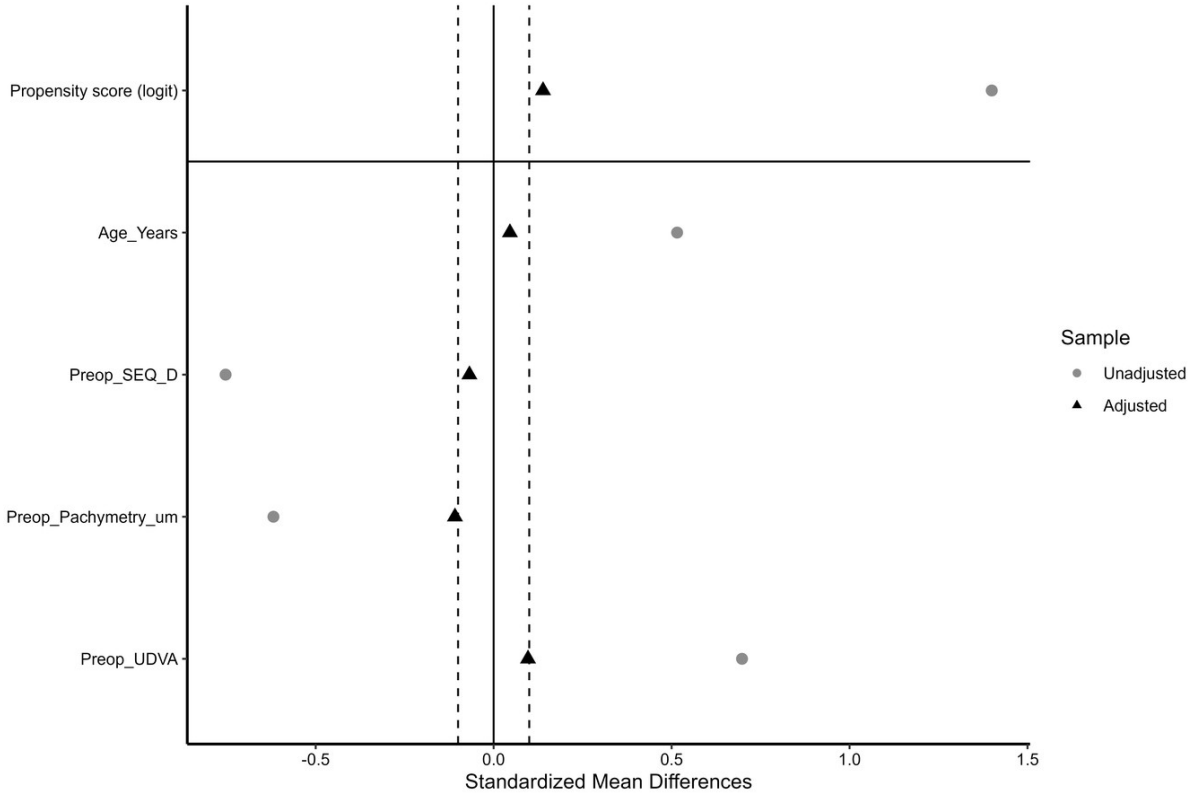
Semicircle vector plots

The Alpins TIA, SIA and DV as single-angle semicircle plots with all cohorts superimposed, vector means computed in double-angle space and drawn at the plus axis (Example 07). They are kept for familiarity and are not part of the JCRS 2025 standard.



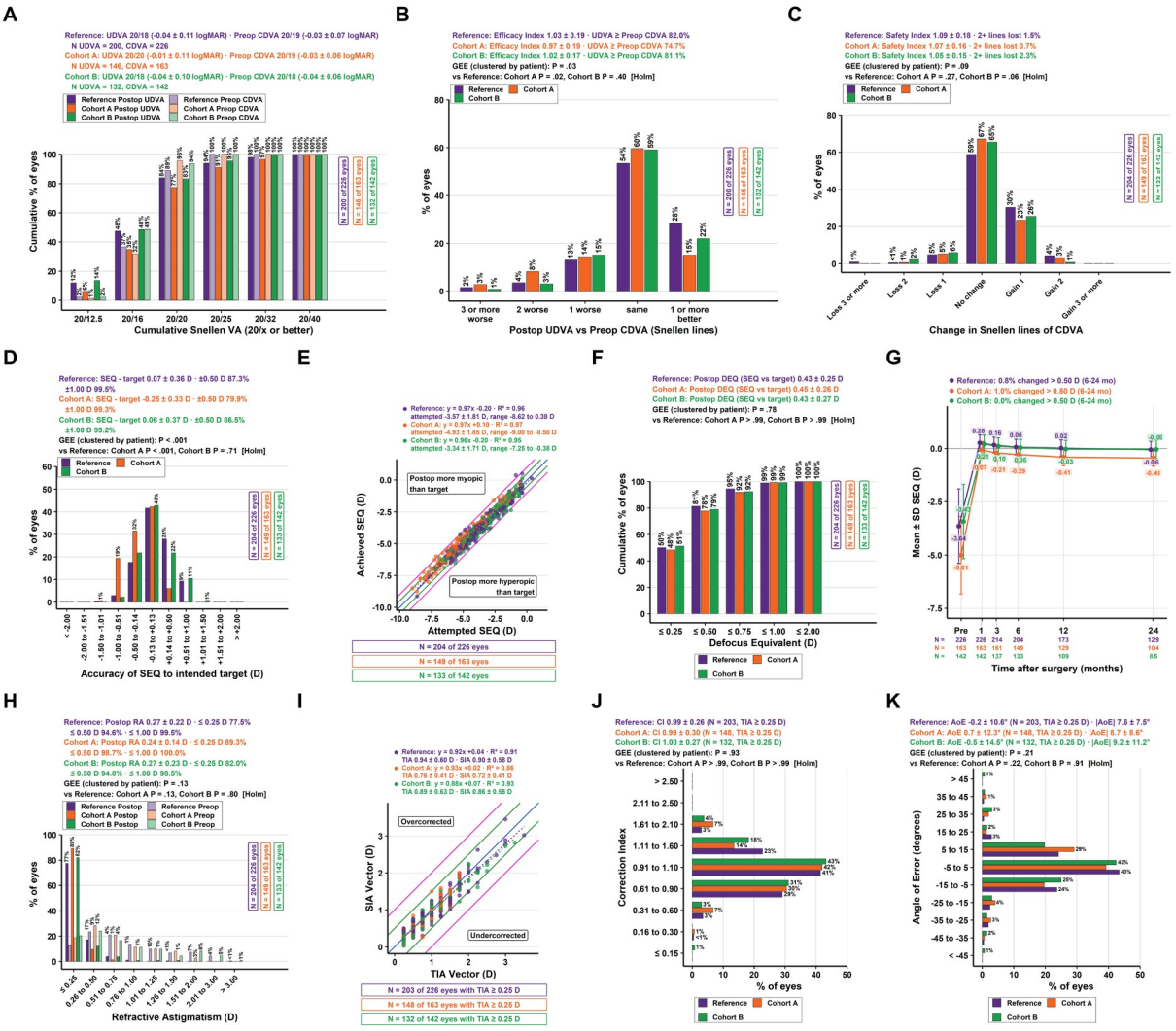
Covariate balance

Produced for each matched pairing (Example 02, Cohort A vs Reference). Before matching, Cohort A differs from the Reference cohort in all four covariates (SMD: age 0.52, preoperative SEQ −0.75, pachymetry −0.62, UDVA 0.70; propensity score 1.40). After matching, age (0.05), SEQ (−0.07) and UDVA (0.10) are inside the 0.1 threshold, but pachymetry is still −0.11. That is why the matched cohort is also adjusted for pachymetry (A.11). In the PRK subset of Example 04, preoperative SEQ (−0.15) and UDVA (0.18) remain above the threshold after matching. Read this plot before trusting a matched comparison.



The composite with three cohorts

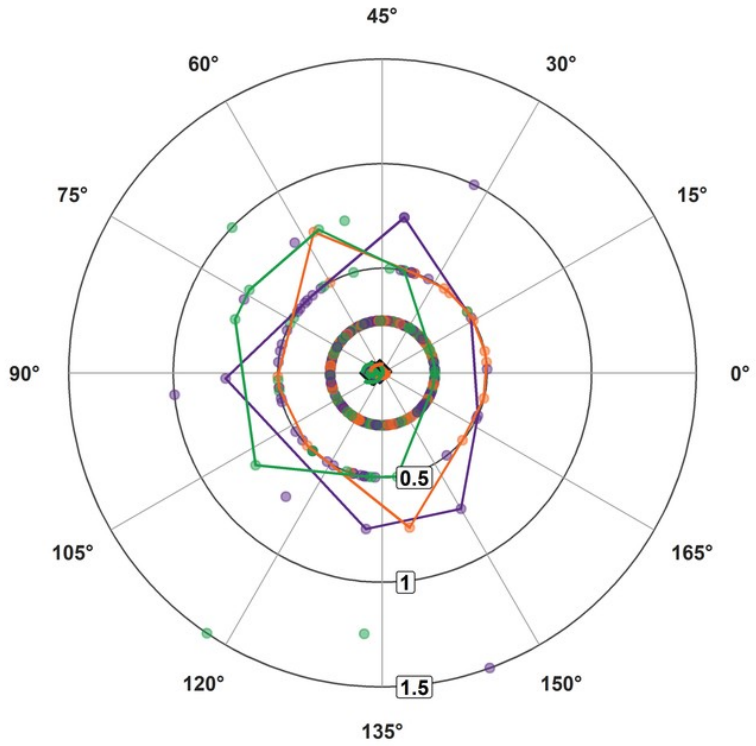
All cohorts share every panel (Example 07). Each header prints the omnibus test and each cohort against the reference, Holm-adjusted and tagged [Holm].



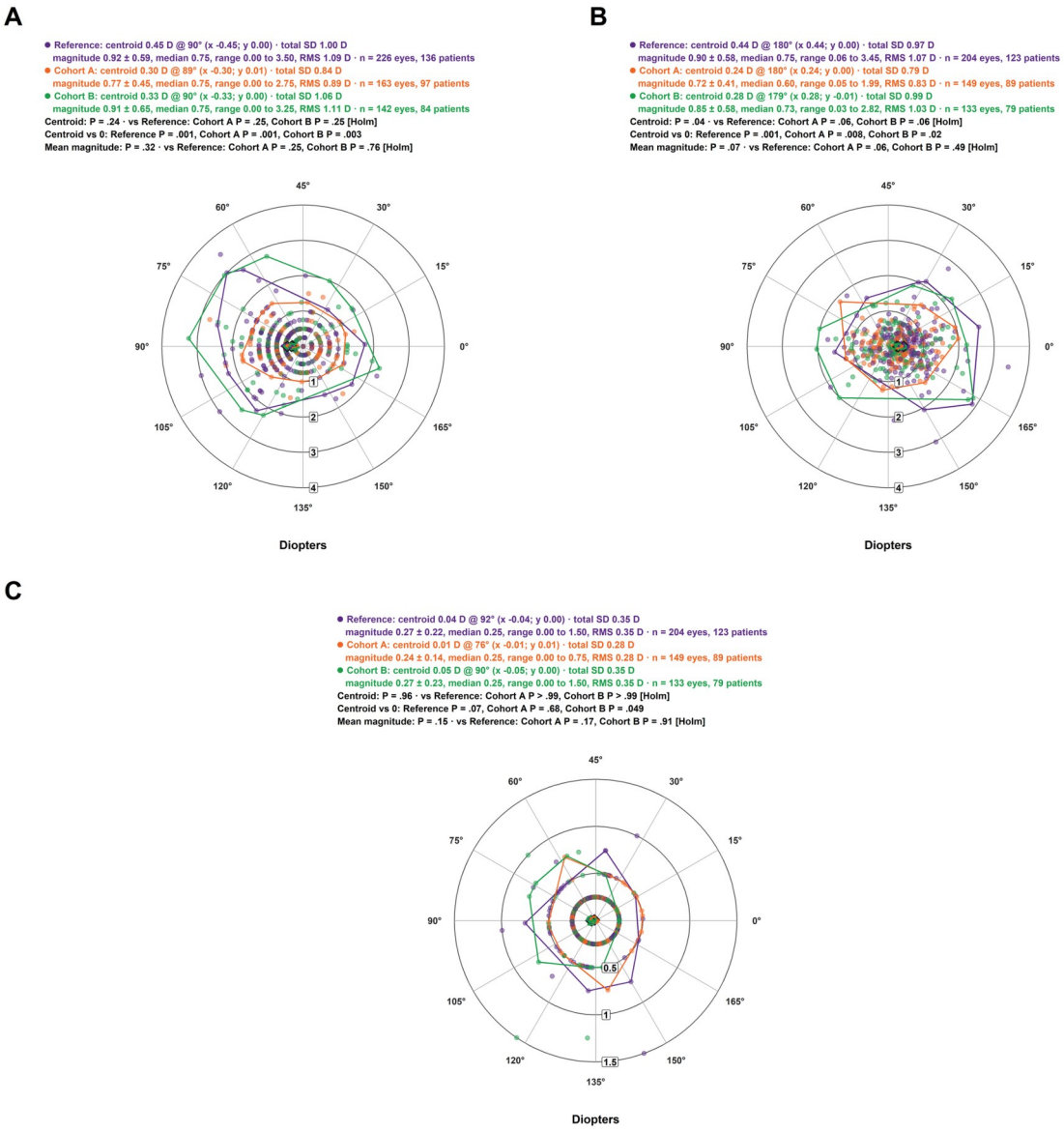
The JCRS double-angle plots

The prediction error of Example 07 (target plano, so PE = POST). Meridians are doubled, so with-the-rule eyes lie on the left and against-the-rule eyes on the right instead of splitting across the horizontal. Each cohort has its centroid (a diamond), its 95% depth-based convex polygon (Reference 197 of 204 eyes, Cohort A 149 of 149, Cohort B 128 of 133; in Cohort A the depth rule removed no eye) and the dashed bootstrap convex polygon of its centroid. The centroids are 0.04 D @ 92° (Reference), 0.01 D @ 76° (Cohort A) and 0.05 D @ 90° (Cohort B), close to the simulated true centroids of about 0.035 D @ 90° in every cohort; the omnibus centroid comparison gives $P = .96$ (robust 20%-trimmed T^2, 999 resamples of patients). The mean-magnitude comparison ran untrimmed, because the quarter-diopter magnitudes leave no spread after winsorizing; the label and the legend file say so.

● Reference: centroid 0.04 D @ 92° (x -0.04; y 0.00) · total SD 0.35 D
magnitude 0.27 ± 0.22, median 0.25, range 0.00 to 1.50, RMS 0.35 D · n = 204 eyes, 123 patients
● Cohort A: centroid 0.01 D @ 76° (x -0.01; y 0.01) · total SD 0.28 D
magnitude 0.24 ± 0.14, median 0.25, range 0.00 to 0.75, RMS 0.28 D · n = 149 eyes, 89 patients
● Cohort B: centroid 0.05 D @ 90° (x -0.05; y 0.00) · total SD 0.35 D
magnitude 0.27 ± 0.23, median 0.25, range 0.00 to 1.50, RMS 0.35 D · n = 133 eyes, 79 patients
Centroid: P = .96 · vs Reference: Cohort A P > .99, Cohort B P > .99 [Holm]
Centroid vs 0: Reference P = .07, Cohort A P = .68, Cohort B P = .049
Mean magnitude: P = .15 · vs Reference: Cohort A P = .17, Cohort B P = .91 [Holm]

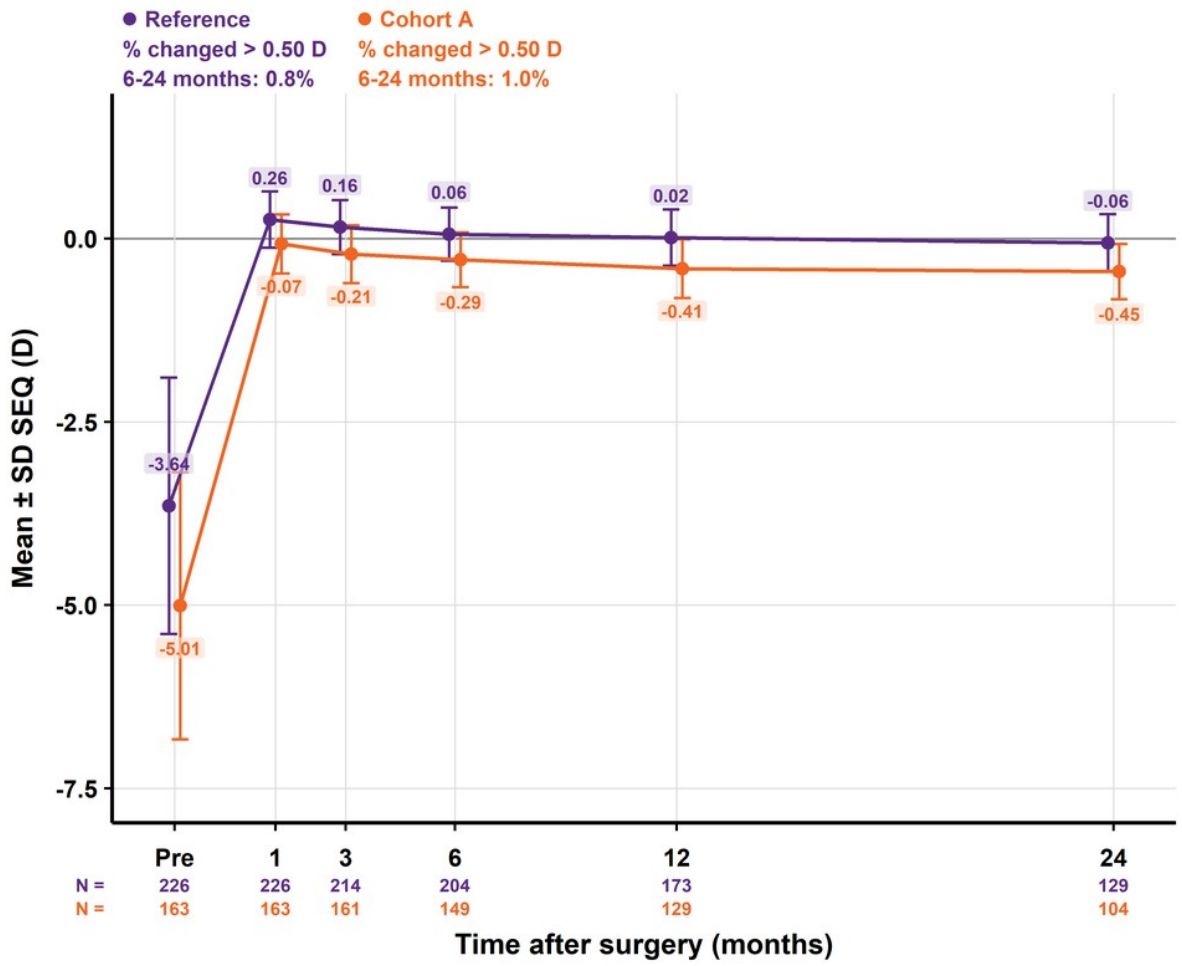


The plotted vectors, PRE, SIA and PE, are also composed into one figure:



Stability over time

The stability curve as its own panel (Example 05, the same cohorts and visits as Example 02): mean ± SD SEQ at the preoperative visit and at 1, 3, 6, 12 and 24 months, the number of eyes beneath the axis (Reference 226 preoperatively and 129 at 24 months), and the paired percentage changing by more than 0.50 D between 6 and 24 months.



B.5 The output files and sheets

The files a run produces

File / folder	Contents
<prefix>_<run>_<time stamp>.xlsx	The **main workbook** of the run (All without subsets, else the subset's name, or Pooled).
<prefix>_<run>_MatchedOnly_<time stamp>.xlsx	The matched rows of every pairing in *Matched_Data* (a *Pairing* column says which; MatchIt's columns *distance*, *weights* and *subclass* at the end), with a Manifest that counts the eyes per pairing and as distinct eyes.
<prefix>_<run>_Astigmatism_<time stamp>.csv	The tables of the astigmatism sheets (A.14).
Graphs_<prefix>_<run>_<time stamp>\	Every figure in each chosen format with its _legend.txt, for the whole cohort and for each matched pairing; the WHWK input sub-folder; the _preview sub-folder of PNG copies.
LovePlot_<prefix>_<run>_<pairing>_<time stamp>.<format>	The balance plot of each matched pairing, with its legend file, e.g. LovePlot_Ex02_Full_All_Cohort A_vs_Reference_<time stamp>.tiff.
Filtered_Dataset_<time stamp>.csv	The analyzed rows (A.15): 389 rows in Example 02, 531 in Example 07, 341 in Example 04 (PRK and LASIK).

Example 02 writes one main workbook, one matched-only workbook (234 matched rows: 137 Reference and 97 Cohort A eyes), 41 TIFF figures (40 in the figure folder and the balance plot) and 13 files in `WHWK input` (PRE, SIA and PE for the whole cohort and for the matched pairing, and the README).

Contents and Manifest

The workbook opens on **Contents** (one row per sheet: a link, its full name, as Excel cuts sheet names at 31 characters, and a description) and **Manifest** (A.15). In Example 02 the Manifest records StandardEyes 1.1.0, build 2026-10-04-se.10, 389 rows analyzed (Reference = 226; Cohort A = 163), the matching "nearest-neighbor on the propensity score (MatchIt)" with "caliper 0.2 SD of the logit of the propensity score", matching weights "non-unit (0.706-1.41)", and every setting as JSON.

Overview

Cohort description and quality control: numbers of patients (233) and eyes (389), age per patient (35.92 ± 8.23, range 19–56), follow-up per eye (median 181 days), sex per patient, the group sizes and how each source value was assigned to a group. A note names the acuity columns analyzed in logMAR. Read this first: if a group is the wrong size, the reason is visible here.

Notes

Note

Visual acuity analyzed in logMAR (lower = better): Preop_UDVA, Preop_CDVA, Postop_UDVA, Postop_CDVA - converted from the recorded notation.

Initial Summary

Description	Value
Number of Patients	233
Number of Eyes	389
Mean Age (per patient)	35.92
SD Age (per patient)	8.23
Age Range (per patient)	19–56
Mean Follow-up Time (per eye)	180.52
SD Follow-up Time (per eye)	14.70
Median Follow-up Time [IQR] (per eye)	181.00 [169.00–193.00]

Sex (per patient)

Sex	Count	Percentage
F	135	57.9
M	98	42.1

Groups

Group	Count	Percentage
Reference	226	58.1
Cohort A	163	41.9

QC - group assignment by source value

Value in Cohort	Group	N
Reference	Reference	226
Cohort A	Cohort A	163

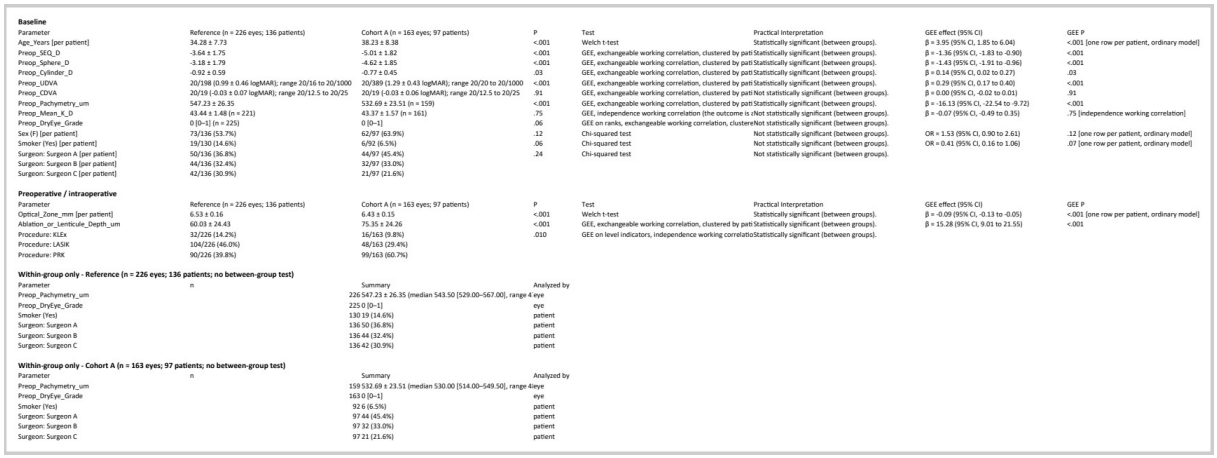
Baseline & Preop Summary

Every baseline and preoperative variable, one column per group, with the *P* value, **the test that produced it** and its interpretation, followed by the GEE columns. The headers give both denominators, `Reference (n = 226 eyes; 136 patients)` and `Cohort A (n = 163 eyes; 97 patients)`. The rows show each rule of A.4:

- per-eye rows get the patient-clustered GEE: preoperative SEQ −3.64 ± 1.75 against −5.01 ± 1.82 D, *P* < .001, *GEE, exchangeable working correlation, clustered by patient (robust SE (Mancl-DeRouen), t reference)*;
- the ordinal dry-eye grade is tested by a GEE on ranks (*P* = .06), the three-level procedure by the joint test on its level indicators (*P* = .010, *F* reference);

- mean keratometry, almost constant within patients, was refitted with an independence working correlation, which the Test column and the GEE column ("[independence working correlation]") say;
- the [per patient] rows (age, sex, smoking, optical zone, surgeon) get the classical tests on one row per patient, and their GEE column the ordinary per-patient model ("[one row per patient, ordinary model]");
- acuity reads in Snellen with logMAR: preoperative CDVA `20/19 (-0.03 ± 0.07 logMAR); range 20/12.5 to 20/25`.

The within-group sections close the sheet, one block per group with no between-group test, and say whether each row was analyzed per eye or per patient.



Baseline

Parameter	Reference (n = 226 eyes; 136 patients)	Cohort A (n = 163 eyes; 97 patients)	P	Test	Practical Interpretation	GEE effect (95% CI)	GEE P
Age_Years [per patient]	34.28 ± 7.73	38.23 ± 8.38	<.001	Welch t-test	Statistically significant (between groups).	β = 3.95 (95% CI, 1.85 to 6.04)	<.001 [one row per patient, ordinary model]
Preop_SEQ_D	-3.64 ± 1.75	-5.01 ± 1.82	<.001	GEE, exchangeable working correlation, clustered by pati	Statistically significant (between groups).	β = -1.36 (95% CI, -1.83 to -0.90)	<.001
Preop_Sphere_D	-3.18 ± 1.79	-4.62 ± 1.85	<.001	GEE, exchangeable working correlation, clustered by pati	Statistically significant (between groups).	β = -1.43 (95% CI, -1.91 to -0.96)	<.001
Preop_Cylinder_D	-0.92 ± 0.59	-0.77 ± 0.45	.03	GEE, exchangeable working correlation, clustered by pati	Statistically significant (between groups).	β = 0.14 (95% CI, 0.02 to 0.27)	.03
Preop_UDVA	20/198 (0.99 ± 0.46 logMAR); range 20/16 to 20/1000	20/389 (1.29 ± 0.43 logMAR); range 20/20 to 20/1000	<.001	GEE, exchangeable working correlation, clustered by pati	Statistically significant (between groups).	β = 0.29 (95% CI, 0.17 to 0.40)	<.001
Preop_CDVA	20/19 (-0.03 ± 0.07 logMAR); range 20/12.5 to 20/25	20/19 (-0.03 ± 0.06 logMAR); range 20/12.5 to 20/25	.91	GEE, exchangeable working correlation, clustered by pati	Not statistically significant (between groups).	β = 0.00 (95% CI, -0.02 to 0.01)	.91
Preop_Pachymetry_um	547.23 ± 26.35	532.69 ± 23.51 (n = 159)	<.001	GEE, exchangeable working correlation, clustered by pati	Statistically significant (between groups).	β = -16.13 (95% CI, -22.54 to -9.72)	<.001
Preop_Mean_K_D	43.44 ± 1.48 (n = 221)	43.37 ± 1.57 (n = 161)	.75	GEE, independence working correlation (the outcome is n	Not statistically significant (between groups).	β = -0.07 (95% CI, -0.49 to 0.35)	.75 [independence working correlation]
Preop_DryEye_Grade	0 [0–1] (n = 225)	0 [0–1]	.06	GEE on ranks, exchangeable working correlation, cluster	Not statistically significant (between groups).		
Sex (F) [per patient]	73/136 (53.7%)	62/97 (63.9%)	.12	Chi-squared test	Not statistically significant (between groups).	OR = 1.53 (95% CI, 0.90 to 2.61)	.12 [one row per patient, ordinary model]
Smoker (Yes) [per patient]	19/130 (14.6%)	6/92 (6.5%)	.06	Chi-squared test	Not statistically significant (between groups).	OR = 0.41 (95% CI, 0.16 to 1.06)	.07 [one row per patient, ordinary model]
Surgeon: Surgeon A [per patient]	50/136 (36.8%)	44/97 (45.4%)	.24	Chi-squared test	Not statistically significant (between groups).		
Surgeon: Surgeon B [per patient]	44/136 (32.4%)	32/97 (33.0%)					
Surgeon: Surgeon C [per patient]	42/136 (30.9%)	21/97 (21.6%)					

Preoperative / intraoperative

Parameter	Reference (n = 226 eyes; 136 patients)	Cohort A (n = 163 eyes; 97 patients)	P	Test	Practical Interpretation	GEE effect (95% CI)	GEE P
Optical_Zone_mm [per patient]	6.53 ± 0.16	6.43 ± 0.15	<.001	Welch t-test	Statistically significant (between groups).	β = -0.09 (95% CI, -0.13 to -0.05)	<.001 [one row per patient, ordinary model]
Ablation_or_Lenticule_Depth_um	60.03 ± 24.43	75.35 ± 24.26	<.001	GEE, exchangeable working correlation, clustered by pati	Statistically significant (between groups).	β = 15.28 (95% CI, 9.01 to 21.55)	<.001
Procedure: KLEx	32/226 (14.2%)	16/163 (9.8%)	.010	GEE on level indicators, independence working correlatio	Statistically significant (between groups).		
Procedure: LASIK	104/226 (46.0%)	48/163 (29.4%)					
Procedure: PRK	90/226 (39.8%)	99/163 (60.7%)					

Within-group only - Reference (n = 226 eyes; 136 patients; no between-group test)

Parameter	n	Summary	Analyzed by
Preop_Pachymetry_um		226 547.23 ± 26.35 (median 543.50 [529.00–567.00], range 4	eye
Preop_DryEye_Grade		225 0 [0–1]	eye
Smoker (Yes)		130 19 (14.6%)	patient
Surgeon: Surgeon A		136 50 (36.8%)	patient
Surgeon: Surgeon B		136 44 (32.4%)	patient
Surgeon: Surgeon C		136 42 (30.9%)	patient

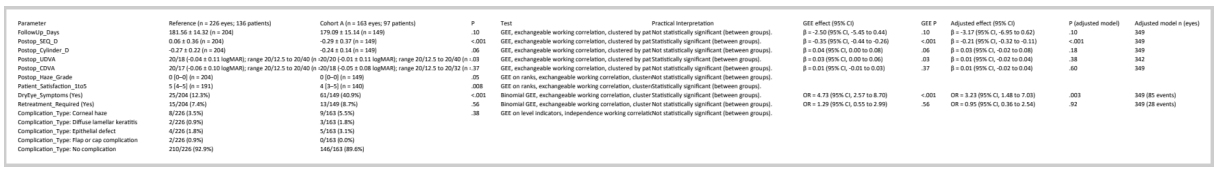
Within-group only - Cohort A (n = 163 eyes; 97 patients; no between-group test)

Parameter	n	Summary	Analyzed by
Preop_Pachymetry_um		159 532.69 ± 23.51 (median 530.00 [514.00–549.50], range 4	eye
Preop_DryEye_Grade		163 0 [0–1]	eye
Smoker (Yes)		92 6 (6.5%)	patient
Surgeon: Surgeon A		97 44 (45.4%)	patient
Surgeon: Surgeon B		97 32 (33.0%)	patient
Surgeon: Surgeon C		97 21 (21.6%)	patient

Outcome Summary

The same layout for the outcomes, with the GEE columns and the adjusted model merged in: *Adjusted effect (95% CI)*, *P (adjusted model)* and *Adjusted model n (eyes)*. Crude and adjusted estimates sit on one line, so the effect of adjustment is visible at once, and the example's true effects show what it does:

- **Postoperative SEQ** (true effect −0.20 D): crude β = −0.35 (95% CI, −0.44 to −0.26), adjusted β = −0.21 (95% CI, −0.32 to −0.11), *P* < .001 both, the adjusted model on 349 eyes.
- **Dry-eye symptoms** (true odds ratio 2.0): 12.3% against 40.9%; crude OR = 4.73 (95% CI, 2.57 to 8.70), adjusted OR = 3.23 (95% CI, 1.48 to 7.03), *P* = .003.
- **Postoperative cylinder** (no true effect): *P* = .06 crude, .18 adjusted.
- **Postoperative UDVA** reads `20/18 (-0.04 ± 0.11 logMAR)` against `20/20 (-0.01 ± 0.11 logMAR)` (*P* = .03 crude, .38 adjusted).
- **Complication type** is tested by the joint GEE on its level indicators (*P* = .38), although Cohort A has no flap or cap complication.



Parameter	Reference (n = 226 eyes; 136 patients)	Cohort A (n = 163 eyes; 97 patients)	P	Test	Practical Interpretation	GEE effect (95% CI)	GEE P	Adjusted effect (95% CI)	P (adjusted model)	Adjusted model n (eyes)
FollowUp_Days	181.50 ± 14.32 (n = 204)	179.09 ± 15.14 (n = 149)	.10	GEE, exchangeable working correlation, clustered by pat	Not statistically significant (between groups).	β = -2.50 (95% CI, -5.45 to 0.44)	.10	β = -3.17 (95% CI, -6.95 to 0.62)	.10	349
Postop_SEQ_D	0.06 ± 0.36 (n = 204)	-0.29 ± 0.37 (n = 149)	<.001	GEE, exchangeable working correlation, clustered by pat	Statistically significant (between groups).	β = -0.35 (95% CI, -0.44 to -0.26)	<.001	β = -0.21 (95% CI, -0.32 to -0.11)	<.001	349
Postop_Cylinder_D	-0.27 ± 0.22 (n = 204)	-0.24 ± 0.14 (n = 149)	.06	GEE, exchangeable working correlation, clustered by pat	Not statistically significant (between groups).	β = 0.04 (95% CI, 0.00 to 0.08)	.06	β = 0.03 (95% CI, -0.02 to 0.08)	.18	349
Postop_UDVA	20/18 (-0.04 ± 0.11 logMAR); range 20/12.5 to 20/40 (n	-20/20 (-0.01 ± 0.11 logMAR); range 20/12.5 to 20/40 (n =	.03	GEE, exchangeable working correlation, clustered by pat	Statistically significant (between groups).	β = 0.03 (95% CI, 0.00 to 0.06)	.03	β = 0.01 (95% CI, -0.02 to 0.04)	.38	342
Postop_CDVA	20/17 (-0.06 ± 0.10 logMAR); range 20/12.5 to 20/40 (n	-20/18 (-0.05 ± 0.08 logMAR); range 20/12.5 to 20/32 (n =	.37	GEE, exchangeable working correlation, clustered by pat	Not statistically significant (between groups).	β = 0.01 (95% CI, -0.01 to 0.03)	.37	β = 0.01 (95% CI, -0.02 to 0.04)	.60	349
Postop_Haze_Grade	0 [0–0] (n = 204)	0 [0–0] (n = 149)	.05	GEE on ranks, exchangeable working correlation, cluster	Not statistically significant (between groups).					
Patient_Satisfaction_1to5	5 [4–5] (n = 191)	4 [3–5] (n = 140)	.008	GEE on ranks, exchangeable working correlation, cluster	Statistically significant (between groups).					
DryEye_Symptoms (Yes)	25/204 (12.3%)	61/149 (40.9%)	<.001	Binomial GEE, exchangeable working correlation, cluster	Statistically significant (between groups).	OR = 4.73 (95% CI, 2.57 to 8.70)	<.001	OR = 3.23 (95% CI, 1.48 to 7.03)	.003	349 (85 events)
Retreatment_Required (Yes)	15/204 (7.4%)	13/149 (8.7%)	.56	Binomial GEE, exchangeable working correlation, cluster	Not statistically significant (between groups).	OR = 1.29 (95% CI, 0.55 to 2.99)	.56	OR = 0.95 (95% CI, 0.36 to 2.54)	.92	349 (28 events)
Complication_Type: Corneal haze	8/226 (3.5%)	9/163 (5.5%)	.38	GEE on level indicators, independence working correlatio	Not statistically significant (between groups).					
Complication_Type: Diffuse lamellar keratitis	2/226 (0.9%)	3/163 (1.8%)								
Complication_Type: Epithelial defect	4/226 (1.8%)	5/163 (3.1%)								
Complication_Type: Flap or cap complication	2/226 (0.9%)	0/163 (0.0%)								
Complication_Type: No complication	210/226 (92.9%)	146/163 (89.6%)								

Publication Table

One manuscript-ready sheet: baseline, preoperative and outcome blocks under section headers, group columns, *P* values and the GEE and adjusted columns, without the Test and Interpretation columns. The

footnotes say what the cells hold ("Values: … mean ± SD …; Visual acuity: the Snellen geometric mean (logMAR mean ± SD), then the range …; β = difference from Reference …") and list the **tests behind the *P* values** with their variables. A test that could not be performed prints **NE** with a footnote naming the variable and the reason: in the PRK workbook of Example 04, "NE = not estimable (no P value): variable is constant in this cohort: Procedure". The table of Example 02 has no NE cell.

Parameter	Reference (n = 226 eyes; 136 patients)	Cohort A (n = 163 eyes; 97 patients)	P	GEE effect (95% CI)	GEE P	Adjusted effect (95% CI)	P (adjusted model)	Adjusted model n (eyes)
Baseline characteristics								
Age_Years [per patient]	34.28 ± 7.73	38.23 ± 8.38	<.001	β = 3.95 (95% CI, 1.85 to 6.04)	<.001 [one row per patient, ordinary model]			
Preop_SEQ_D	-3.64 ± 1.75	-5.01 ± 1.82	<.001	β = -1.36 (95% CI, -1.83 to -0.90)	<.001			
Preop_Sphere_D	-3.18 ± 1.79	-4.62 ± 1.85	<.001	β = -1.43 (95% CI, -1.91 to -0.96)	<.001			
Preop_Cylinder_D	-0.92 ± 0.59	-0.77 ± 0.45	.03	β = 0.14 (95% CI, 0.02 to 0.27)	.03			
Preop_UDVA	20/198 (0.99 ± 0.46 logMAR); range 20/16 to 20/1000	20/389 (1.29 ± 0.43 logMAR); range 20/20 to 20/1000	<.001	β = 0.29 (95% CI, 0.17 to 0.40)	<.001			
Preop_CDVA	20/19 (-0.03 ± 0.07 logMAR); range 20/12.5 to 20/25	20/19 (-0.03 ± 0.06 logMAR); range 20/12.5 to 20/25	.91	β = 0.00 (95% CI, -0.02 to 0.01)	.91			
Preop_Pachymetry_um	547.23 ± 26.35	532.69 ± 23.51 (n = 159)	<.001	β = -16.13 (95% CI, -22.54 to -9.72)	<.001			
Preop_Mean_K_D	43.44 ± 1.48 (n = 221)	43.37 ± 1.57 (n = 161)	.75	β = -0.07 (95% CI, -0.49 to 0.35)	.75 [independence working correlation]			
Preop_DryEye_Grade	0 [0–1] (n = 225)	0 [0–1]	.06					
Sex (F) [per patient]	73/136 (53.7%)	62/97 (63.9%)	.12	OR = 1.53 (95% CI, 0.90 to 2.61)	.12 [one row per patient, ordinary model]			
Smoker (Yes) [per patient]	19/130 (14.6%)	6/92 (6.5%)	.06	OR = 0.41 (95% CI, 0.16 to 1.06)	.07 [one row per patient, ordinary model]			
Surgeon: Surgeon A [per patient]	50/136 (36.8%)	44/97 (45.4%)	.24					
Surgeon: Surgeon B [per patient]	44/136 (32.4%)	32/97 (33.0%)						
Surgeon: Surgeon C [per patient]	42/136 (30.9%)	21/97 (21.6%)						
Preoperative / intraoperative								
Optical_Zone_mm [per patient]	6.53 ± 0.16	6.43 ± 0.15	<.001	β = -0.09 (95% CI, -0.13 to -0.05)	<.001 [one row per patient, ordinary model]			
Ablation_or_Lenticule_Depth_um	60.03 ± 24.43	75.35 ± 24.26	<.001	β = 15.28 (95% CI, 9.01 to 21.55)	<.001			
Procedure: KLEx	32/226 (14.2%)	16/163 (9.8%)	.010					
Procedure: LASIK	104/226 (46.0%)	48/163 (29.4%)						
Procedure: PRK	90/226 (39.8%)	99/163 (60.7%)						
Outcomes								
FollowUp_Days	181.56 ± 14.32 (n = 204)	179.09 ± 15.14 (n = 149)	.10	β = -2.50 (95% CI, -5.45 to 0.44)	.10	β = -3.17 (95% CI, -6.95 to 0.62)	.10	349
Postop_SEQ_D	0.06 ± 0.36 (n = 204)	-0.29 ± 0.37 (n = 149)	<.001	β = -0.35 (95% CI, -0.44 to -0.26)	<.001	β = -0.21 (95% CI, -0.32 to -0.11)	<.001	349
Postop_Cylinder_D	-0.27 ± 0.22 (n = 204)	-0.24 ± 0.14 (n = 149)	.06	β = 0.04 (95% CI, 0.00 to 0.08)	.06	β = 0.03 (95% CI, -0.02 to 0.08)	.18	349
Postop_UDVA	20/18 (-0.04 ± 0.11 logMAR); range 20/12.5 to 20/40 (n = 20/20 (-0.01 ± 0.11 logMAR); range 20/12.5 to 20/40 (n = .03			β = 0.03 (95% CI, 0.00 to 0.06)	.03	β = 0.01 (95% CI, -0.02 to 0.04)	.38	342
Postop_CDVA	20/17 (-0.06 ± 0.10 logMAR); range 20/12.5 to 20/40 (n = 20/18 (-0.05 ± 0.08 logMAR); range 20/12.5 to 20/32 (n = .37			β = 0.01 (95% CI, -0.01 to 0.03)	.37	β = 0.01 (95% CI, -0.02 to 0.04)	.60	349
Postop_Haze_Grade	0 [0–0] (n = 204)	0 [0–0] (n = 149)	.05					
Patient_Satisfaction_1to5	5 [4–5] (n = 191)	4 [3–5] (n = 140)	.008					
DryEye_Symptoms (Yes)	25/204 (12.3%)	61/149 (40.9%)	<.001	OR = 4.73 (95% CI, 2.57 to 8.70)	<.001	OR = 3.23 (95% CI, 1.48 to 7.03)	.003	349 (85 events)
Retreatment_Required (Yes)	15/204 (7.4%)	13/149 (8.7%)	.56	OR = 1.29 (95% CI, 0.55 to 2.99)	.56	OR = 0.95 (95% CI, 0.36 to 2.54)	.92	349 (28 events)
Complication_Type: Corneal haze	8/226 (3.5%)	9/163 (5.5%)	.38					
Complication_Type: Diffuse lamellar keratitis	2/226 (0.9%)	3/163 (1.8%)						
Complication_Type: Epithelial defect	4/226 (1.8%)	5/163 (3.1%)						
Complication_Type: Flap or cap complication	2/226 (0.9%)	0/163 (0.0%)						
Complication_Type: No complication	210/226 (92.9%)	146/163 (89.6%)						

Values:
- Values are mean ± SD, median [interquartile range], n/N (%) - eyes (or patients) with the level among those with data.
- Visual acuity: the Snellen geometric mean (logMAR mean ± SD), then the range of the eyes in Snellen.
- (n = …) in a cell: the eyes (patients, for a row marked [per patient]) with data, where fewer than the column's n.
- β = difference from Reference in the variable's units (logMAR for acuity), with its 95% CI.
- OR = odds ratio of the level named in the row vs Reference, with its 95% CI.
- Adjusted model n (eyes): the complete-case eyes of the adjusted model.

Tests behind the P values:
- Welch t-test: Age_Years; Optical_Zone_mm.
- GEE, exchangeable working correlation, clustered by patient (robust SE (Mancl-DeRouen), t reference): Preop_SEQ_D; Preop_Sphere_D; Preop_Cylinder_D; Preop_UDVA; Preop_CDVA; Preop_Pachymetry_um; Ablation_or_Lenticule_Depth_um; FollowUp_Days; Postop_SEQ_D; Postop_Cylinder_D; Postop_UDVA; Postop_CDVA.
- GEE, independence working correlation (the outcome is almost constant within patients), clustered by patient (robust SE (Mancl-DeRouen), t reference): Preop_Mean_K_D.
- GEE on ranks, exchangeable working correlation, clustered by patient (robust SE (Mancl-DeRouen), t reference): Preop_DryEye_Grade; Postop_Haze_Grade; Patient_Satisfaction_1to5.
- Chi-squared test: Sex; Smoker; Surgeon.
- GEE on level indicators, independence working correlation, clustered by patient (robust SE (Mancl-DeRouen), F reference): Procedure; Complication_Type.
- Binomial GEE, exchangeable working correlation, clustered by patient (robust SE (Mancl-DeRouen), t reference): DryEye_Symptoms; Retreatment_Required.

PSM sheet (one per pairing)

Everything about the matching in one place: the settings, with the propensity-score scale and the caliper on both scales; the covariates; the balance before and after matching, with the verdict against 0.1 (A.8 and B.4); and the logit of the propensity score and the matching weight of every matched row.

PSM settings

Setting	Value
Matching method	nearest
Distance	propensity score, logit scale (MatchIt link linear.logit)
Propensity score scale	logit of the propensity score
Ratio	2
Replacement	no
Caliper	0.2 SD of the logit of the propensity score (= 0.321 on that scale)
Standardized caliper	yes
Matching order	largest
Random seed (matching)	20250601
Reference group	Reference
Patient ID (clustering for GEE only; not used in matching)	PatientID
Matched analyses	Matched analyses use an independence working correlation with robust standard errors clustered by patient; matched sets (pairs) are not used as clusters.

Matching covariates

Matching covariate
Age_Years
Preop_SEQ_D
Preop_Pachymetry_um
Preop_UDVA

Covariate balance (cobalt; standardized differences for every covariate)

Variable	Type	Standardized difference before matching	Standardized difference after matching	Balanced (|SMD| < 0.1)
Propensity score (logit)	Distance	1.400	0.139	
Age_Years	Continuous	0.516	0.046	yes
Preop_SEQ_D	Continuous	-0.753	-0.068	yes
Preop_Pachymetry_um	Continuous	-0.619	-0.109	no
Preop_UDVA	Continuous	0.698	0.097	yes

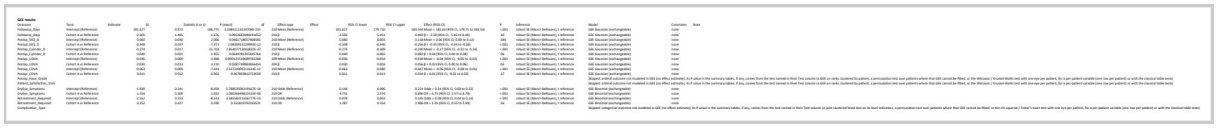
Propensity-score distribution (logit of the propensity score; matched rows)

PatientID	Group	Propensity score (logit)	Matching weight
P0004	Cohort A	1.086	1.000
P0004	Cohort A	1.864	1.000
P0006	Reference	-0.598	0.706
P0006	Reference	-0.450	1.412
P0008	Reference	-0.611	0.706
P0008	Reference	0.340	1.412
P0009	Reference	0.980	1.412
P0010	Reference	0.255	1.412
P0010	Reference	1.455	1.412
P0011	Reference	0.036	1.412
P0011	Reference	-0.816	0.706
P0013	Cohort A	0.426	1.000
P0015	Reference	-0.507	0.706
P0016	Reference	-0.616	0.706
P0017	Cohort A	-0.294	1.000
P0023	Cohort A	-0.652	1.000

GEE sheets

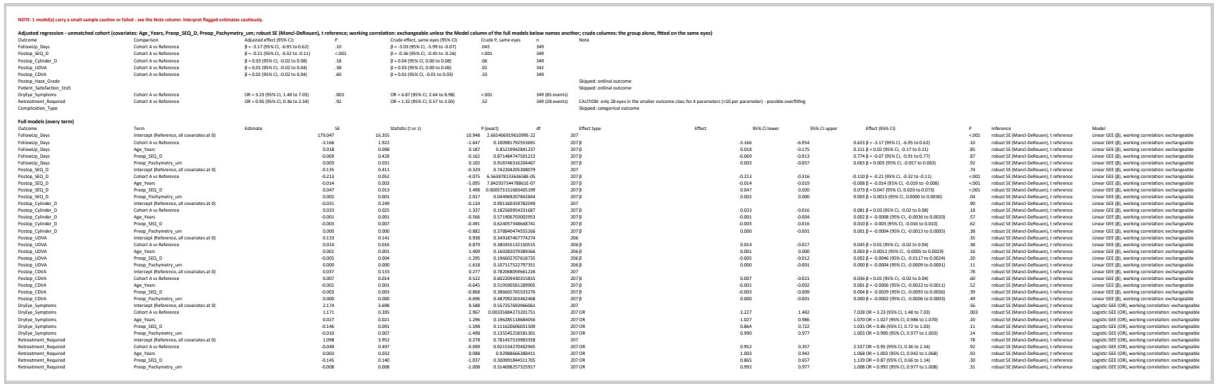
The clustered models: for each outcome and term the estimate, SE, statistic, exact *P*, df, the effect with its 95% CI, the formatted *P*, the inference ("robust SE (Mancl-DeRouen), t reference"), the model and its working correlation, and a Note column. The intercept of the main model is the reference group's mean or odds. Separate sheets hold the main and adjusted models of the unmatched cohort, one per table section (`GEE_Main_Unmatched_Base`, `_Preop`, `_Out`, and `GEE_Adj_Unmatched_…`), and of each matched pairing (`GEE_M_Cohort A_vs_Reference`, `GEE_Adj_M_Cohort A_vs_Reference`). The note row counts the per-patient models and the independence refits. In `GEE_Main_Unmatched_Base`, age is "Linear regression, per patient (one row per patient), 233 patients", sex and smoking are logistic regressions per patient, and mean keratometry is "GEE Gaussian, independence working correlation …

- exchangeable fit not accepted: estimated working correlation 0.984 (0.95 or more)". The df of the unmatched models of baseline variables is 231 = 233 patients − 2 coefficients.



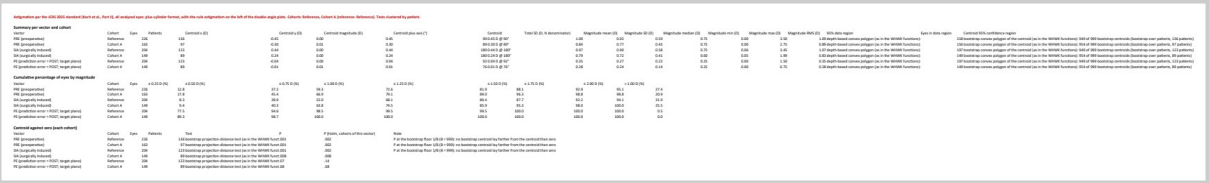
Regression (adjusted)

The adjusted table (outcome, comparison, adjusted effect and *P*, the crude effect and *P* on the same eyes, n, note), then the full models with every term. In Example 02 the crude estimate on the 349 complete-case eyes of the SEQ model is β = −0.36 against the adjusted −0.21; the retreatment model carries the caution "only 28 eyes in the smaller outcome class for 4 parameters (<10 per parameter) - possible overfitting".



Astigmatism (JCRS 2025)

The astigmatism statistics of A.14 for PRE, SIA and PE. In Example 02 the preoperative centroid is 0.45 D @ 90° in the Reference cohort (with-the-rule, x = −0.45) and 0.30 D @ 89° in Cohort A; the prediction-error centroids are 0.04 D @ 92° and 0.01 D @ 76°, with total SD 0.35 and 0.28 D. Each centroid is tested against zero: PRE and the Reference SIA reach the bootstrap floor (*P* = .001, said so in the Note column), the Reference PE gives *P* = .07 and Cohort A PE *P* = .68. Between the cohorts, the PE centroid (*P* = .82) and precision (*P* = .67) use the trimmed tests, while the mean magnitude ran untrimmed (*P* = .08). For the PE intervals from ≤ 0.75 D, every Cohort A eye is within, so the binomial GEE cannot be fitted and the permutation test over patients runs; from ≤ 1.50 D every eye of both cohorts is within, and nothing is tested. The McNemar table compares preoperative and postoperative refractive astigmatism of the same eyes at each interval: in the Reference cohort 132 eyes improved and none worsened at ≤ 0.25 D (cluster-adjusted McNemar, *P* < .001). The right- and left-eye centroids, the vertices of every polygon and the method notes close the sheet.



Astigmatism vectors

One row per eye and vector (PE, SIA, PRE, POST, TIA, DV): group, patient ID, row label, magnitude, plus axis, x and y.

Group	Patient ID	Row label	Vector	Magnitude (D)	Plus axis (°)	x (D)	y (D)
Reference	P0001	P0001-OD	PE	0.250	117.0	-0.147	-0.202
Reference	P0001	P0001-OS	PE	0.250	165.0	0.217	-0.125
Reference	P0002	P0002-OD	PE	0.500	168.0	0.457	-0.203
Cohort A	P0004	P0004-OD	PE	0.250	85.0	-0.246	0.043
Cohort A	P0004	P0004-OS	PE	0.250	107.0	-0.207	-0.140
Reference	P0006	P0006-OD	PE	0.250	156.0	0.167	-0.186
Reference	P0006	P0006-OS	PE	0.750	41.0	0.104	0.743
Cohort A	P0007	P0007-OD	PE	0.000	10.0	0.000	0.000
Reference	P0008	P0008-OD	PE	0.250	2.0	0.249	0.017
Reference	P0008	P0008-OS	PE	0.250	33.0	0.102	0.228
Reference	P0009	P0009-OD	PE	0.250	66.0	-0.167	0.186
Reference	P0010	P0010-OD	PE	0.250	156.0	0.167	-0.186
Reference	P0010	P0010-OS	PE	0.250	127.0	-0.069	-0.240
Reference	P0011	P0011-OD	PE	0.250	72.0	-0.202	0.147
Reference	P0011	P0011-OS	PE	0.250	138.0	0.026	-0.249
Reference	P0012	P0012-OD	PE	0.000	114.0	0.000	0.000
Cohort A	P0013	P0013-OD	PE	0.250	46.0	-0.009	0.250
Cohort A	P0013	P0013-OS	PE	0.250	138.0	0.026	-0.249
Cohort A	P0014	P0014-OD	PE	0.500	91.0	-0.500	-0.017
Cohort A	P0014	P0014-OS	PE	0.250	76.0	-0.221	0.117
Reference	P0015	P0015-OS	PE	0.500	121.0	-0.235	-0.441
Reference	P0016	P0016-OS	PE	0.000	131.0	0.000	0.000
Cohort A	P0017	P0017-OD	PE	0.500	3.0	0.497	0.052
Cohort A	P0017	P0017-OS	PE	0.250	7.0	0.243	0.060
Reference	P0018	P0018-OD	PE	0.000	84.0	0.000	0.000
Reference	P0018	P0018-OS	PE	0.000	42.0	0.000	0.000
Reference	P0019	P0019-OD	PE	0.500	127.0	-0.138	-0.481
Reference	P0019	P0019-OS	PE	0.250	74.0	-0.212	0.132
Reference	P0020	P0020-OD	PE	0.500	82.0	-0.481	0.138
Reference	P0020	P0020-OS	PE	0.250	27.0	0.147	0.202
Reference	P0022	P0022-OD	PE	0.250	105.0	-0.217	-0.125
Reference	P0022	P0022-OS	PE	0.250	71.0	-0.197	0.154

mEYEstro and AstigMATIC tabs

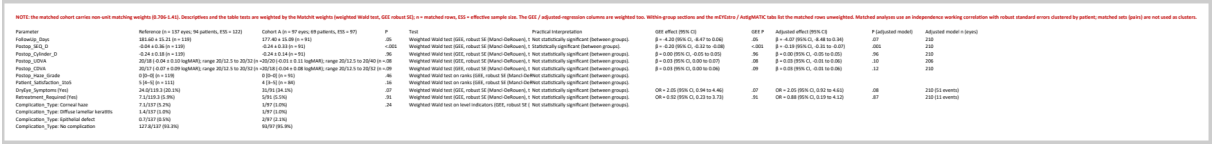
Worksheets laid out for direct entry into those tools, one sheet per group, for users who wish to reproduce the figures in the original software (A.15). The values are already standardized: minus cylinder, Snellen acuity in the mEYEstro tab, OD / OS in the AstigMATIC tab. The matched pairings have their own tabs, `mEYEstro_M1_<group>` and `AstigMATIC_M1_<group>` (M2 for a second pairing), whose full names the Contents sheet gives.

GroupLabel	Pre_BCVA	Post_UCVA	Post_BCVA	Pre_Sph	Pre_Cyl	Pre_Axis	Preop_MR_Vertex_Distance	Preop_CDVA	Intended_Postop_MR_Sphere	Intended_Postop_MR_Cylinder	Intended_Postop_MR_Axis	Intended_Postop_MR_Vertex_Distance	Post_Sph	Post_Cyl	Post_Axis	Postop_MR_Vertex_Distance	Postop_CDVA	Postop_UDVA	Post_SEQ
Reference	0.8	0.8	0.8	-4.5	-0.5	139	12 25		0	0	0	12	0.25	-0.25	27	12 25	25		0.125
Reference	0.8	0.8	0.8	-4.75	-0.75	41	12 25		0	0	0	12	0.5	-0.25	75	12 25	25		0.375
Reference	0.8	1	1	-3	-0.75	70	12 25		0	0	0	12	0.25	-0.5	78	12 20	20		0
Reference	0.8	1	1	-4	-0.5	179	12 25		0	0	0	12	0.25	-0.25	66	12 20	20		0.125
Reference	0.8	0.8	1	-4	-1	170	12 25		0	0	0	12	0.5	-0.75	131	12 20	25		0.125
Reference	1	1.25	1.25	-3.5	-0.75	95	12 20		0	0	0	12	0	-0.25	92	12 15-1	15-1		-0.125
Reference	1	1.25	1.25	-4.75	-0.5	106	12 20		0	0	0	12	0.25	-0.25	123	12 15-1	15-1		0.125
Reference	1	1.25	1.25	-3.75	-0.75	4	12 20		0	0	0	12	0	-0.25	156	12 15-1	15-1		-0.125
Reference	1	1	1.25	-5	-1.75	23	12 20		0	0	0	12	-0.5	-0.25	66	12 15-1	20		-0.625
Reference	0.8	1	1	-6.5	-1.5	172	12 25		0	0	0	12	-0.25	-0.25	37	12 20	20		-0.375
Reference	1	1.25	1.25	-5	-1.25	171	12 20		0	0	0	12	0.75	-0.25	162	12 15-1	15-1		0.625
Reference	1	1	1	-4.5	-0.5	22	12 20		0	0	0	12	0.75	-0.25	48	12 20	20		0.625
Reference	1.25	1.6	1.6	-1.25	-0.25	165	12 15-1		0	0	0	12	0.25	0	114	12 12.5	12.5		0.25
Reference	1.25	1	1.25	-5	-0.75	50	12 15-1		0	0	0	12	0.25	-0.5	31	12 15-1	20		0
Reference	1	1.25	1.6	-3	-0.5	85	12 20		0	0	0	12	0.5	0	131	12 12.5	15-1		0.5
Reference	1	1	1	-1.5	-1.5	57	12 20		0	0	0	12	0.5	0	84	12 20	20		0.5
Reference	1	1	1	-4	-1	128	12 20		0	0	0	12	0	0	42	12 20	20		0
Reference	1.25	1.25	1.25	-1	-1.25	15	12 15-1		0	0	0	12	1	-0.5	37	12 15-1	15-1		0.75
Reference	1.25	1.25	1.25	-1.5	-0.5	6	12 15-1		0	0	0	12	0.25	-0.25	164	12 15-1	15-1		0.125
Reference	1.25	1.25	1.25	-2.25	-1.75	23	12 15-1		0	0	0	12	0.5	-0.5	172	12 15-1	15-1		0.25
Reference	1	1.25	1.25	-4	-0.75	149	12 20		0	0	0	12	0	-0.25	117	12 15-1	15-1		-0.125
Reference	1.25	1.25	1.25	-2	-1.75	108	12 15-1		0	0	0	12	0.5	-0.25	15	12 15-1	15-1		0.375
Reference	1.25	1.6	1.6	-1.25	-1	67	12 15-1		0	0	0	12	0.75	-0.25	161	12 12.5	12.5		0.625
Reference	1	1	1	-2	-1	109	12 20		0	0	0	12	0.25	0	159	12 20	20		0.25
Reference	1	1	1	-5.5	-0.25	14	12 20		0	0	0	12	-0.25	-0.25	84	12 20	20		-0.375
Reference	1			-4	-0.5	179	12 20		0	0	0	12				12			
Reference	1			-4.5	-0.5	176	12 20		0	0	0	12				12			
Reference	1.25	1.6	1.6	-2	-1.25	2	12 15-1		0	0	0	12	0.25	-0.25	157	12 12.5	12.5		0.125
Reference	1	1	1	-1.75	-1.5	176	12 20		0	0	0	12	0.5	-0.25	20	12 20	20		0.375
Reference	1	1	1	-1.5	-1	21	12 20		0	0	0	12	0	-0.25	52	12 20	20		-0.125
Reference	1.25	1.6	1.6	-2.5	-1	167	12 15-1		0	0	0	12	0.25	-0.25	142	12 12.5	12.5		0.125
Reference	0.8			-6	-0.75	116	12 25		0	0	0	12				12			
Reference	1			-4	-1	59	12 20		0	0	0	12				12			
Reference	1	1.25	1.25	-3	-1	101	12 20		0	0	0	12	0	0	58	12 15-1	15-1		0
Reference	1.25		1.6	-2.75	-1	83	12 15-1		0	0	0	12	0.25	-0.25	165	12 12.5			0.125
Reference	1.25	1.25	1.25	-2.25	-1	170	12 15-1		0	0	0	12	0.25	-0.25	70	12 15-1	15-1		0.125
Reference	1	1	1	-2	-1.25	170	12 20		0	0	0	12	0.25	-0.25	108	12 20	20		0.125
Reference	1	1	1	-3.75	-1.5	160	12 20		0	0	0	12	0.75	-0.5	106	12 20	20		0.5
Reference	1.25	1.6	1.6	-4	-0.5	11	12 15-1		0	0	0	12	0	0	111	12 12.5	12.5		0
Reference	1.25	1.25	1.25	-3.25	-0.75	19	12 15-1		0	0	0	12	0.5	-0.5	43	12 15-1	15-1		0.25
Reference	1	1	1	-4.75	-0.5	150	12 20		0	0	0	12	0.5	-0.25	148	12 20	20		0.375
Reference	1	1	1	-4	-2.5	173	12 20		0	0	0	12	0	-0.25	156	12 20	20		-0.125
Reference	1.25	1	1	-3.75	-1.5	4	12 15-1		0	0	0	12	-0.25	0	178	12 20	20		-0.25
Reference	0.8	0.8	0.8	-8	-1.25	44	12 25		0	0	0	12	0.75	-0.5	4	12 25	25		0.5
Reference	0.8	0.8	0.8	-7.5	-0.5	142	12 25		0	0	0	12	0	-0.25	175	12 25	25		-0.125
Reference	1.25	1	1.25	-1.5	-0.5	122	12 15-1		0	0	0	12	0.5	-0.25	85	12 15-1	20		0.375
Reference	1	1	1.25	-0.75	-0.5	11	12 20		0	0	0	12	0.25	-0.25	120	12 15-1	20		0.125
Reference	1	1	1.25	-0.25	-0.75	153	12 20		0	0	0	12	0.75	-0.25	87	12 15-1	20		0.625
Reference	1	1	1	-0.75	-2.25	27	12 20		0	0	0	12	0	-0.25	178	12 20	20		-0.125
Reference	1	1	1	-0.25	-2.25	152	12 20		0	0	0	12	0	-0.25	172	12 20	20		-0.125
Reference	1	1.25	1.25	-7.25	-0.75	165	12 20		0	0	0	12	0	-0.25	133	12 15-1	15-1		-0.125
Reference	1	1	1	-6.25	-2	24	12 20		0	0	0	12	0	-0.5	24	12 20	20		-0.25
Reference	1	1.25	1.25	-4	-1	21	12 20		0	0	0	12	0.5	-0.5	156	12 15-1	15-1		0.25
Reference	1	1.6	1.6	-1.75	-1.75	164	12 20		0	0	0	12	0.5	-0.5	24	12 12.5	12.5		0.25
Reference	1			-3	-0.75	9	12 20		0	0	0	12				12			
Reference	1	1.25	1.25	-3.75	-0.5	174	12 20		0	0	0	12	0.25	0	51	12 15-1	15-1		0.25
Reference	0.8	1	1	-5	-0.5	2	12 25		0	0	0	12	0.5	0	99	12 20	20		0.5
Reference	1.25	1.25	1.25	-3.5	-0.5	18	12 15-1		0	0	0	12	0	-0.25	37	12 15-1	15-1		-0.125
Reference	1.25	1.25	1.25	-4	-0.75	171	12 15-1		0	0	0	12	0	-0.25	2	12 15-1	15-1		-0.125
Reference	1	1.25	1.6	-5.5	-1	156	12 20		0	0	0	12	0.5	-0.5	126	12 12.5	15-1		0.25
Reference	1.25	1.6	1.6	-5.75	-1.25	7	12 15-1		0	0	0	12	0.25	-0.25	35	12 12.5	12.5		0.125
Reference	1	0.5	0.5	-2.25	-0.25	52	12 20		0	0	0	12	0.5	-0.25	144	12 40	40		0.375
Reference	1	0.5	0.5	-3.5	-0.25	133	12 20		0	0	0	12	-0.25	-0.25	19	12 40	40		-0.375
Reference	1	1.25	1.25	-1	-0.5	176	12 20		0	0	0	12	0.5	-0.25	123	12 15-1	15-1		0.375
Reference	1.25	1.25	1.25	-0.25	-0.5	11	12 15-1		0	0	0	12	0.5	-0.25	61	12 15-1	15-1		0.375
Reference	1.25	1.25	1.25	-2.5	-1	100	12 15-1		0	0	0	12	0.25	-0.25	161	12 15-1	15-1		0.125
Reference	1.25	1.25	1.25	-4	-0.75	86	12 15-1		0	0	0	12	-0.25	0	52	12 15-1	15-1		-0.25
Reference	1			-4.5	-0.75	177	12 20		0	0	0	12				12			
Reference	1			-5.25	-0.5	19	12 20		0	0	0	12				12			
Reference	1	1	1	-1.25	-3	158	12 20		0	0	0	12	0.5	-0.5	122	12 20	20		0.25
Reference	1	1.25	1.25	-1	-1.75	24	12 20		0	0	0	12	0.25	-0.25	52	12 15-1	15-1		0.125
Reference	1.25	1.25	1.25	-5.25	-1.25	179	12 15-1		0	0	0	12	0	-0.25	16	12 15-1	15-1		-0.125
Reference	1.25	1.6	1.6	-3.25	-0.5	18	12 15-1		0	0	0	12	0.25	-0.25	141	12 12.5	12.5		0.125
Reference	1.25	1.25	1.25	-3.75	-1	162	12 15-1		0	0	0	12	0	0	45	12 15-1	15-1		0
Reference	1.25			-2.5	-0.25	36	12 15-1		0	0	0	12				12			
Reference	1			-3	-0.5	145	12 20		0	0	0	12				12			
Reference	1	0.63	0.8	-1	-1.75	179	12 20		-0.5	0	0	12	-0.25	-0.5	127	12 25	30-1		-0.5
Reference	1	1	1	-2.5	-2	1	12 20		0	0	0	12	0	-0.25	71	12 20	20		-0.125

Eye	preoperative cylinder	preoperative axis	postoperative cylinder	postoperative axis
OD	-0.5	139	-0.25	27
OS	-0.75	41	-0.25	75
OD	-0.75	70	-0.5	78
OD	-0.5	179	-0.25	66
OS	-1	170	-0.75	131
OD	-0.75	95	-0.25	92
OS	-0.5	106	-0.25	123
OD	-0.75	4	-0.25	156
OD	-1.75	23	-0.25	66
OS	-1.5	172	-0.25	37
OD	-1.25	171	-0.25	162
OS	-0.5	22	-0.25	48
OD	-0.25	165	0	114
OS	-0.75	50	-0.5	31
OS	-0.5	85	0	131
OD	-1.5	57	0	84
OS	-1	128	0	42
OD	-1.25	15	-0.5	37
OS	-0.5	6	-0.25	164
OD	-1.75	23	-0.5	172
OS	-0.75	149	-0.25	117
OD	-1.75	108	-0.25	15
OS	-1	67	-0.25	161
OD	-1	109	0	159
OS	-0.25	14	-0.25	84
OD	-0.5	179		
OS	-0.5	176		
OS	-1.25	2	-0.25	157
OD	-1.5	176	-0.25	20
OS	-1	21	-0.25	52
OD	-1	167	-0.25	142
OD	-0.75	116		

Matched-cohort sheets

The summary, publication, GEE, regression and astigmatism sheets are repeated for each matched pairing (`BasePreop_M_…`, `Out_M_…`, `Pub_M_…`, `GEE_M_…`, `GEE_Adj_M_…`, `Reg_M_…`, `Astig_M_…`). In Example 02 the matching weights are not all 1, so a note row says the tables are weighted, the headers add the effective sample size, `Reference (n = 137 eyes; 94 patients, ESS = 122)`, and every *P* value is a *Weighted Wald test (GEE, robust SE (Mancl-DeRouen), t reference)*. Matching brings the SEQ difference close to the true −0.20 D (β = −0.20, 95% CI, −0.32 to −0.08; adjusted −0.19), the dry-eye odds ratio to 2.05 (95% CI, 0.94 to 4.46), and the cylinder difference to 0.00 (*P* = .96). Comparing the unmatched and matched outcome sheets side by side is the clearest way to see what matching changed.

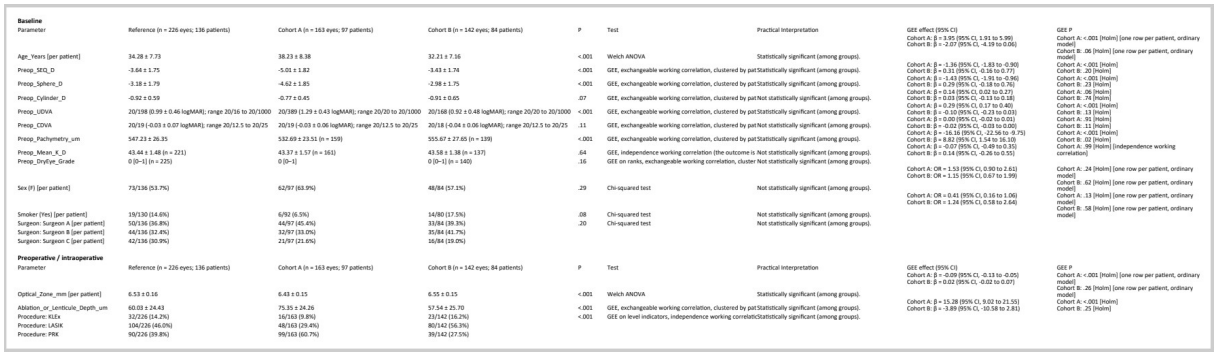

NOTE: the matched cohort carries non-unit matching weights (0.706–1.41). Descriptives and the table tests are weighted by the MatchIt weights (weighted Wald test, GEE robust SE); n = matched rows, ESS = effective sample size. The GEE / adjusted-regression columns are weighted too. Within-group sections and the mEYEstro / AstigMATIC tabs list the matched rows unweighted. Matched analyses use an independence working correlation with robust standard errors clustered by patient; matched sets (pairs) are not used as clusters.

Parameter	Reference (n = 137 eyes; 94 patients, ESS = 122)	Cohort A (n = 97 eyes; 69 patients, ESS = 97)	P	Test	Practical Interpretation	GEE effect (95% CI)	GEE P	Adjusted effect (95% CI)	P (adjusted model)	Adjusted model n (eyes)
Followup_Days	181.60 ± 15.21 (n = 119)	177.40 ± 15.09 (n = 91)	.05	Weighted Wald test (GEE, robust SE (Mancl-DeRouen), t	Not statistically significant (between groups).	β = -4.20 (95% CI, -8.47 to 0.06)	.05	β = -4.07 (95% CI, -8.48 to 0.34)	.07	210
Postop_SEQ_D	-0.04 ± 0.36 (n = 119)	-0.24 ± 0.33 (n = 91)	<.001	Weighted Wald test (GEE, robust SE (Mancl-DeRouen), t	Statistically significant (between groups).	β = -0.20 (95% CI, -0.32 to -0.08)	<.001	β = -0.19 (95% CI, -0.31 to -0.07)	.001	210
Postop_Cylinder_D	-0.24 ± 0.18 (n = 119)	-0.24 ± 0.14 (n = 91)	.96	Weighted Wald test (GEE, robust SE (Mancl-DeRouen), t	Not statistically significant (between groups).	β = 0.00 (95% CI, -0.05 to 0.05)	.96	β = 0.00 (95% CI, -0.05 to 0.05)	.96	210
Postop_UDVA	20/38 (-0.04 ± 0.10 logMAR); range 20/12.5 to 20/32 (n =20/20 (-0.01 ± 0.11 logMAR); range 20/12.5 to 20/40 (n =.08			Weighted Wald test (GEE, robust SE (Mancl-DeRouen), t	Not statistically significant (between groups).	β = 0.03 (95% CI, 0.00 to 0.07)	.08	β = 0.03 (95% CI, -0.01 to 0.06)	.10	206
Postop_CDVA	20/17 (-0.07 ± 0.09 logMAR); range 20/12.5 to 20/32 (n =20/18 (-0.04 ± 0.08 logMAR); range 20/12.5 to 20/32 (n =.09			Weighted Wald test (GEE, robust SE (Mancl-DeRouen), t	Not statistically significant (between groups).	β = 0.03 (95% CI, 0.00 to 0.06)	.09	β = 0.03 (95% CI, -0.01 to 0.06)	.12	210
Postop_Haze_Grade	0 [0–0] (n = 119)	0 [0–0] (n = 91)	.46	Weighted Wald test on ranks (GEE, robust SE (Mancl-DeRouen), t	Not statistically significant (between groups).					
Patient_Satisfaction_Visit	5 [4–5] (n = 111)	4 [4–5] (n = 84)	.16	Weighted Wald test on ranks (GEE, robust SE (Mancl-DeRouen), t	Not statistically significant (between groups).					
Ocular_Symptoms (Yes)	24 (21.0 [13.2,30.1%)	31 (51 [34.1%)	.07	Weighted Wald test (GEE, robust SE (Mancl-DeRouen), t	Not statistically significant (between groups).	OR = 2.05 (95% CI, 0.94 to 4.46)	.07	OR = 2.05 (95% CI, 0.92 to 4.61)	.08	210 (11 events)
Retreatment_Required (Yes)	7 (1/110.3 (1.3%)	5/91 (1.1%)	.91	Weighted Wald test (GEE, robust SE (Mancl-DeRouen), t	Not statistically significant (between groups).	OR = 0.92 (95% CI, 0.23 to 3.73)	.91	OR = 0.88 (95% CI, 0.19 to 4.12)	.87	210 (11 events)
Complication_Type: Corneal haze	7 (1/137 (5.2%)	1/97 (1.0%)	.24	Weighted Wald test on level indicators (GEE, robust SE (	Not statistically significant (between groups).					
Complication_Type: Diffuse lamellar keratitis	1.4/137 (1.0%)	1/97 (1.0%)								
Complication_Type: Epithelial defect	0.7/137 (0.5%)	2/97 (2.1%)								
Complication_Type: No complication	127.8/137 (93.3%)	93/97 (95.9%)								

What changes with three cohorts

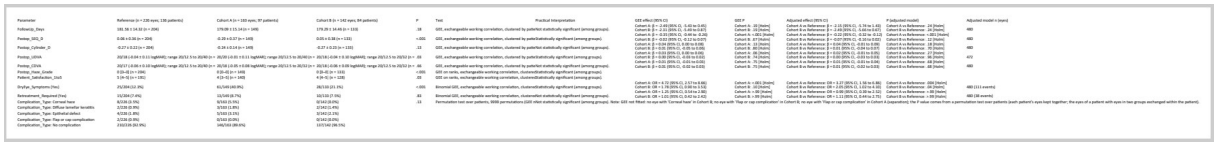
The baseline sheet of Example 07 carries three group columns and a single **omnibus** *P* value per variable: the joint GEE test with an *F* reference for the per-eye rows, Welch ANOVA or χ^2 for the per-patient rows. The

interpretation reads "among groups". The GEE column holds one line per comparison against the reference, Holm-adjusted: for preoperative SEQ, "Cohort A: <.001 [Holm]" and "Cohort B: .20 [Holm]".



Baseline Parameter	Reference (n = 226 eyes; 136 patients)	Cohort A (n = 163 eyes; 97 patients)	Cohort B (n = 142 eyes; 84 patients)	P	Test	Practical Interpretation	GEE effect (95% CI)	GEE P
							Cohort A: β = 3.95 (95% CI, 1.91 to 5.99) Cohort B: β = -2.07 (95% CI, -4.19 to 0.06)	Cohort A: <.001 [Holm] [one row per patient, ordinary model] Cohort B: .06 [Holm] [one row per patient, ordinary model]
Age_Years [per patient]	34.28 ± 7.73	38.23 ± 8.38	32.21 ± 7.16	<.001	Welch ANOVA	Statistically significant (among groups).		
Preop_SEQ_D	-3.64 ± 1.75	-5.01 ± 1.82	-3.43 ± 1.74	<.001	GEE, exchangeable working correlation, clustered by pat	Statistically significant (among groups).	Cohort A: β = -1.36 (95% CI, -1.83 to -0.90) Cohort B: β = 0.31 (95% CI, -0.16 to 0.77)	Cohort A: <.001 [Holm] Cohort B: .20 [Holm]
Preop_Sphere_D	-3.18 ± 1.79	-4.62 ± 1.85	-2.98 ± 1.75	<.001	GEE, exchangeable working correlation, clustered by pat	Statistically significant (among groups).	Cohort A: β = -1.43 (95% CI, -1.91 to -0.96) Cohort B: β = 0.29 (95% CI, -0.18 to 0.76)	Cohort A: <.001 [Holm] Cohort B: .23 [Holm]
Preop_Cylinder_D	-0.92 ± 0.59	-0.77 ± 0.45	-0.91 ± 0.65	.07	GEE, exchangeable working correlation, clustered by pat	Not statistically significant (among groups).	Cohort A: β = 0.14 (95% CI, 0.02 to 0.27) Cohort B: β = 0.03 (95% CI, -0.13 to 0.18)	Cohort A: .06 [Holm] Cohort B: .74 [Holm]
Preop_UDVA	20/198 (0.99 ± 0.46 logMAR); range 20/16 to 20/1000	20/389 (1.29 ± 0.43 logMAR); range 20/20 to 20/1000	20/168 (0.92 ± 0.48 logMAR); range 20/20 to 20/1000	<.001	GEE, exchangeable working correlation, clustered by pat	Statistically significant (among groups).	Cohort A: β = 0.29 (95% CI, 0.17 to 0.40) Cohort B: β = -0.10 (95% CI, -0.23 to 0.03)	Cohort A: <.001 [Holm] Cohort B: .13 [Holm]
Preop_CDVA	20/19 (-0.03 ± 0.07 logMAR); range 20/12.5 to 20/25	20/19 (-0.03 ± 0.06 logMAR); range 20/12.5 to 20/25	20/18 (-0.04 ± 0.06 logMAR); range 20/12.5 to 20/25	.11	GEE, exchangeable working correlation, clustered by pat	Not statistically significant (among groups).	Cohort A: β = 0.00 (95% CI, -0.02 to 0.01) Cohort B: β = -0.02 (95% CI, -0.03 to 0.00)	Cohort A: .91 [Holm] Cohort B: .11 [Holm]
Preop_Pachymetry_um	547.23 ± 26.35	532.69 ± 23.51 (n = 159)	555.67 ± 27.65 (n = 139)	<.001	GEE, exchangeable working correlation, clustered by pat	Statistically significant (among groups).	Cohort A: β = -16.16 (95% CI, -22.56 to -9.75) Cohort B: β = 8.82 (95% CI, 1.54 to 16.10)	Cohort A: <.001 [Holm] Cohort B: .02 [Holm]
Preop_Mean_K_D	43.44 ± 1.48 (n = 221)	43.37 ± 1.57 (n = 161)	43.58 ± 1.38 (n = 137)	.64	GEE, independence working correlation (the outcome is N	Not statistically significant (among groups).	Cohort A: β = -0.07 (95% CI, -0.49 to 0.35) Cohort B: β = 0.14 (95% CI, -0.26 to 0.55)	Cohort A: .99 [Holm] [independence working correlation]
Preop_DryEye_Grade	0 [0–1] (n = 225)	0 [0–1]	0 [0–1] (n = 140)	.16	GEE on ranks, exchangeable working correlation, cluster	Not statistically significant (among groups).		
Sex (F) [per patient]	73/136 (53.7%)	62/97 (63.9%)	48/84 (57.1%)	.29	Chi-squared test	Not statistically significant (among groups).	Cohort A: OR = 1.53 (95% CI, 0.90 to 2.61) Cohort B: OR = 1.15 (95% CI, 0.67 to 1.99)	Cohort A: .24 [Holm] [one row per patient, ordinary model] Cohort B: .62 [Holm] [one row per patient, ordinary model]
Smoker (Yes) [per patient]	19/130 (14.6%)	6/92 (6.5%)	14/80 (17.5%)	.08	Chi-squared test	Not statistically significant (among groups).	Cohort A: OR = 0.41 (95% CI, 0.16 to 1.06) Cohort B: OR = 1.24 (95% CI, 0.58 to 2.64)	Cohort A: .13 [Holm] [one row per patient, ordinary model] Cohort B: .58 [Holm] [one row per patient, ordinary model]
Surgeon: Surgeon A [per patient]	50/136 (36.8%)	44/97 (45.4%)	31/84 (39.3%)	.20	Chi-squared test	Not statistically significant (among groups).		
Surgeon: Surgeon B [per patient]	44/136 (32.4%)	32/97 (33.0%)	35/84 (41.7%)					
Surgeon: Surgeon C [per patient]	42/136 (30.9%)	21/97 (21.6%)	16/84 (19.0%)					
Preoperative / intraoperative Parameter	Reference (n = 226 eyes; 136 patients)	Cohort A (n = 163 eyes; 97 patients)	Cohort B (n = 142 eyes; 84 patients)	P	Test	Practical Interpretation	GEE effect (95% CI)	GEE P
							Cohort A: β = -0.09 (95% CI, -0.13 to -0.05) Cohort B: β = 0.02 (95% CI, -0.02 to 0.07)	Cohort A: <.001 [Holm] [one row per patient, ordinary model] Cohort B: .26 [Holm] [one row per patient, ordinary model]
Optical_Zone_mm [per patient]	6.53 ± 0.16	6.43 ± 0.15	6.55 ± 0.15	<.001	Welch ANOVA	Statistically significant (among groups).		
Ablation_or_Lenticule_Depth_um	60.03 ± 24.43	75.35 ± 24.26	57.54 ± 25.70	<.001	GEE, exchangeable working correlation, clustered by pat	Statistically significant (among groups).	Cohort A: β = 15.28 (95% CI, 9.02 to 21.55) Cohort B: β = -3.89 (95% CI, -10.58 to 2.81)	Cohort A: <.001 [Holm] Cohort B: .25 [Holm]
Procedure: KLEx	32/226 (14.2%)	16/163 (9.8%)	23/142 (16.2%)	<.001	GEE on level indicators, independence working correlati	Statistically significant (among groups).		
Procedure: LASIK	104/226 (46.0%)	48/163 (29.4%)	80/142 (56.3%)					
Procedure: PRK	90/226 (39.8%)	99/163 (60.7%)	39/142 (27.5%)					

In the outcome sheet the omnibus *P* is still one value, while the GEE and adjusted cells hold one line per comparison. Postoperative SEQ differs among the groups (*P* < .001); the adjusted comparisons are β = −0.22 for Cohort A (*P* < .001 [Holm]) and β = −0.07 for Cohort B (*P* = .12 [Holm]), against true values of −0.20 and −0.10 D.

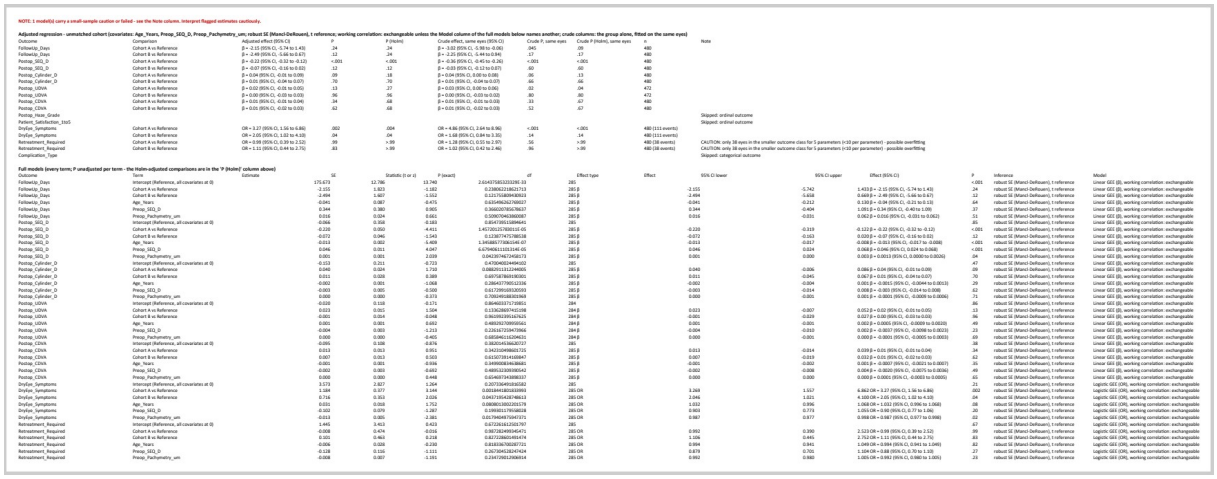


Parameter	Reference (n = 226 eyes; 136 patients)	Cohort A (n = 163 eyes; 97 patients)	Cohort B (n = 142 eyes; 84 patients)	P	Test	Practical Interpretation	GEE effect (95% CI)	GEE P	Adjusted effect (95% CI)	P (adjusted model)	Adjusted model n (eyes)
Follow-Up_Days	181.56 ± 14.32 (n = 204)	179.09 ± 15.14 (n = 149)	179.29 ± 14.46 (n = 133)	.18	GEE, exchangeable working correlation, clustered by pat	Statistically significant (among groups).	Cohort A: β = -2.49 (95% CI, -5.43 to 0.45) Cohort B: β = -2.31 (95% CI, -5.49 to 0.87) Cohort A: β = -0.35 (95% CI, -0.44 to -0.26) Cohort B: β = -0.02 (95% CI, -0.12 to 0.07) Cohort A: β = 0.04 (95% CI, 0.00 to 0.08) Cohort B: β = 0.01 (95% CI, -0.05 to 0.06) Cohort A: β = 0.03 (95% CI, 0.00 to 0.06) Cohort B: β = 0.00 (95% CI, -0.03 to 0.03) Cohort A: β = 0.01 (95% CI, -0.02 to 0.03) Cohort B: β = 0.01 (95% CI, -0.02 to 0.03)	Cohort A: .19 [Holm] Cohort B: .19 [Holm] Cohort A: <.001 [Holm] Cohort B: .67 [Holm] Cohort A: .13 [Holm] Cohort B: .80 [Holm] Cohort A: .06 [Holm] Cohort B: .74 [Holm] Cohort A: .40 [Holm] Cohort B: .75 [Holm]	Cohort A vs Reference: β = -2.15 (95% CI, -5.74 to 1.43) Cohort B vs Reference: β = -2.49 (95% CI, -5.66 to 0.67) Cohort A vs Reference: β = -0.22 (95% CI, -0.32 to -0.13) Cohort B vs Reference: β = -0.07 (95% CI, -0.16 to 0.02) Cohort A vs Reference: β = -0.05 (95% CI, -0.09 to -0.01) Cohort B vs Reference: β = 0.01 (95% CI, -0.04 to 0.07) Cohort A vs Reference: β = 0.02 (95% CI, -0.01 to 0.05) Cohort B vs Reference: β = 0.00 (95% CI, -0.03 to 0.04) Cohort A vs Reference: β = 0.02 (95% CI, -0.01 to 0.05) Cohort B vs Reference: β = 0.01 (95% CI, -0.02 to 0.03)	Cohort A vs Reference: .24 [Holm] Cohort A vs Reference: .24 [Holm] Cohort A vs Reference: <.001 [Holm] Cohort A vs Reference: .12 [Holm] Cohort A vs Reference: .18 [Holm] Cohort A vs Reference: .70 [Holm] Cohort A vs Reference: .27 [Holm] Cohort A vs Reference: .96 [Holm] Cohort A vs Reference: .68 [Holm] Cohort A vs Reference: .68 [Holm]	480 480 472 480
Postop_SEQ_D	0.06 ± 0.36 (n = 204)	-0.29 ± 0.37 (n = 149)	0.05 ± 0.38 (n = 133)	<.001							
Postop_Cylinder_D	-0.27 ± 0.22 (n = 204)	-0.24 ± 0.14 (n = 149)	-0.27 ± 0.23 (n = 133)	.13							
Postop_UDVA	20/18 (-0.04 ± 0.11 logMAR); range 20/12.5 to 20/40 (n = 20/20 (-0.01 ± 0.11 logMAR); range 20/12.5 to 20/40 (n = 20/18 (-0.04 ± 0.10 logMAR); range 20/12.5 to 20/32 (n = 69										
Postop_CDVA	20/17 (-0.06 ± 0.10 logMAR); range 20/12.5 to 20/40 (n = 20/18 (-0.05 ± 0.08 logMAR); range 20/12.5 to 20/32 (n = 20/18 (-0.06 ± 0.09 logMAR); range 20/12.5 to 20/32 (n = 66										
Postop_Haze_Grade	0 [0–0] (n = 204)	0 [0–0] (n = 149)	0 [0–0] (n = 133)	<.001							
Patient_Satisfaction_1to5	5 [4–5] (n = 191)	4 [3–5] (n = 140)	4 [4–5] (n = 128)	.03							
DryEye_Symptoms (Yes)	25/204 (12.3%)	61/149 (40.9%)	29/133 (21.1%)	<.001	Binomial GEE, exchangeable working correlation, cluster	Statistically significant (among groups).	Cohort A: OR = 4.72 (95% CI, 2.57 to 8.66) Cohort B: OR = 1.78 (95% CI, 0.92 to 3.53) Cohort A: OR = 1.25 (95% CI, 0.54 to 2.90) Cohort B: OR = 1.01 (95% CI, 0.42 to 2.42)	Cohort A: <.001 [Holm] Cohort B: .10 [Holm] Cohort A: >.99 [Holm] Cohort B: >.99 [Holm]	Cohort A vs Reference: OR = 3.27 (95% CI, 1.56 to 6.86) Cohort B vs Reference: OR = 2.05 (95% CI, 1.02 to 4.10) Cohort A vs Reference: OR = 0.98 (95% CI, 0.39 to 2.52) Cohort B vs Reference: OR = 1.11 (95% CI, 0.44 to 2.75)	Cohort A vs Reference: .004 [Holm] Cohort B vs Reference: .04 [Holm] Cohort A vs Reference: >.99 [Holm] Cohort B vs Reference: >.99 [Holm]	480 (111 events) 480 (38 events)
Retreatment_Required (Yes)	15/204 (7.4%)	13/149 (8.7%)	10/133 (7.5%)	.83	Binomial GEE, exchangeable working correlation, cluster	Statistically significant (among groups).	Cohort A: OR = 1.20 (95% CI, 0.54 to 2.66) Cohort B: OR = 1.00 (95% CI, 0.42 to 2.40)	Cohort A: >.99 [Holm] Cohort B: >.99 [Holm]	Cohort A vs Reference: OR = 1.01 (95% CI, 0.39 to 2.52) Cohort B vs Reference: OR = 1.11 (95% CI, 0.44 to 2.75)	Cohort A vs Reference: >.99 [Holm] Cohort B vs Reference: >.99 [Holm]	480 (38 events)
Complication_Type: Corneal haze	8/226 (3.5%)	9/163 (5.5%)	0/142 (0.0%)	.13							
Complication_Type: Diffuse lamellar keratitis	2/226 (0.9%)	3/163 (1.8%)	2/142 (1.4%)								
Complication_Type: Epithelial defect	4/226 (1.8%)	5/163 (3.1%)	3/142 (2.1%)								
Complication_Type: Flap or cap complication	2/226 (0.9%)	0/163 (0.0%)	0/142 (0.0%)								
Complication_Type: No complication	210/226 (92.9%)	146/163 (89.6%)	137/142 (96.5%)								

Values:
- Values are mean ± SD, median [interquartile range], n/N (%) - eyes (or patients) with the level among those with data.
- Visual acuity: the Snellen geometric mean (logMAR mean ± SD), then the range of the eyes in Snellen.
- (n = …) in a cell: the eyes (patients, for a row marked [per patient]) with data, where fewer than the column's n.
- β = difference from Reference in the variable's units (logMAR for acuity), with its 95% CI.
- OR = odds ratio of the level named in the row vs Reference, with its 95% CI.
- [Holm] = P Holm-adjusted across the comparisons with the reference group; the 95% CIs are not adjusted, so a CI can exclude the null while the adjusted P is .05 or more.
- Adjusted model n (eyes): the complete-case eyes of the adjusted model.

Tests behind the P values:
- Welch ANOVA: Age_Years; Optical_Zone_mm.
- GEE, exchangeable working correlation, clustered by patient (robust SE (Mancl-DeRouen), F reference): Preop_SEQ_D; Preop_Sphere_D; Preop_Cylinder_D; Preop_UDVA; Preop_CDVA; Preop_Pachymetry_um; Ablation_or_Lenticule_Depth_um; FollowUp_Days; Postop_SEQ_D; Postop_Cylinder_D; Postop_UDVA; Postop_CDVA.
- GEE, independence working correlation (the outcome is almost constant within patients), clustered by patient (robust SE (Mancl-DeRouen), F reference): Preop_Mean_K_D.
- GEE on ranks, exchangeable working correlation, clustered by patient (robust SE (Mancl-DeRouen), F reference): Preop_DryEye_Grade; Postop_Haze_Grade; Patient_Satisfaction_1to5.
- Chi-squared test: Sex; Smoker; Surgeon.
- GEE on level indicators, independence working correlation, clustered by patient (robust SE (Mancl-DeRouen), F reference): Procedure.
- Binomial GEE, exchangeable working correlation, clustered by patient (robust SE (Mancl-DeRouen), F reference): DryEye_Symptoms; Retreatment_Required.
- Permutation test over patients, 9999 permutations (GEE not fitted): Complication_Type.

The regression sheet lists one comparison per group, with *P (Holm)* and the crude estimates on the same eyes. It also shows why true effects and sampling variation matter: the adjusted dry-eye odds ratio of Cohort B is 2.05 (95% CI, 1.02 to 4.10; *P* = .04, Holm .04), although the true value is 1.0 (B.1).



B.6 Reading the output well

1. **Check the Overview first.** Wrong group sizes are almost always a grouping or filter problem, and everything downstream inherits the error.
2. **Read the Test column (or the Publication Table's test footnote).** It tells you, and your reviewer, exactly what produced each *P* value.
3. **If fellow eyes are present, keep the clustered tests.** Eye-level *P* values are anticonservative in that situation.
4. **Check post-matching balance before believing the matched comparison**, and adjust for what is still above 0.1.
5. **Read the Holm-adjusted *P* values when several groups are compared**, and pre-specify a primary outcome.
6. **Do not report an odds ratio from a zero-event cell.** Give the counts and the exact test instead.
7. **Keep the settings file (*Download config JSON*), the Manifest and the filtered CSV with the manuscript.** Together they make the analysis exactly reproducible.

How to cite

If StandardEyes contributes to a publication, please cite:

> Noyman DBE, Mimouni M, Safir M. StandardEyes: free software for standardized comparative analysis and outcomes reporting in refractive surgery. Version 1.1.0. Manuscript in preparation (2026).

StandardEyes 1.1.0 · By Noyman DBE, Mimouni M, Safir M